

Baseline Study on the State of Early Childhood Development in the Cities of Lusaka and Chongwe, Zambia



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List of Acronyms and Abbreviations

Acronym	Full Meaning
AMALI	Africa Mayoral Leadership Initiative
CBO	Community-Based Organization
CREDI	Caregiver-Reported Early Development Index
CSO	Central Statistical Office / Civil Society Organization (<i>context-dependent</i>)
ECD	Early Childhood Development
ECE	Early Childhood Education
ECDAN	Early Childhood Development Action Network
ECDAN-Z	Early Childhood Development Action Network – Zambia
FGDs	Focus Group Discussions
HH	Household
HIV	Human Immunodeficiency Virus
HOME	Home Observation for Measurement of the Environment (Inventory)
IDELA	International Development and Early Learning Assessment
IRB	Institutional Review Board
KII	Key Informant Interview
MELE	Measuring Early Learning Environments
M&E	Monitoring and Evaluation (<i>implied</i>)
NGO	Non-Governmental Organization
SSA	Sub-Saharan Africa
UNZAHSS REC	University of Zambia Humanities and Social Sciences Research Ethics Committee
UNICEF	United Nations Children’s Fund
ZAMSTAT	Zambia Statistics Agency
ZMK	Zambian Kwacha
ZDHS	Zambia Demographic and Health Survey

Executive Summary

Background

Early Childhood Development (ECD) in sub-Saharan Africa, including Zambia, faces persistent challenges due to limited government leadership, fragmented service provision, and constrained resources. In Zambia, the mayors of Lusaka and Chongwe with support from international partners such as Big Win Philanthropy—through initiatives like the Africa Mayoral Leadership Initiative (AMALI)—have committed to strengthening ECD through a holistic and city-led approach. This baseline study was commissioned to assess the current status of ECD in these two cities, laying a foundation for evidence-based planning and future evaluation.

Methodology

The study adopted a cross-sectional design employing both quantitative and qualitative methods across Lusaka and Chongwe. Stratified random sampling was used to select a representative sample of 1,086 households from low-, medium-, and high-income residential areas. Additionally, 84 ECD centers were observed, 13 focus group discussions (FGDs) were conducted with caregivers, and 15 key informant interviews (KIIs) with stakeholders from education, health, civil society, and local governance institutions were held.

Quantitative tools included the Caregiver-Reported Early Development Index (CREDI) for children aged 0–3, the International Development and Early Learning Assessment (IDELA) for children 4–6, anthropometric measures, and the HOME Inventory. Qualitative insights were analyzed thematically. Ethical clearance was obtained from the University of Zambia Research Ethics Committee.

Key Findings by Thematic Area

1. Early Childhood Development Outcomes

- **Developmental Performance:** in both cities, children aged 0–17 months in both cities performed above the normative reference, but developmental scores dropped significantly after 17 months. Children in both planned and unplanned settlements scored below global norms after this age.
- **Early learning opportunities:** Only 10% of children aged 0–3 participated in any ECD activity, whereas over 67% of children aged 4–6 attended preschool. Participation was higher in urban areas and among families in planned settlements.
- **School readiness:** About 58% of children in Lusaka and 55% of children in Chongwe are school ready. Developmental progression on IDELA indicates that 28%-46 % of 3.5 to 6 years old children in Lusaka and Chongwe are struggling while 35% to 42% are in the merging stage. Mastery rates are critically low ranging from 21.8% to 25.4% across the domains. Motor skills are comparatively stronger were as literacy is the weakest among the children.
- **Predictors of Development:** A better **home environment**—especially access to learning materials, positive discipline, and responsive parenting—were strongly associated with child development. Neither settlement type nor household vulnerability independently predicted child outcomes.
- **Nutrition:** Nutritional status was poor, with **30% stunted, 11% underweight, and 9% wasted**—higher than national averages. Stunting was more prevalent in Chongwe (38%) than Lusaka (27%) and was associated with low caregiver education and high vulnerability.

2. Household Caregiving Practices

- **Caregiver Roles:** Over 94% of primary caregivers were female, with most being biological mothers. Fathers were primary financial providers (65.8%).
- **Knowledge and Attitudes:** Caregiver knowledge on child development was suboptimal, especially in unplanned and rural settlements. The “Caregiver Knowledge of Child Development” tool yielded a low reliability score ($\alpha = .41$), indicating potential gaps in understanding.
- **Caregiving Practices:** Many children under 3 were left in the care of older siblings, grandmothers, or neighbors while mothers worked. Engagement in **responsive parenting** and availability of learning resources varied widely across settlements.
- **Discipline and Protection:** Though responsive parenting was associated with better development outcomes, physical punishment and psychological aggression were still reported. Caregivers in unplanned settlements were more likely to employ harsh disciplinary methods.
- **Nutrition and Breastfeeding:** Most children were introduced to solid foods early, and exclusive breastfeeding rates appeared low. While over 50% of caregivers received nutrition counseling, willingness to pay for improved nutrition services remained limited.

3. Early Childhood Development Services

- **Availability and Access:** ECD centers were unevenly distributed, with most located in high- and medium-cost urban areas. Very few functional centers existed in low-income or rural communities.
- **Quality of Services:** Observations of 84 centers revealed significant disparities in infrastructure, staffing, and service quality. Many centers lacked stimulating environments, trained caregivers, and essential learning materials.
- **Service Providers:** The ECD workforce was largely composed of underqualified caregivers. Despite government mandates requiring early childhood units in primary schools, these remained under-resourced and inconsistently operational.
- **Utilization Patterns:** Usage of services was lower in Chongwe and among low-income households, largely due to affordability, distance, and lack of awareness.

4. Structural and Policy Landscape

- **Policy Environment:** Zambia has a robust but fragmented ECD policy framework. Key policies span health, nutrition, education, and child protection, including the 2022 Children’s Code Act and the National Health and Education Policies.
- **Coordination:** Multisectoral coordination exists but is weak, especially at the local level. Most ECD programs remain donor-driven, with limited integration into national systems or budgets.
- **Barriers and Enablers:** Key barriers include inadequate financing, poor infrastructure, and lack of trained personnel. However, growing political will and community-level support present strong opportunities for reform.
- **Urban-Rural Disparities:** Services implementation were significantly better in Lusaka than in Chongwe, particularly in unplanned and rural settlements, exacerbating developmental inequities.

Recommendations

The following recommendations have been proposed to improve child development in the two cities. The recommendations are organized around four thematic areas:

Early Childhood Development Outcomes

- **Prioritize developmental support for children especially after 17 months:** Tailor interventions to focus on the critical drop in development scores observed after 17 months of age by intensifying stimulation and responsive caregiving during this window while maintaining momentum for the 1000 critical days.
- **Integrate developmental screening into routine child health visits:** Ensure regular developmental assessments in child delivery services to identify at-risk children early and guide targeted support.
- **Expand direct school readiness programs:** Increase structured ECD programs for children aged 3–6 to promote early cognitive, motor, and socio-emotional skills, particularly in underserved areas.
- **Invest in interventions targeting stunting and undernutrition:** Design nutrition-sensitive ECD programs—such as feeding schemes, parenting education on infant diets, and micronutrient supplementation—especially for children under five, where stunting affects up to 38% in Chongwe.
- **Strengthen household-level awareness of child milestones:** Develop community-based caregiver education campaigns to increase recognition of developmental milestones and the importance of early intervention.

Household Caregiving Practices

- **Promote responsive caregiving:** Train and support parents and secondary caregivers (siblings, grandmothers, neighbors) on positive parenting and stimulation using home-based learning materials.
- **Caregiver education on child development:** Provide simplified parenting materials and community support programs for caregivers with limited knowledge on child development.
- **Promote positive discipline methods:** Scale up parenting programs that teach alternatives to physical and psychological punishment, especially in unplanned and rural settlements where such practices are more prevalent.
- **Leverage fathers' provider role for caregiving awareness:** Engage fathers in ECD messaging to expand their role beyond economic provision to include stimulation and interaction with children.
- **Support informal caregivers:** Recognize the role of siblings and extended family in childcare by creating community-based learning hubs or playgroups that engage these caregivers.
- **Promote exclusive breastfeeding and nutrition practices:** Increase household-level nutrition education and breastfeeding promotion using local health and community workers, especially where early introduction to solids is common.

Early Childhood Development Services

- **Increase access in low-income and rural areas:** Construct low-cost, community-run ECD centers with support from local government and NGOs, prioritizing underserved areas.
- **Improve quality of ECD centers:** Sustain providing training for ECD teachers and caregivers, ensuring centers meet minimum standards for learning materials, safety, and stimulation.
- **Play and recreational facilities:** Construct play and recreational facilities in communities using locally available and sustainable materials. Designate ECD week in which various activities that support ECD will be championed and implemented using a community driven approach.
- **Establish workplace-linked daycare programs:** Incentivize employers to offer or co-fund ECD centers as part of workplace benefits, particularly in urban settings.

Structural and Policy Landscape

- **Strengthen multisectoral coordination:** Institutionalize inter-ministerial ECD coordination at national and city levels, integrating health, education, nutrition, child protection, and local governance.
- **Increase public funding for ECD:** Advocate for increased ECD-specific budget allocations, particularly for children under 3 years, whose funding is currently disproportionately low.
- **Decentralize ECD policy implementation:** Empower local governments with funds and technical support to operationalize ECD policies tailored to local needs and vulnerabilities.
- **Monitor policy enforcement:** Strengthen accountability and enforcement of child protection laws (e.g., Children’s Code Act 2022) and ECE policies across all settlements.
- **Support policy integration and data systems:** Develop an integrated ECD information system that tracks service coverage, workforce training, and child outcomes from 0- 6 years to inform planning and evaluation.
- **Promote private and community partnerships:** Encourage partnerships with faith-based, private, and community organizations to scale ECD services while ensuring adherence to quality standards.

Conclusion

The baseline study highlights critical developmental deficits among children in Lusaka and Chongwe, particularly after the age of 17 months. These are compounded by weak service delivery platforms, limited caregiver capacity, and systemic policy gaps. While political commitment exists, there is an urgent need to expand and integrate ECD services equitably across settlement types, with a focus on caregiver education, infrastructure, and policy enforcement. The findings lay a strong foundation for targeted interventions and monitoring frameworks to enhance developmental outcomes for Zambia’s youngest citizens.

Background

In Africa, ECD initiatives are limited but steadily growing in an unevenly distributed manner. “Although it is mandate of governments to implement ECD activities, most ECD programmes are developed and implemented by the private sector, non-governmental organisations (NGOs) and international organisations”.¹ In most African countries, the government's role is limited to providing policy direction. As a result, the ECD sector, like many others, remains underdeveloped, particularly in areas where civil society organizations (CSOs) are not present to support its growth. For instance, evidence suggests that in the Sub-Saharan African (SSA) context, many children under the age of five face irreversible disabilities or fail to reach their full developmental potential, often because intervention comes too late.² Children still experience poor health, malnutrition and impaired mental, social and emotional development in this region. High poverty levels and lack of leadership in the region have significantly contributed to the state of ECD. Therefore, the key challenge in SSA is to find effective ways to lead, organise and finance the ECD. International partners such as the Big Win Philanthropy have realised the potential city leadership has in developing and meeting ECD needs. Thus, through its flagship program, the Africa Mayoral Leadership Initiative (AMALI), Big Win has been supporting some participating leaders with an exclusive, tailored initiative to hone the ECD visions for their cities.

In Zambia, the mayors of Lusaka and Chongwe have committed to promoting early childhood development (ECD) in their cities. To ensure children in these areas receive the necessary experiences during their formative years, the cities plan to implement a holistic ECD program targeting children under six. This initiative will be delivered in collaboration with central government ministries and directorates, including health and nutrition, education, local government, labor and social affairs, and urban development. To guide the program’s design and identify key strategic interventions that can positively impact developmental outcomes for children under six, a baseline study has been commissioned to assess the current state of ECD in Lusaka and Chongwe. This inception report outlines the plan for conducting the baseline study, following the consultant's first-hand insights into the program and its objectives, gathered from discussions with Big Win, the two mayors, and the technical committee.

The State of ECD in Zambia

This section outlines the state of ECD in Zambia, aiming to review key literature and connect the problem statement to the study. A detailed literature synthesis report will be provided separately.

Lusaka and Chongwe

Lusaka and Chongwe districts are located in Lusaka province. Lusaka is predominately urban while Chongwe is mostly rural. Lusaka province has a population of 3,093,617. According to this 2022 census, children aged 0 – 4 years are 378,783.³

Lusaka district

Lusaka is located centrally in Zambia and relies on its network of roads and railways to connect it to regional markets.⁴ According to the 2022 Census, Lusaka has a population of 2,212,301. The current

¹ Association for the development of Education in Africa. Working Group on Early Childhood Development (WGECD). Available from: <http://www.adeanet.org/en/working-groups/early-childhood-development#:~:text=Making%20early%20childhood%20development%20a, and%20social%20and%20emotional%20development>.

² Colletta, Nat J.; Balachander, Jayshree; Liang, Xiaoyan. 1997. Early Childhood Development in Sub-Saharan Africa: Issues and Strategies. Africa Region Findings & Good Practice Info briefs; No. 96. World Bank, Washington, DC. © World Bank. <https://openknowledge.worldbank.org/handle/10986/9916> License: CC BY 3.0 IGO

³ Zambia 2022 Census of Population and Housing

⁴ Infrastructure & Cities for Economic Development. Lusaka Inclusive Urban Economic Development Diagnostic

CSO summary report does not include statistics on the number of children. However, the 2010 census reports that Lusaka had a population of 475,703 for children aged 0-9 years.⁵ Lusaka serves as Zambia's capital and the country's commercial center. Lusaka is all urban. Lusaka is the main commercial and manufacturing hub of Zambia. It accounts for a large proportion of the national economy and has the highest concentration of firms per capita.⁴ It is divided into seven constituencies, namely: In addition, these constituencies are divided into wards. Lusaka has 33 wards.

Whilst poverty in Lusaka province (not city) is extremely low (27% compared to national average of 48%), in 2022⁶, living costs are also extremely high, which stood at ZMK10, 307.01⁷ as of February of 2024. This cost is higher than the average income of ZMK 7,000 among Zambian.

Chongwe district

Chongwe has a population of 315,121 of which 111,597 live in the rural part of this district. There are approximately 51,344 children aged 0-5 in Chongwe.⁸ The district is predominately rural. The district has one Constituency and 21 wards.

Chongwe is an agricultural district. It is believed to be one of the two largest suppliers of fresh vegetables to Lusaka.⁹ Agricultural activities include crop production, horticultural production and livestock production.¹⁰ Despite its agricultural potential, poverty is widespread, compounded by lack of land ownership. Much of the land in Chongwe is under traditional tenure.¹¹

Approximately 350 million children, or roughly 40% of children under primary school age, lack sufficient access to childcare and educational services. Consequently, many of these children are compelled to spend their crucial developmental stage in unsafe environments, devoid of appropriate cognitive and social stimulation facilities (Devercelli & Beaton-Day, 2020). Thus, the absence of suitable and affordable childcare not only hinders the parents' ability to engage in productive economic activities, but also has a detrimental impact on the socio-emotional and cognitive development of the children.

As per the 2022 census population projection, Zambia has a population of around 21.3 million individuals, with a median age of 17.7 years, making it one of the youngest populations worldwide (World Bank, 2024). Projections indicate a substantial increase in this figure in the near future (ZAMSTAT, 2022). Zambia boasts one of the most youthful populations globally, with over 50% of its inhabitants under 14 years of age. Approximately 2.5 million children, accounting for 16% of the national population, fall within the 0–3-year age bracket. Expanding the age range to include 5-year-olds raises the number of young children to approximately 3.8 million. The growing number of young children increases the demand for good nutrition, healthcare, child security, and childcare services for this age group. The following section gives an analysis of the nutrition, health, security, and childcare status of children in Zambia.

⁵2010 Census of Population and Housing: National Analytical Report

⁶ Zambia Statistics Agency. Highlights of the 2022 Poverty Assessment in Zambia

⁷ JCTR (2024). Urgent Call for Solidarity with the Poor as the Cost of Living Continues to Rise. Available from: <https://repository.jctr.org.zm/handle/20.500.14274/1830>

⁸ Zamstats. Population and Demographic projections. Available from: <https://zambia.opendataforafrica.org/ZMPHC2015/population-and-demographic-projections-2011-2035?country=1000390-lusaka-district>

⁹ HealthyFoodAfrica. Fostering vegetable production for a sustainable food system that supplies the city. Available from: <https://healthyfoodafrica.eu/food-system-labs/zambia-chongwe-district/>

¹⁰ Inonge Milupi, D., Moonga, M. S., Chileshe, B., & Monde, P. (2020). Women and Natural Resource Conservation: A Study of Chongwe District, Zambia. International Journal of Humanities Social Sciences and Education (IJHSSE), 7(3), 22-30.

¹¹ HIVOS. Urban Futures. Available from: <https://hivos.org/program/urban-futures/cities/chongwe/>

Poverty

Children in Zambia are faced with a lot of nutritional, health, and security challenges, which negatively affect their social and cognitive development. Poverty is the biggest driver of most problems affecting child development, as 54.4% of the population is classified as poor. Rural areas experience higher levels of poverty, with 76% of rural dwellers living in poverty and of these 40% classified as extremely poor (UNICEF, 2019).

Health Status

In 2022, Zambia's birth rate was 35.2 births per 1000 individuals. The country faces significant childhood challenges, with infant and under-5 mortality rates of 39 and 56 deaths per 1000 live births, respectively, and a neonatal mortality rate of 24 per 1000. While 97% of expectant mothers receive care from qualified medical personnel in antenatal clinics, only 84% give birth at health centers. Additionally, only 78% of children receive the necessary vaccines between 12 and 23 months after birth (Brudevold-Newman et al., 2018). There are over 1.2 million people living with HIV in Zambia; 66,000 of these individuals are children under the age of 15.¹² As of 2023, approximately, 3200 [2000 - 4100] children 0-14 years were newly infected with HIV.¹³ Unfortunately, the proportion of children attending under-5 clinics is decreasing, despite government efforts (ZAMSTAT et al., 2019).

According to the Zambia Demographic and Health Survey report, a significant number of children experience frequent episodes of diarrhea (15%), fever (16%), and acute respiratory infections (2%). However, the report also indicates that treatment-seeking behavior is quite low (ZAMSTAT et al., 2019). This significant disease burden among young children greatly hinders their cognitive, physical, and psychosocial development (Matafwali & Zingel, 2017).

Nutritional status

A significant proportion of children do not meet the minimum daily intake (Mulenga, 2014; UNICEF, 2021), and malnutrition is responsible for approximately 40% of childhood mortality in Zambia. FAO (2009) states that malnutrition is directly linked to high levels of illness, limited access to healthcare services, poverty, and food insecurity. Children in their first 36 months of life face a higher risk of severe malnutrition (Matsunaga & Yamauchi, 2020), leading to stunting, wasting, and underweight conditions.

Currently, 35% of 18- to 23-month-old children suffer from stunting, indicating their heightened vulnerability. Wasting is less prevalent among Zambian children, with only 4% of children under 5 years old displaying obvious signs. Eight percent of children aged 6-71 months are underweight. However, 5% of children under 5 are overweight and obese (ZAMSTAT et al., 2019). Typically, malnutrition, stunting, wasting, and being underweight are primarily observed in rural areas, while being overweight is more prevalent in urban areas.

Child Safety, Protection and Justice

Zambia acknowledges the pivotal role of child protection, safety, and justice in fostering optimal child development. These tenets serve as the bedrock for safeguarding children's rights and mitigating unwarranted transgressions. The nation has achieved noteworthy advancements in combating violations and bolstering the welfare of children, both through domestic initiatives and international accords. Nevertheless, despite these commendable endeavors, children in Zambia persistently endure detrimental living circumstances.

Children in Zambia are subjected to various forms of violence, such as physical, emotional, and sexual abuse, as well as neglect (Nakamba & Kaani, 2023). Moreover, child labor and child marriages act as

¹² The Elizabeth Glaser Pediatric AIDS Foundation. Zambia. <https://pedaids.org/country/zambia/>

¹³ UNAIDS. Zambia 2023. Available from: <https://www.unaids.org/en/regionscountries/countries/zambia>

further obstacles to children's development. Despite the fact that these issues may not directly impact the children under study, being raised by parents who have experienced these challenges could have an impact on their future development. As stated by the Ministry of Youth, Sport and Child Development et al. (2018), around 1.3 million children are involved in some form of child labor, and child marriages are widespread, particularly in rural areas where the enforcement of laws against these practices is difficult.

Moreover, a considerable number of minors in Zambia face the peril of being unregistered, impeding the ability to locate them in cases of trafficking. Findings from a 2014 survey indicate that merely 14% of children aged 0-5 possess legitimate birth certificates. It is imperative to establish a family-oriented childcare framework to cater to a substantial proportion of these individuals (Ministry of Youth, Sport and Child Development et al., 2018).

Early Childhood Education (ECE)

There is a clear difference in the number of children under the age of 5 who are enrolled and not enrolled in school in Zambia. The proportion of Grade 1 entrants with ECE exposure has significantly improved, moving from only 16% in 2004 to 41% in 2021, reflecting a positive trend in expanding access to ECE.¹⁴ Only about one in three children successfully completes their early childhood education, indicating a significant problem with both access and retention in the system. It is also important to note that early childhood education centres in Zambia mainly serve children between the ages of 3 and 5, while day cares are almost non-existent.

Types of Schools: Due to inadequate and disorganized access to ECD in Zambia, the country has implemented policy frameworks to address this issue. These frameworks aim to ensure that a range of institutions are involved in providing childcare and education services for children in the pre-primary category. The main providers of ECE and childcare include government primary schools, community-based childcare centres, and privately and church-owned schools.

All government primary schools are now required to have an early childhood centre within their premises. These centres provide child development stimulation and school-readiness education to children before they are enrolled in grade 1. Community-based childcare centres are operated mainly by town councils in collaboration with relevant ministries such as education, community development and social services, health, and agriculture. Private and faith-based schools have historically been the primary providers of ECE in Zambia. However, the quality and cost of these institutions vary significantly, and access depends on parents' ability to pay for the services.

Recent Development in ECE Development in Zambia: Despite a plethora of challenges affecting the Early Childhood Development (ECD) in Zambia, there has seen significant strides in recent years. Stakeholders, including the government have embarked on strengthening the ECE-specific Policy Framework to enhance ECE Development. In 2013, Zambia introduced the National Early Childhood Education Policy. This policy marked a significant step towards formalizing ECE in Zambia and led to the creation of the Teaching Council of Zambia in 2014 to ensure that ECE teachers are part of main education through professional accreditation and regulation of teacher education (UNICEF, 2023). ECE teachers are now expected to have a diploma qualification as a minimum standard (Kaunda & Ailwood, 2021).

Additionally, there has been Increased Awareness among parents and communities, which increased Enrolment in newly established ECE centres. Unfortunately, the increase in enrolment has mainly been achieved in urban areas, while rural areas are still struggling to increase numbers. Kaunda and Ailwood (2021) stated that the Government of Zambia has enlisted the help of NGOS to promote ECD programmes through targeted campaigns and programmes.

¹⁴ UNICEF, 2022. 2022 Education Budget Brief: Dealing with new demands emanating from the “education for all” policy

Challenges in the Development ECD Sector: Developments mentioned above have generated enthusiasm among stakeholders in the ECD sub-sector. However, ECD still faces a number of challenges in order to be fully effective. The most prominent of these challenges is insufficient funding, as more funds are directed towards older children compared to the 0–5-year age group. Between 2016 and 2020, only ZMW 397 per child was spent on children aged 2 years, compared to ZMW 2123 on the 18–22-year cohort (UNICEF, 2023). Such low funding significantly affects children's access to quality ECE services, leading to low achievement and high dropout rates in primary schools.

The second challenge to ECE development in Zambia is poor and limited infrastructure. Many centres and schools lack adequate and appropriate classrooms and teaching materials to support ECD. Additionally, there are notable disparities in access and quality between urban and rural schools, with the latter being severely disadvantaged.

Thirdly, due to long-term neglect, there is a severe shortage of qualified teachers, resulting from limited training institutions and ECE-specific programmes. This problem is worsened by low teacher motivation due to low salaries and limited professional development opportunities. Finally, the ECD sub-sector lacks regular monitoring to provide up-to-date data for identifying gaps in ECE provision, which is crucial for effective planning and budget allocation. Therefore, this study is conducted at an optimal time to make this information available.

Mitigation Measures to Improve ECD in Zambia

Many national programs in Zambia have been introduced to address health, nutrition, security, safety, and early learning issues that impact ECD. Unfortunately, these initiatives have been largely ineffective due to poor implementation and a lack of reliable data to inform their development and execution. Several policies and legislation have been formulated and implemented, which can be categorized into three groups: 1) promoting good health, nutrition, and responsive caregiving, 2) ensuring security and safety, and 3) providing opportunities for early learning. The most recent legislation, the Children's Code of 2022, consolidates existing policies. Table 1 below illustrates the policies within each of the three categories.

Table 1: ECD Policies and Legislation in Zambia

Good health, nutrition, responsive caregiving	Security and Safety	Opportunities for Early Learning
• National health policy of 2013 • Antenatal clinics • Under 5 clinics • The First 1000 Most Critical Days of Life Programme • Public Health Act, CAP 295 of the Laws of Zambia	• National youth policy • Juvenile Act • Prisons Act	• Nurseries Act of 1957 • Education Act of 2011 • Zambia Curriculum Framework • Early childhood development action networks (ECDAN)

However, despite these efforts, the well-being of children in Zambia has not shown substantial improvements over the years. Children continue to suffer from poor health, with no significant changes in their nutritional status. Complete eradication of child marriages and child labour has not been achieved. Many children still lack birth registration certificates, and child trafficking remains widespread. A considerable portion of children lack access to suitable childcare and education (UNICEF, 2023). There is a significant demand for an inclusive ECD reform to ensure that all eligible children are adequately supported. The reasons for these challenges include fragmented and lack of policy enforcement, and funding to ECD program.¹⁵

¹⁵ UNICEF. Investing in Early Childhood to Advance Human Capital Development: Early Childhood Development Budget Brief.

Objective of the Study

The main objective of this baseline study was to assess the state of ECD in Lusaka and Chongwe. To this end, the specific objective and the associated research questions were as follow:

Thematic area 1: Early childhood development outcomes

1. What is the status of children aged 0-3 reaching their developmental milestones in all domains in Zambia? How does this differ by vulnerability and type of settlement?
2. What is the status of children aged 4-6 in terms of school readiness and learning outcomes in Zambia? How does this differ by vulnerability and type of settlement?

Thematic area 2: Household caregiving practices

1. What is the status of 0-3-year-old children's parental and caregivers' knowledge, attitudes, behaviours, and practices that promote or hinder children's development and learning in Zambia?
2. To what extent is responsive parenting practiced at home?
3. What do Zambian households consider to be, nurturing care and early stimulation of children?
4. To what extent do caregivers practice responsive social protection and ensure child safeguarding in Zambia?
5. What proportion of children experience psychological aggression or physical punishment in Zambia?
6. How can nurturing care be promoted among parents and caregivers in Zambia?
7. How do socio-economic factors impact caregiving practices and child development outcomes?

Thematic area 3: Early childhood development services

1. What is the status of ECD service provision in Zambia?
2. Who are the primary ECD service providers in Zambia and who comprises providers' ECD workforce?

Thematic area 4: Structural and policy landscape

1. What is the policy and regulatory landscape of ECD in Zambia? How does this interact with the wider country policy ecosystem (if applicable)?
2. How do these policies and regulations serve as either barriers or enablers of ECD programming?
3. How is coordination of ECD done at the multisectoral level?

Methodology

This section describes the methodology implementation approach that was applied in this baseline study on the state of ECD in the cities of Lusaka and Chongwe.

Design

This study was designed as a baseline evaluation to inform the development of evidence-based interventions aimed at improving child outcomes in the two cities. Additionally, it served as a reference point for future intervention evaluations. A cross-sectional design was used to describe systematically the state and characteristics of the sample and outcomes of interest (household caregiving practices, ECD service delivery platforms, ECD outcomes, and, structural and value perception).

Study Sites and Population

This study was conducted in Chongwe and Lusaka cities. The study targeted households domiciled in high, medium and low-income communities as designated by the Zambia Statistics Agency. Within each of the stratum, the started targeted primary caregivers of children between 0-6 years and their children of this same age.

Sampling Frame

According to the 2023 ZamStats Child Poverty report, 70.6% of Zambian children are multidimensionally poor, meaning that they suffer from deprivations in 2 or more dimensions of wellbeing simultaneously.¹⁶ Thus, to ensure that this baseline had a representative sample of the characteristics of children in Zambia, a sample size formula based on proportion of characteristic of interest was used as follows;

$$n = \frac{z^2 * p(1 - p)}{e^2}$$

n = Sample size for the stratum (low, medium, and high cost)¹⁷

Z = critical z-score for 95% confidence level (1.96)

P = Proportion of the population with the characteristic of interest 78%¹⁸ (0.78)

e = Margin of error 5%, (0.05)

1. $1-p=1-0.78=0.22$
2. $z^2=1.96^2=3.8416$
3. $3.8416 \times 0.78 \times 0.22=0.66182496$
4. $e^2=0.05^2=0.0025$
5. $n = 0.66182496/0.0025 = 264$

Thus, the proposed sample for each stratum was 264

Based on the formula and assumptions, the initial sample size for each stratum was 264 household heads (HH) respondents. To account for non-response (10%) and the design effect (1.25), the sample size per stratum was increased to 363 HH respondents. Unlike simple random sampling, the precision

¹⁶ This index of child poverty looks at child wellbeing across 7 dimensions: nutrition, health, education, water, sanitation, housing and access to information.

¹⁷ The Zamba urban settlement areas are categorized as low, medium and high-cost residence. Thus, these will be the strata used in this baseline study. See https://housingfinanceafrica.org/app/uploads/Final-Quantitative-Zambia_sm.pdf

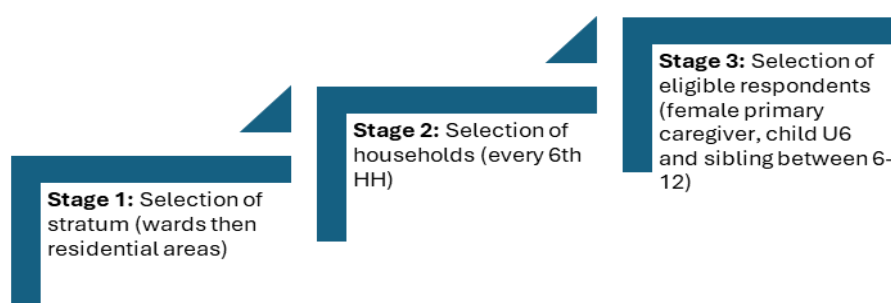
¹⁸ Early childhood Development in Zambia Budget Brief 2023. Available from: https://www.unicef.org/zambia/media/3496/file/UNICEF%20ZAMBIA%20Budget%20brief_Early%20Childhood%20Development%20Brief%202023%20Final.pdf.pdf

of estimates in a stratified sample design is influenced by intra-class correlation. The design effect represented the degree of variance inflation caused by stratified sampling.

According to the 2022 Census of Population and Housing, Lusaka district, with a population of 2,212,301, is approximately seven times larger than Chongwe district (315,121)¹⁹. To ensure proportional representation of both cities, sampling was done based on population size. From the 363 respondents in each stratum (Low, Medium, and High Cost), 30% of the sample was drawn from Chongwe and 70% from Lusaka. This translated in 109 respondents from each stratum in Chongwe and 254 respondents from each stratum in Lusaka, making a total sample size of **1,086 household respondents (762 from Lusaka and 324 from Chongwe)**.

Sampling method

Stratified random sampling approach to select population-representative samples from the two cities and minimize selection bias in the estimates from the survey was applied. Each eligible household and individual in the catchment area had an equal chance of being selected. The study employed probability sampling procedures. This was a 3-stage sampling process, as shown below:



The study used the ZamStats sampling frames, which is already categorized into the relevant strata and include household numbers, facilitating the application of systematic random sampling methods. Based on these sampling frames, the following steps were taken:

1. We mapped the wards in both cities and their corresponding residential areas.
2. According to the categorization of residential areas (high, medium, and low cost), we used the total number of households to calculate the proportional representation of households from the overall population.
3. Using the proportional representation, we calculated the sample size for each identified residential area from the total sample size of 762 for Lusaka and 324 for Chongwe.
4. Chongwe's rural wards and residential areas were excluded from the sampling frame as they do not meet the criteria for classification as a city.

¹⁹ Zambia Statistics Agency (Zamstats). (2022). *Zambia Census of Population and Housing: Population Summary Report*. www.zamstats.gov.zm

Qualitative Sample

Early Childhood Development (ECD) Center Observations

Selected ECD centres were observed to collect data on type and quality of services in communities. Centres were purposively selected in each of the sampled wards distributed by low, medium and high-cost residential type. A variety of centres were selected for observation.

Eighty-four (84) centres in total were observed in the two cities. Fifty-nine (59) ECD centres in Lusaka and 25 in Chongwe were assessed for type, quality, and adequacy of services. This sample size was determined using a statistical power sampling method. The following parameters were used in G*Power software:

- Effect size = 0.6
- α err prob = 0.05
- Power = 0.80
- Allocation Ratio = 1:1

The distribution across the two cities was proportion to the population distribution. Chongwe city's sample size represents 30% of the total sample size for this study. Thus, 30% (25) of 84 ECD centres were sampled in Chongwe.

City	Residential Classification			Total
	High	Medium	Low	
Lusaka	20	22	17	59
Chongwe	4	10	11	25
Total	24	32	29	84

Inclusion Criteria

There is currently no comprehensive list of ECD centers in Lusaka and Chongwe. Therefore, a purposive sampling approach was used to select ECD centers from each of the following categories:

Category	Inclusion Criteria
High-density Residential areas	Centers located in compounds such as Missis, Ngombe, and Kampasa. The cost of services per term should range from ZMW 0 to 1,500.
Medium-density Residential areas	Centers in areas like Chilenje, Libala, Emmasdale, Matero, and Lubuko. The cost of services should range from ZMW 1,501 to 3,000.
Residential Low-density areas	Centers located in Kabulonga, Rhodes Park, Silverest, and Ndeke. The cost of services should be above ZMW 3,000.

Observation Procedure

In each of the selected Early Childhood Education (ECE) schools or center was purposefully selected, informed consent and permission were sought from the Ministry of Education first and then from the ECD center head or manager and the teacher. Following approval, the quality of the environment, teacher-child interactions, type of activities, and services were assessed for 3 to 4 hours depending on the duration of the children's activities. Government ECD centers typically operate for 3 hours, while private centers run for up to 5 hours.

Caregiver Focus Groups (FGs)

Twelve 12 focus groups with parents of children aged 0-5 years comprising at least 10 members each translating to a total of 120 participants were conducted in the two cities. One focus group discussion from each residential stratum (high, medium and low) will be conducted as follows:

Table 2: Number of Focus Groups Conducted by Residential Classification

City	Males			Females			Total
	High	Medium	Low/Rural	High	Medium	Low	
Lusaka	1	1	1	1	1	1	6
Chongwe	1	1	1	1	1	2	7
Total	2	2	2	2	2	2	13

Participants were selected in collaboration with the two city's community workers, health facilities, and church institutions. Participants were selected based on their familiarity with ECD or past participation in ECD programs, and ability to participate in a group discussion. Participation in the focus groups was voluntary.

Key Informant Interviews (KIIs)

Key Informant interviews were conducted with different informants that provided a multisectoral view of the state of ECD in Chongwe and Lusaka. These were carefully selected to ensure diversity and depth of their responses to cover all domains of child development including nutrition and health, early learning opportunities, safety and security, and responsive parenting. Table 3 presents a list of informants that were interviewed.

Table 3: Types of Key Informants Targeted for Interviews

Respondent	No.	Gender	City
City Mayors	2	F and M	Lusaka and Chongwe
Zambia Enhancement of Early Learning project Coordinator	1	M	National level
District Health Director	1	M	Chongwe
ECE Specialist - World Vision	1	F	National
ECD representative -UNICEF	1	F	National
Garden and parks manager	1	M	Lusaka
Early Childhood Development Network of Zambia (ECDAN-Z) coordinator	1	F	National
Ward Development Committees Chairperson	4	3 Ms	Chongwe and Lusaka
ECE teachers	4	F	Chongwe and Lusaka

Final sample sizes

Based on the described sampling frame and procedures, the final samples sizes for each of the categories was as described in the Figure below:

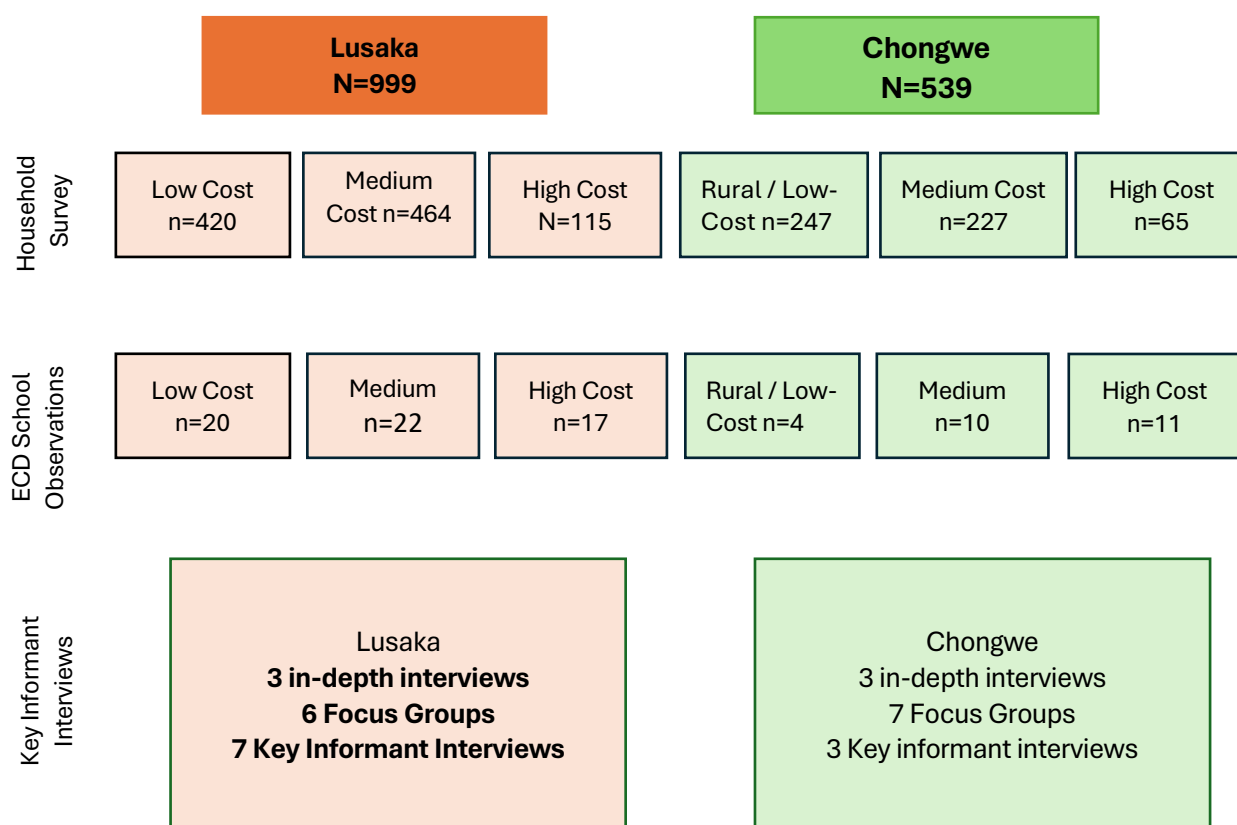


Figure 1: Sample(s) Distribution by Districts per design Category

Data Collection Tools and Methods

The study triangulated different methods in order to enhance the validity and credibility of the findings. All quantitative methods of data collection were programmed in Kobo Collect on tablets. The methods and tools used are well validated well-established tools. Where necessary the team adjusted the tools to ensure they were culturally relevant to the context and to the study objectives. The table below summarises the methods and tools used in the baseline study:

Method	Targeted population	Description
Anthropometric (Weight scale and height board) ²⁰	Children (0-5 years)	Height board and weight scale will be used to support anthropometric assessment, particularly to measure stunting and underweight.
CREDI	Children (0-3 years)	The Caregiver Reported Early Development Index (CREDI) is a population-level measure of early childhood development (ECD) for children from birth to age three and provides an overall developmental status of a particular population of interest.

²⁰ Anthropometric measurement in children are measured using weight scales, MAUCs and height boards. For detail refer to: <https://www.ncbi.nlm.nih.gov/books/NBK537315/>

Method	Targeted population	Description
The International Development and Early Learning Assessment (IDELA)	Children (3.5 to 6 years)	This direct child assessment tool will be used to assess measures of the early learning and development of children on motor development (gross and fine motor skills); Early language and literacy (Emergent literacy; early numeracy/problem-solving); Emergent numeracy including shape identification) and; socio-emotional skills (Social-emotional skills)
MELE	Children 0- 6 years	This will contain interviews and observation of ECD centres for children 3-6 years old
Sibling Relationship Inventory	Younger and older Siblings of targeted children (6-12 years)	The SIB assesses older siblings' feelings toward the targeted child. It is a parent-reported measure. The SIB measures: companionship, empathy, teaching, rivalry, aggression, and avoidance.
The HOME Inventory	Households	To understand household caregiving practices, the HOME inventory assessing the quality of parenting and the home environment provided for the child will be used.
Early Parenting Attitudes and Belief	Household heads of children 0-5	Measure attitudes and beliefs of parents on parenting. It covers attachment, learning, rules and respect
Caregiver Knowledge of Child Development	Household heads of young children	Assess caregivers' knowledge of early childhood development
Household Vulnerability Assessment tool	Households	Measures household vulnerability. The measure used in Zanzibar has been adapted to use in Zambia
Mapping tool	Community of the targeted children	Geo-mapping of ECE related facilities
Willingness to pay	Household heads	To assess willingness to pay for services and the amount parents are willing to pay for ECD service
Desk review	Systems	This method was used to assess the ECD context and landscape situation in Lusaka and Chongwe
FGD	Households	This tool was used to explore caregiver perceptions on factors including ECD including caregiving practices, playful parenting ECD outcomes and value perception of ECD
KII	Government, community leaders, policymakers and key civil servants	This tool was used to explore stakeholders' views and perceptions on different ECD elements such as ECD service delivery platforms, and structural and value perceptions of ECD across the two cities.

Reliability of the tools

Reliability of tools is very important because it ensures the consistency and stability of findings. The general rule of thumb is that $\geq .70$ is acceptable but $.60$ is considered adequate.²¹ The tools used in

²¹ Morera, O. F., & Stokes, S. M. (2016). Coefficient α as a measure of test score reliability: Review of 3 popular misconceptions. *American journal of public health*, 106(3), 458-461.

this study showed that for most tools reliability as measured by Cronbach Alpha was good and, in few cases, adequate except for one tool which was below .60. For details see the table:

Table 4: Cronbach Alpha Coefficient Reliability Test

Name of Measure	Cronbach Alpha Coefficient
CREDI	.78
0-5	.82
6-11	.71
12-17	.87
18-23	.86
24-29	.81
30-36	.66
IDEALA	.85
HOME inventory	.81
Caregiver Knowledge of Child Development	.41
Early Parenting Attitudes and Beliefs	.62
Sibling Relationship	.88

Data analysis

The analysis involved triangulating quantitative and qualitative data on key indicators aligned with the objectives. Below we describe the overarching analysis used around each objective:

Objective 1: Assess the state of ECD in Lusaka and Chongwe. For this objective, descriptive statistics will be used to analyze data from surveys, observations, and interviews. The analysis will include measures of central tendency (mean, median) and frequency distributions to describe the current state of ECD services, infrastructure, and accessibility in Lusaka and Chongwe. Geographic and socio-demographic differences in ECD service provision will also be explored through cross-tabulations and chi-square tests. A weight-for-length/height z-score (WHZ) will be used to compare a children's weight to the weight of a child of the same length/height and sex to classify nutritional status. This will help us to estimate the proportion of children (0-5 years) who are stunted or underweight

Objective 3: Analyze demand for and engagement in ECD service platforms. Regression analysis will be employed to determine the factors that influence demand for ECD services, such as caregiver income, education, and location. Engagement levels with ECD platforms will be analyzed using frequency and participation rates. Additional statistical tests, such as ANOVA, may be used to compare engagement across different demographic groups (low, medium and high).

Willingness to Pay data will be summarized as the willingness by service users to pay for ECD services if they were to be made available to them. This includes additional services over and above the existing services such as social and cognitive developmental services in addition to the usual services such as Growth Monitoring services which are normally provided for free in Government health facilities.

Objective 4: Examine caregiver practices and norms influencing child development Caregiver practices will be analyzed using both qualitative and quantitative methods. Surveys and focus group data will be subject to content analysis to identify prevalent caregiving norms. Correlational analysis will be used to explore the relationship between specific practices and child development outcomes, using developmental assessment scores from children in the sample.

Objective 4: Structural and policy landscape. Data from institutional assessments and stakeholder interviews will be analyzed to diagnose the effectiveness, coverage, and sustainability of existing ECD platforms. This will involve content analysis for qualitative data and thematic coding to identify common

challenges and best practices. Quantitative measures will examine platform utilization, staffing levels, and resource allocation. Qualitative analysis followed the following steps²²:

1. In the first step, transcripts were read repeatedly in order to become familiar with the words of the interviewee/informant to develop a holistic sense of the material. At this stage, the researcher reflected on his preunderstanding of the content and how it might influence the understanding of the interviews.
2. The second step was to identify the meaning units. Meaning units are sentences or phrases relevant to the study's aim.
3. After identifying meaning units, in step three, condensed meaning units (summarized meaning units) were developed in the researcher's own words but without distorting the original core meaning.
4. In the fourth step, sentences or paragraphs related to their content or context were constructed into categories (group the codes with similar content into a category). In step five, common categories in meaning were grouped into themes.

Ethics and Child Safeguarding

This study was ethically cleared by the University of Zambia Humanities and Social Sciences Research Ethics Board (UNZAHSS REC) protocol no. HSSREC IRB No. 00006464. Enumerators were trained to observe and adhere to ethical practices including obtaining consent and reporting factual data.

²² Graneheim, U. H., & Lundman, B. (2004). Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. *Nurse Education Today*, 24(2), 105-112.

Study Findings

The presentation and discussion of the findings will be done in accordance with the study objectives and the associated research questions. The report begins with a description of the sociodemographic characteristics of the sample used in this study.

Sample Distribution of the Two Cities

Understanding the socioeconomic and demographic characteristics of households is essential in promoting effective early childhood development (ECD). Residence, classification of residential areas, and the type and sex of caregiver characteristics were collected. These characteristics influence children's growth, learning and overall well-being of children.

Notably, Lusaka is predominately urban with no rural setting, while Chongwe is a mixed setting with a significant rural population, in this study, about 45% (242) of the children in Chongwe were drawn from rural settings and 55% (297) from urban areas. In terms of classification of the residential areas, 42% of respondents (children and caregivers) were from unplanned residential areas whereas 58% of children were from planned settlements. Mothers or caregivers of children in Lusaka and Chongwe had secondary-level education; 62% and about 50% of mothers/primary caregivers in Lusaka and Chongwe respectively had secondary-level education. This shows that both Lusaka and Chongwe caregivers have modest level of education which may barely be functional for best childhood development outcomes.

Table 5: Selected background Characteristics, Residence and Type of Settlement

Variable	Lusaka (N=999)		Chongwe (N=539)		Overall (N=1538)	
	Count	%	Count	%	Count	%
Residence						
Rural	na	na	242	44.9	242	15.7
Urban	999	100.0	297	55.1	1296	84.3
Type of Settlement						
Planned	579	58.0	292	54.2	871	56.6
Unplanned	420	42.0	247	45.8	667	43.4

Sociodemographic Characteristics of Households

Majority of caregivers are female, with majority of children being cared for by their biological parents, 88% and 89.1% in Lusaka and Chongwe respectively. Age distribution indicates that majority of caregivers in Lusaka were aged between 30 and 34 years (24%), while in Chongwe, about 29% of caregivers were between 25-29 years. This shows that relatively, Chongwe has younger mothers/caregiver compared with Lusaka.

The results further shows that the financial responsibilities fall to the fathers, who contribute to household expenses at 64.2% in Lusaka and 68.8% in Chongwe (Table x). This shows a patriarchal role of fathers in household responsibilities within the communities.

Table 6: Selected Caregivers Characteristics, Lusaka and Chongwe

Variable	Lusaka (N=999)		Chongwe (N=539)		Overall (N=1538)	
	Count	%	Count	%	Count	%
Sex of Caregiver (Age 15 and above)						
Male	43	4.3	37	6.9	80	5.2
Female	955	95.7	502	93.1	1457	94.8
Relationship of Primary Caregiver to Child (0-3)^a						
Biological Parent	710	88	401	89.1	1111	88.4
Foster Parent	9	1.1	1	0.2	10	0.8
Other Relatives (Aunty/Uncle)	39	4.8	10	2.2	49	3.9
Older Sibling	13	1.6	8	1.8	21	1.7
Step Parent	1	0.1	3	0.7	4	0.3
Neighbour	1	0.1	2	0.4	3	0.2
Others	34	4.2	25	5.7	59	4.7
Age of Caregiver^b						
Mean age (Standard Deviation)	32.5 (9.9)		31.0 (8.9)		32 (9.6)	
< 14	6	0.6	1	0.2	7	0.5
15-19	42	4.2	31	5.8	73	4.8
20-24	133	13.3	85	15.8	218	14.2
25-29	236	23.7	154	28.6	390	25.4
30-34	239	24	122	22.6	361	23.5
35-39	161	16.3	69	12.8	230	15
40-44	83	8.3	39	7.2	122	7.9
45+	98	9.8	38	7.1	136	8.9
Caregiver's Level of Education^c						
No Education/Primary	192	19.2	186	34.5	378	24.6
Secondary	621	62.2	268	49.7	889	57.8
Higher	185	18.5	85	15.8	270	17.6
Missing						
Work Looks after child (0-3), Natural Mother at work^d						
Work place daycare	25	3.1	15	3.3	40	3.2
Mostly child at Pre School	7	0.9	7	1.6	14	1.1
Neighbor	42	5.2	29	6.4	71	5.7
Older sibling	191	23.7	85	18.9	276	22
Grand mother	194	24	99	22	293	23.3
Other relatives	188	23.3	101	22.4	289	22.9
Others	160	19.8	114	25.3	274	21.8
Who pays for Household Expenses^e						
Father	641	64.2	371	68.8	1012	65.8
Mother	183	18.3	90	16.7	273	17.8
Relative	34	3.4	12	2.2	46	3.0
Grand or Elderly Parent	129	12.9	59	11	188	12.2
Child (6-17 years)	11	1.1	7	1.3	18	1.2

Note: na=Not applicable, a= 281 missing, b= 1 missing (age not stated), c= 1 missing (level of educated not stated) d=1 missing, e=1 missing

Demographic Characteristics of Children (0-6)

One thousand two hundred and fifty-seven (1257) 0–3-year-old children and caregivers participated in the study, representing 807 children in Lusaka and 450 in Chongwe. Of these, only 9.4% and 12% in Lusaka and Chongwe respectively reported that children were attending an early childhood development activity educational programme. Most of the children 0–3-years were looked after by an older sibling, grandmothers, and other relatives when their natural mothers are at work, a similar pattern is observed for Lusaka and Chongwe.

Among children 4-6-years old, the study captured 593 with 432 and 161 children 4-6 years in Lusaka and Chongwe respectively. Unlike children 0-3 years, school attendance or participation in ECD was relatively high among children 4-6 with slightly over two-thirds of children in Lusaka (67.6%) and Chongwe (66.5%) attending school or participating in ECD activities. This shows that younger children are less likely to be taken to school or participate to early childhood development activities. See the table 7:

Table 7 : Selected Child Characteristics, Lusaka and Chongwe

Variable	Lusaka (N=999)		Chongwe (N=539)		Overall (N=1538)	
	Count	%	Count	%	Count	%
Sex of child (0-12 years)						
Male	510	51.4	281	52.1	791	51.6
Female	483	48.6	258	47.9	741	48.4
Age in Months^a						
< 6 Months	6	10.5	49	14	116	11.7
6-8 Months	52	8.1	26	7.5	78	7.9
9-11 Months	46	7.2	25	7.2	71	7.2
12-17 Months	108	16.9	33	9.5	141	14.3
18-23 Months	70	10.9	46	13.2	116	11.7
24-35 Months	142	22.2	101	28.9	243	24.6
36-47 Months	57	8.9	24	6.9	81	8.2
48-59 Months	39	6.1	20	5.7	59	6
>60 Months	59	9.2	25	7.2	84	8.5
School Attendance (Age 0-3 years)^b						
Yes	76	9.4	54	12	130	10.3
Not attending	731	90.6	396	88	1127	89.7
School Attendance (Age 4-6 years)^c						
Yes	292	67.6	107	66.5	399	67.3
Not attending	140	32.4	54	33.5	194	32.7
Sex of child (0-12 years)						
Male	510	51.4	281	52.1	791	51.6
Female	483	48.6	258	47.9	741	48.4

Note: a= 549 missing (child's age not stated), b= 281 missing (children not 0-3 years), c=945 missing (children are not 0-4 years)

Thematic area 1: Early Childhood Development Outcomes

This section will address two research questions related to child development of children aged 0-3 using data from the CREDI and data from the IDEALA measuring school readiness for children 4-6 years:

1. What is the status of children aged 0-3 reaching their developmental milestones in all domains in Zambia? How does this differ by vulnerability and type of settlement?
2. What is the status of children aged 4-6 in terms of school readiness and learning outcomes in Zambia? How does this differ by vulnerability and type of settlement?

Child Development

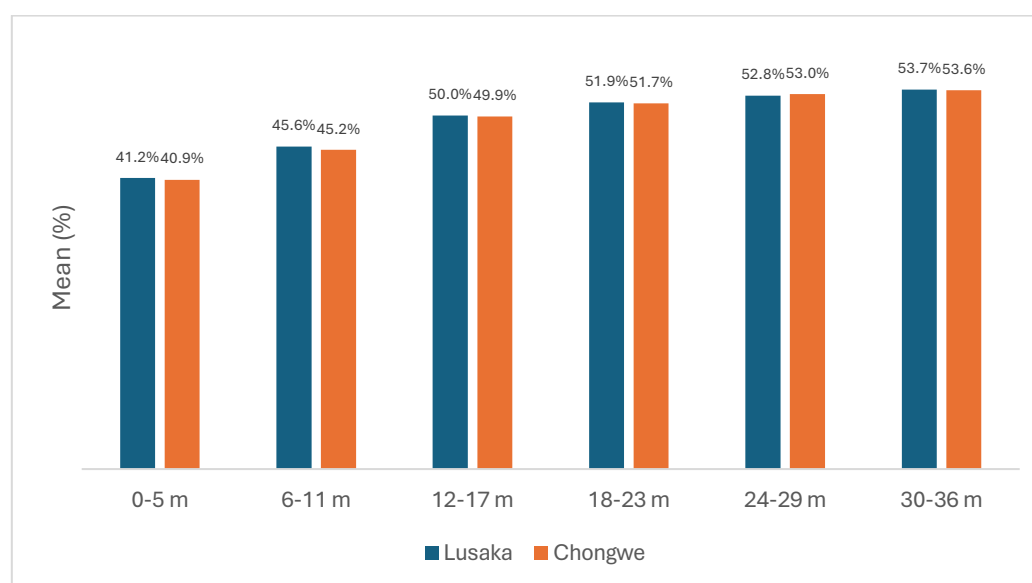
A total sample of 778 of which 495 were from Lusaka city were analysed for the CREDI. Below is the summary of the scores per age group:

Table 8: Summary Number Observations of Children by Age

Age Group	Total Obs	% scored	Score
0-5	116	100%	41.06
6-11	149	100%	45.45
12-17	141	100%	49.99
18-23	116	100%	51.81
24-29	148	100%	52.89
30-36	108	100%	53.63

Table 8 summarizes the total number of observations (children) by age band and indicates what percentage of these observations were able to be scored and the average overall score performance of the children in each band. This information has also been presented in graphical form below. In this graph below the findings show that consistent with child development expectations, older children have consistently higher

raw scaled scores than younger children. Mastery of child development skills across all domains generally improve with age.



Next, we compared these scores to the norm reference to determine if children in the two cities were below or above the norm reference average overall scores and z scores. Findings show that children in both Lusaka and Chongwe aged between 0-17 months start well above the average norm references but the fall below the norms just after the age 17 months. See the Figure below for graphical representation of this developmental trajectory of the children.

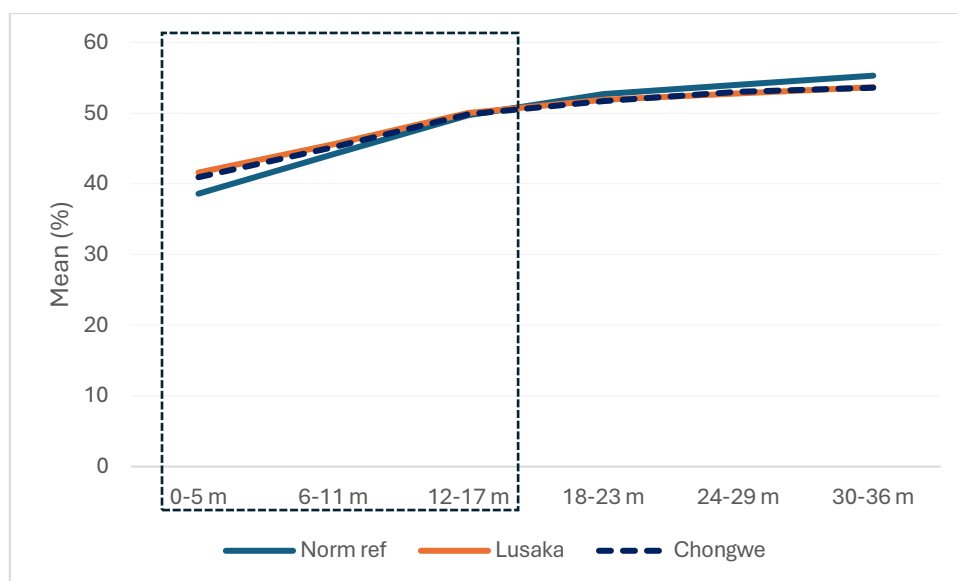


Figure 2: Developmental Trajectory of the Children

One samples *t*-tests were conducted to determine if the observed overall scores statistically differ from the form reference scores provided by the CREDI research team. Findings show that the observed differences from both cities were statistically different from the norm reference across all age bands. However, in Chongwe, the age band 12-17 was not statistically significant with the norm reference group. Children in both Chongwe and Lusaka are significantly below the norm reference groups in child development after the age of 17 months.

Table 9: Developmental Scores of the Children by Age of Child (Grouped)

Age (months)	Norm ref	Lusaka	<i>P</i> cf norm	Chongwe	<i>P</i> cf norm
0-5	38.6±9.5	41.16±1.75	<0.001	40.93±1.84	<0.001
6-11	44.2±8.8	45.59±1.59	<0.01	45.16±1.32	<0.001
12-17	49.7±8.1	50.03±1.49	<0.05	49.86±1.52	>0.05
18-22	52.1±7.9	51.87±1.08	<0.05	51.71±1.01	<0.001
24-29	54.0±7.6	52.80±0.72	<0.001	53.03±0.85	<0.001
30-36	55.3±7.5	53.67±0.63	<0.001	53.59±0.48	<0.001

Further analysis by settlement type in the two cities showed interesting findings. Data showed that regardless of settlement type, child development in the two cities remained the same compared with the norm reference group. For details review the table below:

Table 10: Developmental Scores of the Children by Settlement Type

Age	Norm	Lusaka	Mean (%)	<i>P</i>	Chongwe	Mean (%)	<i>P</i>
0-5	38.6±9.5	Unplanned	41.04±1.90	<0.001	Unplanned	40.54±2.01	<0.001
		Planned	41.24±1.66	<0.001	Planned	41.36±1.60	<0.001
6-11	44.2±8.8	Unplanned	45.81±1.43	<0.001	Unplanned	45.00±1.15	<0.001
		Planned	45.43±1.38	<0.001	Planned	45.32±1.47	<0.001
12-17	49.7±8.1	Unplanned	50.01±1.46	>0.05	Unplanned	50.72±1.43	>0.05
		Planned	50.05±1.54	>0.05	Planned	49.36±1.37	>0.05
18-22	52.1±7.9	Unplanned	51.80±1.12	>0.05	Unplanned	51.73±1.08	>0.05
		Planned	51.90±1.07	>0.05	Planned	51.70±0.96	>0.05
24-29	54.0±7.6	Unplanned	52.72±0.83	<0.001	Unplanned	53.10±0.91	<0.001

		Planned	52.86±0.61	<0.001	Planned	52.98±0.81	<0.001
30-36	55.3±7.5	Unplanned	53.44±0.61	<0.001	Unplanned	53.46±0.40	<0.001
		Planned	53.88±0.66	<0.001	Planned	53.69±0.51	<0.001

In Chongwe the analysis also compared differences between children in urban and rural areas. All were statistically significant compared to the norm reference except for age band 12-17 of rural children compared to the norm reference. See the Figure below:

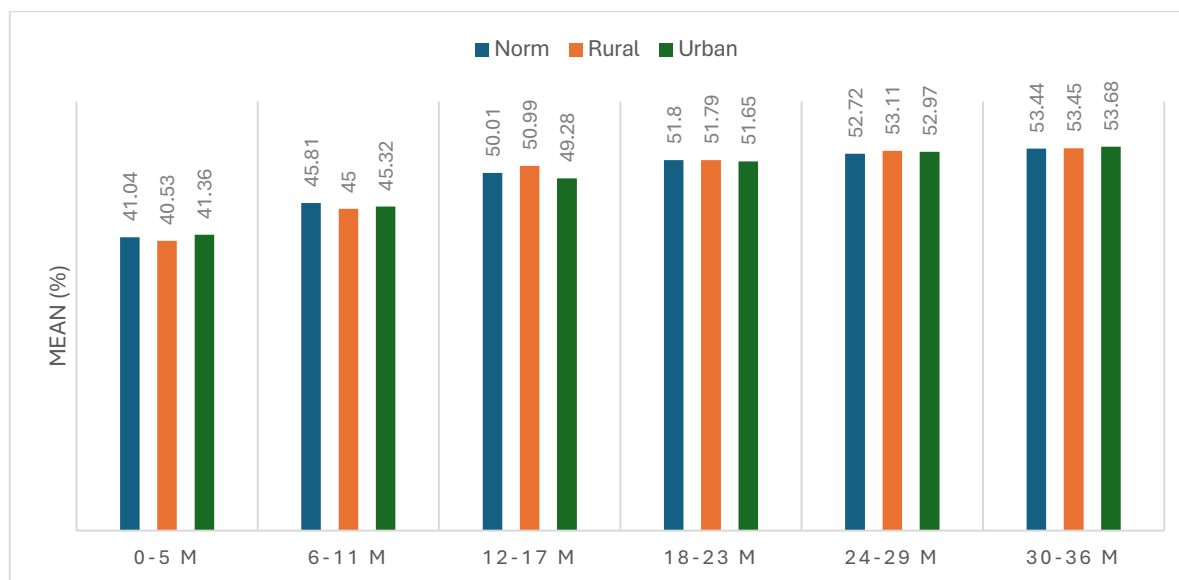


Figure 3: Developmental Scores of the Children by Rural/Urban Residence, Chongwe

The distribution of normed CREDI Z-scores which display the spread of the norm-referenced Z-scores in the sample were created for the two cities and for unplanned and unplanned settlements across the two cities. For Chongwe, distribution for urban and rural sample were also created. In graphs below, we can see that the average scores for children in two cities and settlements were almost evenly distributed with more children above the CREDI reference sample average (0). For details review the graphs below:

Lusaka sample

Below we see that the distribution of score compared to the CREDI reference sample, Lusaka children were slight more above the reference group. This pattern remained when the distribution was compared for children in unplanned and planned settlement. See below

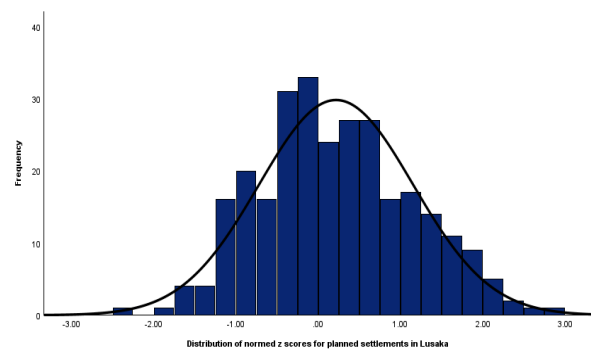
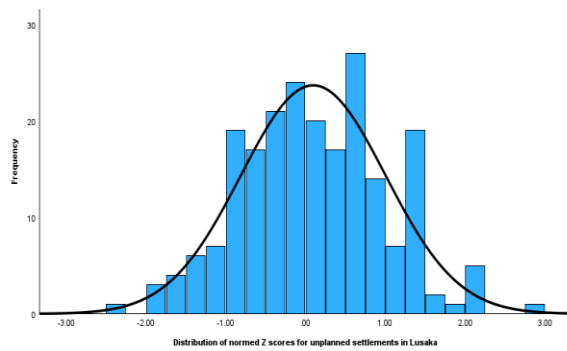
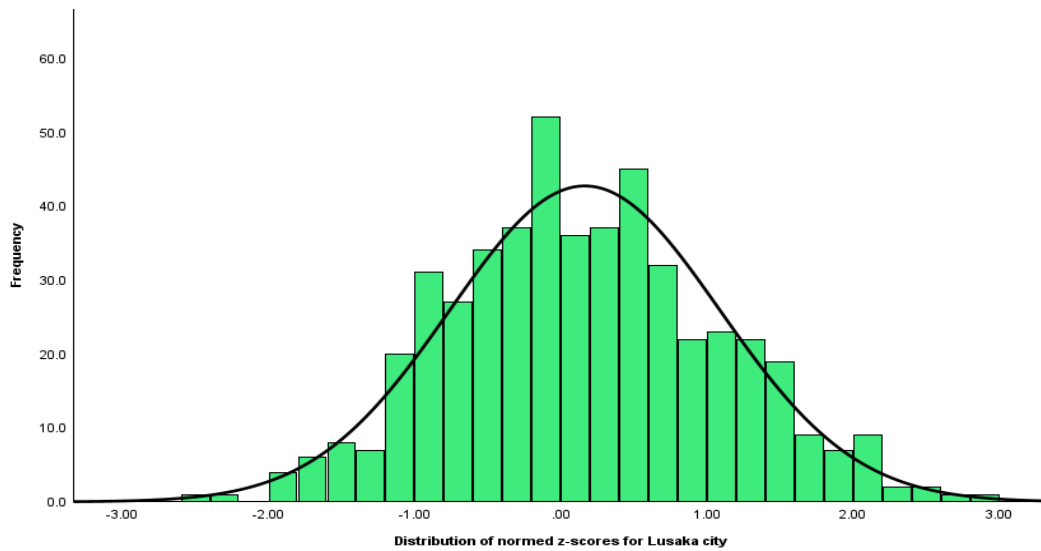
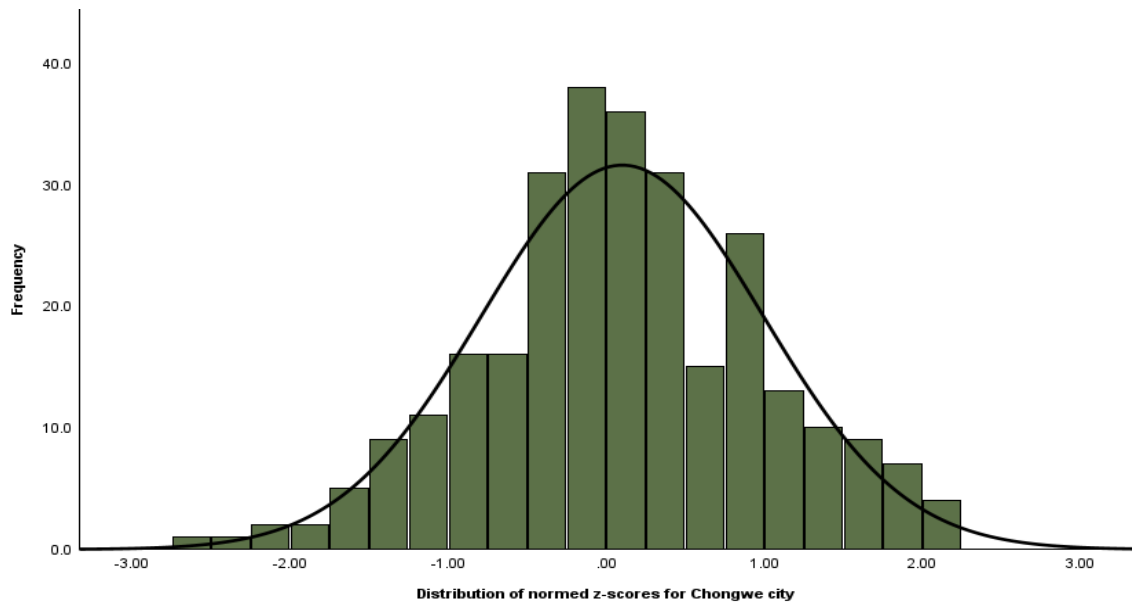


Figure 4: Distribution of score compared to the CREDI reference sample, Lusaka

Chongwe town

In Chongwe, the distribution was not very much different from that of Lusaka. Equally slightly more children were above the CREDI reference sample average (0). See the graph below:



For Chongwe unplanned, far more children are below the norm reference z score while for planned settlement, more children are far more above the norm reference z score. See below for details:

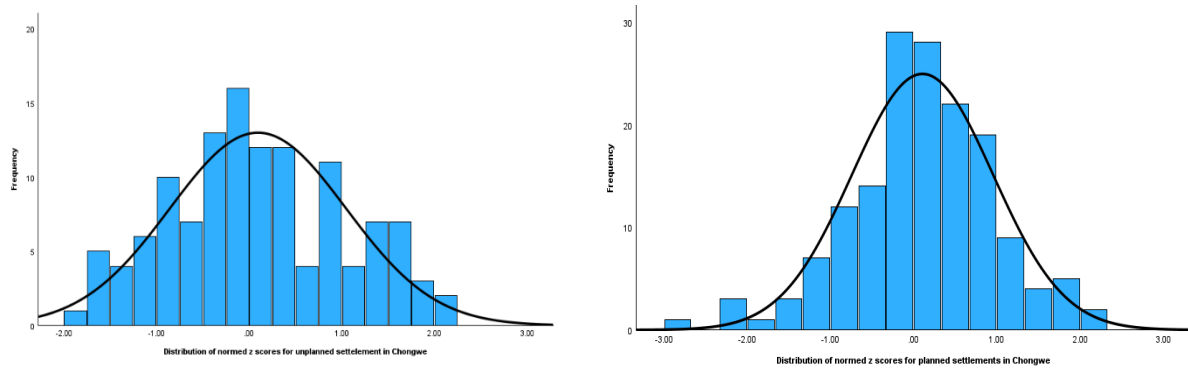
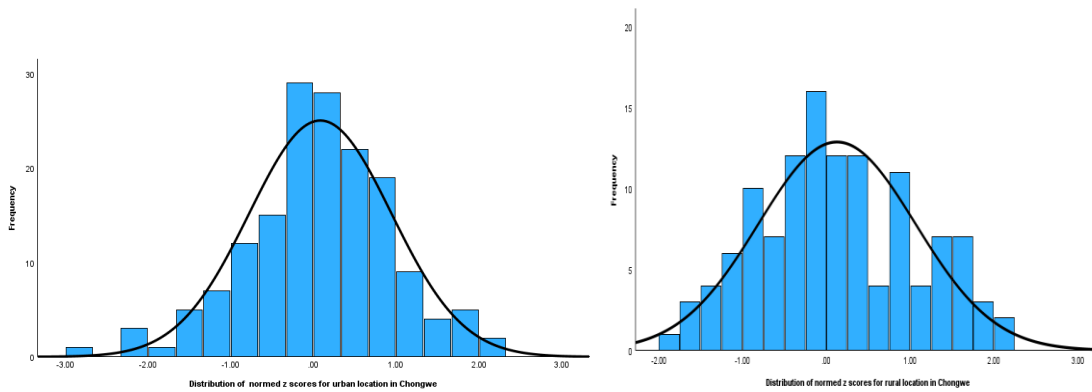


Figure 5: Distribution of score compared to the CREDI reference sample, Chongwe

With regards to rural vs urban, we see more children above the norm reference z score in urban Chongwe than in rural Chongwe. For details review the graphs below.



Predictors of Child Development

Having establishment, the state of child development in the two cities, further analysis was conducted to investigate factors that predict child development in Zambia. Therefore, regression analysis was conducted. In both cities, settlement type (0 – unplanned; 1- planned) and household vulnerability were not statistically significant predictors of children development. However, in both Chongwe and Lusaka, better home environment was a significant predictor of positive child development. All other covariates were not found to be statistically significant of child development in the two cities. For details review the table below:

Table 11: Predictors of Child Development in Chongwe and Lusaka City

Predictors	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
Planned vs unplanned	.62	-.02	-1.44	1.01	.91	.08	-1.03	2.55
HH vulnerability	.07	-.07	-.20	.07	.11	.13	-.06	.37
Model 2								
Planned vs unplanned	.63	-.02	-1.39	1.09	.88	.06	-1.18	2.31
HH vulnerability	.07	-.05	-.18	.09	.11	.07	-.13	.30
HOME inventory	.05	.26***	.09	.29	.07	.33***	.13	.42
Sib Relationship	.01	.01	-.02	.03	.02	.05	-.03	.06
Age of caregiver (yrs)	.04	-.01	-.08	.07	.05	.07	-.06	.15
Caregiver edu level	.10	-.05	-.26	.13	.14	-.12	-.44	.10

Note: * <0.05 ; ** <0.01 ; *** <0.001

Association Between Dimension of the HOME Environment and Child Development

We further explored the nature and strength of the relationship between child development (overall score) and the different dimensions of the HOME inventory. In both cities, positive discipline and structure was associated with better child development outcomes. Similarly, availability of learning materials at home was associated with better child development outcomes.

However, the organization dimension (which measures child's routine, safety, stimulation, and caregiving structure) was not significantly associated with child development outcomes in both cities. The dimension of childcare and involvement (measuring behavioral and interactional aspects) was also positively associated with child development outcomes in both cities. Responsive parenting behaviors were also associated with child development. However, the dimension variety (which focuses on

diverse, frequent, and enriching environment) was negatively associated with child development outcomes. For details review the table below:

Table 12: Correlations between HOME subscales and CREDI Scores, Lusaka and Chongwe

HOME Dimension	CREDI score Lusaka	CREDI score Chongwe
Discipline & structure	.243***	.245***
Organization of home	-.067	-.091
Learning materials	.276***	.210***
Childcare & involvement	.196***	.196***
Variety of stimulating environment	-.178***	-.137*
Responsive parenting	.271***	.266***
HOME total	.256***	.242**

Note: *<0.05; **<0.01; ***<0.001

Anthropometric Assessment of Children

Anthropometry was used to assess child nutritional status, it is used to report on child growth indicators. The state of ECD baseline survey identified 427 children under age 5 with eligible height and weight measurements. The percentages with valid height-for-age, weight-for-height, and weight-for-age were 93%, 100%, and 93%. Overall, 30% of children under age 5 were stunted (short for their age), 11% of children under age 5 were underweight (low weight for their height), and 9% of children under age 5 were wasted (thin for their height). By districts, among children under age 5 in Chongwe about 3 in 10 children were stunted (37.8%), 10% were underweight, and 7% were wasted. In Lusaka, 27% of children under age 5 were stunted, 11% were underweight and 10% were wasted. The results on nutritional status among children under age 5 for Lusaka and Chongwe indicate that all nutritional status indicators were above national average as estimated by 2024 ZDHS. Details are shown in the graph below:

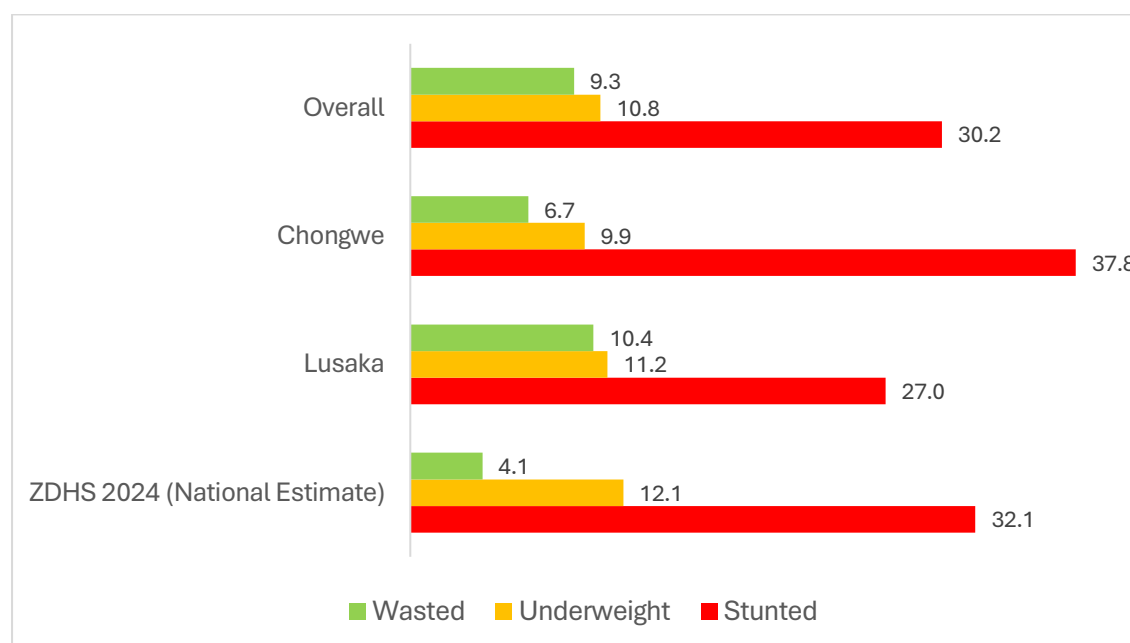


Figure 6: Child Nutrition Assessment, Stunting, Underweight and Wasting; Chongwe and Lusaka

In Lusaka, the three-child nutritional outcomes appear to vary by child's sex and age, caregiver level of education, type of residential area and vulnerability status. The results show that male children have higher proportions of being stunted, underweight, or wasted, and children whose caregivers have higher

levels of education appear to have, those planned settlement and children from households that can graduate on the vulnerability ladder have lower proportions of being stunted, underweight or wasted. However, only a child's age was significantly associated with the child being underweight or wasted ($p<0.05$), and vulnerability status was significantly associated with stunting ($p<0.05$). See the table below:

Table 13 : Selected socio-economic characteristics and Child Nutritional Outcomes, Lusaka District

Characteristic	Nutritional Assessment (Children <5)					
	Stunted (N=278)	P	Underweight (N=296)	P	Wasted (N=279)	P
Sex of Child						
Male	29.7	>0.05	13.6	<0.05	11.5	>0.05
Female	24.3		8.7		9.3	
Age of child						
< 6 Months	21.7	>0.05	29.2	<0.1	30.0	<0.05
6-8 Months	26.3		0.0		21.1	
9-11 Months	27.3		8.3		8.7	
12-17 Months	28.3		10.0		10.5	
18-23 Months	33.3		15.4		11.5	
24-35 Months	21.5		9.1		6.2	
36-47 Months	37.2		6.7		0.0	
48-59 Months	20.7		16.1		14.8	
Caregiver's Level of Education						
No/Primary Education	31.0	>0.05	16.7	>0.05	12.7	>0.05
Secondary	27.4		10.9		10.3	
Tertiary (Higher)	19.5		4.6		7.5	
Type of Settlement						
Planned	25.2	>0.05	10.3	>0.05	11.1	>0.05
Unplanned	29.3		9.8		11.5	
Vulnerability						
Can Graduate	23.0	<0.1	8.9	>0.05	10.5	>0.05
Slightly vulnerable	35.1		15.4		10.3	
Moderately Vulnerable	-		-		-	
Total	27.0		11.2		10.2	

In Chongwe nutritional outcomes for children under age 5 were significantly associated with caregivers' level of education for stunting ($p<0.05$), and the child's age was significantly associated with being underweight and wasting ($p<0.05$). See table below:

Table 14 : Selected Socio-Economic Characteristics and Child Nutritional Outcomes, Chongwe District

Characteristic	Nutritional Assessment (Children <5)					
	Stunted (N=119)	<i>P</i>	Underweight (N=131)	<i>P</i>	Wasted (N=120)	<i>P</i>
Sex of Child						
Male	41.0	>0.05	13.6	<0.05	13.3	>0.05
Female	34.5		6.2		11.5	
Age of child						
< 6 Months	27.3	>0.05	27.3	<0.05	22.2	>0.05
6-8 Months	37.5		10.0		11.1	
9-11 Months	25.0		37.0		25.0	
12-17 Months	60.0		15.4		8.3	
18-23 Months	9.1		0.0		0.0	
24-35 Months	44.7		7.5		2.6	
36-47 Months	33.3		0.0		0.0	
48-59 Months	46.7		5.9		7.1	
Caregiver's Level of Education						
No/Primary Education	40.5	<0.05	7.1	>0.05	10.3	>0.05
Secondary	41.3		8.8		3.3	
Tertiary (Higher)	21.1		19.1		9.5	
Residence						
Rural	30.0	>0.05	12	>0.05	4.1	>0.05
Urban	43.5		7.3		10.1	
Type of Settlement						
Planned	41.5	>0.05	7.7	>0.05	10.8	>0.05
Unplanned	33.3		11.1		3.8	
Vulnerability						
Can Graduate	38.6	>0.05	9.4	>0.05	4.5	>0.05
Slightly vulnerable	35.5		11.4		12.9	
Moderately Vulnerable	-		-		-	
Total	37.8		9.9		7.6	

Further analysis was conducted to assess the state of nutrition in the targeted households. On average, 63% of households reported that nutritious foods for children under the age of five were always available. However, the availability of nutritious food differed significantly between planned and unplanned settlements, with households in planned areas more frequently reporting consistent access to such foods.

The data also revealed that while only about half of the households were willing to pay for nutritious meal programs, approximately 75% expressed a desire to be supported through such programs. Notably, households in unplanned settlements were more likely to express this need compared to those in planned areas. Over 50% of respondents reported having received nutrition counselling. In Lusaka,

there was no significant difference between planned and unplanned settlements. In contrast, in Chongwe, significantly more households in unplanned settlements reported receiving nutrition counselling compared to those in planned areas. Interestingly, the proportion of children aged 0–6 years who were already on solid foods and liquids was high, suggesting that most were not exclusively breastfed. For further details, refer to the table below:

Table 15: Access to Nutritious Foods by Type of Settlement, Lusaka and Chongwe

Variables	All	Lusaka		Chongwe	
		Planned	Unplanned	Planned	Unplanned
Availability of nutritious food for U6:					
Always available	971 (63.1)	430(74.4)	291 (69.3)*	170(58.2)	80 (32.4)**
Sometimes	372 (24.2)	108(18.7)	93 (22.1)	82 (28.1)	89 (36.0)
Rarely	163 (10.6)	36 (6.2)	29 (6.9)	35 (12.0)	63 (25.5)
Not available	31 (2.0)	4 (0.7)	7 (1.7)	5 (1.7)	15 (6.1)
Willing to pay for nutritious meal prog	852 (55.7)	289(50.4)	246 (58.9)**	184(63.0)	132 (53.4)*
Desire nutrition support programs	1,155(75.4)	404(70.5)	341 (81.6)**	210(71.9)	199 (80.6)*
Have received nutrition counselling education before	986 (64.4)	360(62.8)	262 (62.7)	182(62.3)	181 (73.2)**
Child has received essential vitamins and micronutrients (self-report)	1,339(87.4)	495(86.3)	366 (87.6)	65(90.8)	212 (85.8)
Child taking any food or liquid (0-6 months) *inclusive breastfeeding	64 (55.2)	25 (62.5)	13 (48.2) *	11 (45.8)	15 (60) **

**=<0.01, *<0.05

School Readiness

The results are organised by city and evaluated by subscale. Descriptive statistics for each city council are provided and categorised by gender. Additionally, inter-city comparisons are made to assess the comparability, taking into account the socio-economic variations between the Lusaka and Chongwe districts.

Idela Performance for Lusaka District

The results presented in Table 2 show that the overall mean achievement for children in Lusaka is 57.7% ($M = .577$, $SD = .228$). Additionally, girls ($M = .580$, $SD = .229$) out-performed boys ($M = .575$, $SD = .228$), with a statistically significant mean difference ($F(1, 415) = 4.754$, $p < .05$). These findings indicate sub-optimal development among children in the Lusaka district and suggest gender parity in early childhood development. Overall, the low achievement among children also highlights systemic challenges in the evaluated domains.

When results are segregated by subscale, the mean performance for Motor development is 61.6% ($M = .616$, $SD = .299$). After desegregating the results by gender, girls ($M = .627$, $SD = .309$) performed significantly better than boys ($M = .603$, $SD = .288$), and mean difference in motor development is statistically significant [$F(1, 415) = 4.546$, $p < .05$] between genders. On socio-emotional development, the combined mean performance among children in Lusaka district was 64.0% ($SD = .243$), and similarly girls ($M = .643$, $SD = .239$) out-performed their male ($M = .638$, $SD = .248$) counterparts. The socio-emotional mean difference was even smaller and not statistically significant ($p > .05$).

Compared to motor skills and socio-emotional development, children demonstrated slightly lower performance in emergent numeracy ($M = .588$, $SD = .251$) and the lowest in literacy development ($M = .463$, $SD = .278$). Gender analysis revealed that boys ($M = .596$, $SD = .253$) outperformed girls ($M = .581$, $SD = .249$) in emergent numeracy, whereas girls ($M = .467$, $SD = .279$) exhibited superior performance

in emergent literacy compared to boys ($M = .467$, $SD = .279$). Overall, children in Lusaka district displayed low performance across all domains, with minimal gender differences observed, as boys and girls performed similarly. The highest mean score was recorded in socio-emotional development, while emergent literacy received the lowest score.

Table: IDELA Achievement by City and Gender

	LUSAKA						CHONGWE					
	Boys		Girls		Total		Boys		Girls		Total	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Motor Development	.603	.288	.627	.309	.616	.299	.586	.307	.700	.245	.654	.275
Socio-emotional	.638	.248	.643	.239	.640	.243	.512	.284	.619	.240	.576	.262
Emergent Literacy	.463	.278	.467	.279	.465	.278	.337	.271	.473	.286	.419	.286
Emergent Numeracy	.596	.253	.581	.249	.588	.251	.473	.280	.590	.228	.543	.254
Overall Total	.575	.228	.580	.229	.577	.228	.477	.255	.596	.217	.548	.238

IDELA Results for Chongwe District

In Chongwe district, the IDELA overall mean score across all domains was 54.8% ($SD = .238$). Girls scored higher ($M = .596$, $SD = .217$) than boys, whose mean score was 47.7% ($SD = .255$). However, this gender advantage in developmental outcomes was not statistically significant ($p > .05$). When assessed by subscale, the mean score was 65.4% ($SD = .275$), with girls ($M = .700$, $SD = .245$) outperforming boys ($M = .586$, $SD = .307$) in motor skills development. Similarly, girls scored higher ($M = .619$, $SD = .240$) than boys ($M = .576$, $SD = .262$) in socio-emotional development. The mean differences for both motor skills and socio-emotional development were not statistically significant ($p > .05$), with an overall socio-emotional development mean of 57.6% ($SD = .262$).

Children in Chongwe district exhibited the lowest scores in emergent literacy ($M = .419$, $SD = .286$), followed by emergent numeracy ($M = .543$, $SD = .254$). Girls' literacy performance ($M = .473$, $SD = .286$) significantly surpassed that of boys ($M = .337$, $SD = .271$), though the mean difference lacked statistical significance ($p > .05$). Conversely, unlike in Lusaka district, boys ($M = .473$, $SD = .286$) were outperformed by girls, who recorded a mean score of 54.3% ($SD = .254$). Again, the mean difference was not significant ($p > .05$). In conclusion, emergent literacy was the weakest domain in Chongwe district, followed by emergent numeracy development, while motor development was the strongest. The results indicate significant gender disparities across all assessed domains, with girls achieving higher levels of performance than boys.

An inter-district comparative analysis revealed significant findings worth considering. Children in Lusaka ($M = .563$, $SD = .237$) generally achieved higher scores across most domains compared to their counterparts in Chongwe ($M = .527$, $SD = .231$), and the mean difference was statistically significant ($F(1,572) = 6.749$, $p < .05$). Gender differences were more pronounced in Chongwe, where girls consistently outperformed boys in all assessed areas. In contrast, Lusaka exhibited minimal gender disparities, with nearly equal performance between boys and girls. These findings suggest potential regional and gender-based variations in early childhood development outcomes. When desegregated

by subscale, Children in Lusaka performed better in all domains, with the exception of motor development.

IDELA Achievement by Age in Lusaka District

Developmental progression was assessed across age groups to quantify improvements in children aged 3.5 to 6 years. Table 3 displays data for children in the Lusaka district, highlighting significant competencies among 3-year-olds across all evaluated domains. Emergent literacy skills were the lowest ($M = .128$, $SD = .138$), followed by socio-emotional skills ($M = .154$, $SD = .069$). Notably, the most substantial developmental gains occurred between ages 3 and 5, with motor skills more than doubling from .263 to .71. Literacy skills also tripled during this period, increasing from .128 to .535, suggesting that the age range of 3.5 to 5 years is critical for cognitive development.

Motor skills consistently surpassed other competencies, peaking at age 6 ($M = .771$, $SD = .244$). Although emergent literacy proved to be the most challenging skill to acquire at age 6 ($M = .621$), it experienced a remarkable 385% growth from age 3.5 to 6, yet remained the lowest-performing domain at that age. Children demonstrated greater developmental consistency, as indicated by narrowing standard deviations with age, which decreased from .252 to .194 between ages 4 and 6. This trend may be linked to the influence of educational experiences. Children demonstrated a 41% developmental gain from ages 4 to 5, compared to a 13% increase from ages 5 to 6. By age 6, motor development neared mastery ($M = .771$), while literacy reached basic proficiency ($M = .621$), highlighting the necessity for targeted pedagogical strategies for the later.

Table: IDELA Achievement by Age in Lusaka City

Subscales	3 years (> 3.5)		4 years		5 years		6 years	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Motor Development	.263	.156	.427	.284	.710	.220	.771	.244
Socio-emotional	.154	.069	.493	.252	.622	.237	.728	.194
Emergent Literacy	.128	.138	.309	.232	.535	.237	.621	.266
Emergent Numeracy	.221	.146	.443	.225	.650	.222	.714	.218
Overall	.191	.082	.418	.205	.629	.190	.708	.189

IDELA Achievement by Age in Chongwe District

In Chongwe district, as showed in Table 4, children showed clear, significant developmental trajectories between 3.5 years and 6 years across all domains. Children made the most gains between 4 and 5 years, where their motor skills increased from .484 to .681—an increase of more than 41%—and in the emergent numeracy domain, children made gains of over 51%. During the 5 to 6 years period, children made gradual developmental improvements ranging from 4% to 12%.

In more specific terms, the development of motor skills is gradual and steady, peaking at the age of 6 years ($M = .775$), where the lowest variability is also observed ($SD = .180$). Additionally, the developmental trajectories of motor skills are age-related in progression. Emergent literacy was the slowest domain to develop, with an overall rate of 4.0%. Literacy peaks around age 5 and begins to stagnate, with scores reducing from 47.2% to 46.0%. Interestingly, children's literacy scores exhibited the highest variability levels, ranging from .232 to .287. The progression of children's socio-emotional

and numeracy skills in Chongwe district is characterized by intermediate performance. Participants' emergent numeracy skills consistently outperformed their literacy for age-groups under consideration. In summary, 3-year-olds exhibited very low baseline skills but made significantly rapid growth afterward. Motor skills approach mastery by the age of 6, while emergent literacy development lagged behind throughout. Variability in development in all domains reduces considerably with age. There is rapid acceleration in the development of up to 49.5% between 4 and 5 years, but after age 5 to 6 years, their gains are modest, averaging as low as 9.8%.

Table 16: IDELA Achievement by Age in Chongwe City

Subscales	3 years (> 3.5)		4 years		5 years		6 years	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Motor Development	.244	.214	.484	.285	.681	.214	.775	.180
Socio-emotional	.350	.049	.389	.261	.570	.211	.686	.229
Emergent Literacy	.161	.005	.290	.232	.472	.271	.460	.287
Emergent Numeracy	.386	.178	.407	.240	.613	.217	.642	.212
Overall	.285	.038	.392	.228	.584	.195	.641	.193

Comparisons between Lusaka and Chongwe reveal that children have the best performance in motor skills and the weakest in literacy. Motor skills seem to peak at 6 years, while the rate of literacy development is relatively low throughout, with similar age-related progression curves across domains. However, Lusaka children outperformed Chongwe district in all except motor skills—albeit with nearly identical results—urban children seem to have a significant advantage over their rural-based counterparts. One notable divergence is that children in Chongwe appear to have a slight head start in emergent numeracy compared to Lusaka.

Thus, we recommend that interventions leverage motor skills to improve other developmental domains. Thus, implementing age-appropriate activities to enhance emergent literacy at the community level is essential to improve children's literacy skills. These initiatives should encompass reading programs and parent training focused on storytelling, picture books, and phonemic awareness²³ (Neuman & Dickinson, 2011). Additionally, it is crucial to incorporate object labelling and singing rhymes into daily routines to reinforce foundational skills²⁴ (Britto et al., 2017).

IDELA Achievement by Settlement Type in Lusaka and Chongwe

Children's level of achievement was assessed by settlement type as defined planned and unplanned in each city. The results are displayed in Table 5 and presented by city.

IDELA Achievement by Settlement Type in Lusaka

Results show that children from planned settlements exhibited significantly elevated developmental scores across all domains compared to those from unplanned settlements (all $p < .001$) in Lusaka district. The most pronounced disparities were observed in socio-emotional development ($\Delta M = .106$) and numeracy ($\Delta M = .103$), with the smallest difference in motor skills ($\Delta M = .060$). Overall developmental scores were .093 points higher in planned settlements ($M = .602$ vs. $M = .509$), demonstrating moderate variability across domains ($SDs = .228 - .303$).

²³ Neuman, S. B., & Dickinson, D. K. (Eds.). (2011). Handbook of early literacy research (Vol. 3). Guilford Press.

²⁴ Britto et al. (2017). Nurturing care: promoting early childhood development. The Lancet, 389(10064), 91-102. [https://doi.org/10.1016/S0140-6736\(16\)31390-3](https://doi.org/10.1016/S0140-6736(16)31390-3)

Table 17: IDELA Achievement by Settlement Type in Lusaka and Chongwe

Subscale	Settlement Type	Lusaka			Chongwe		
		<i>M</i>	<i>SD</i>	<i>p</i>	<i>M</i>	<i>SD</i>	<i>p</i>
Motor Development	Planned	.635	.303	.050	.707	.240	.001
	Unplanned	.575	.294		.553	.269	
Socio-emotional	Planned	.641	.247	.001	.608	.244	.001
	Unplanned	.535	.254		.460	.256	
Emergent Literacy	Planned	.509	.293	.001	.538	.272	.001
	Unplanned	.406	.255		.265	.198	
Emergent Numeracy	Planned	.624	.251	.001	.657	.225	.001
	Unplanned	.521	.250		.436	.215	
Overall Total	Planned	.602	.236	.001	.628	.218	.001
	Unplanned	.509	.228		.429	.200	

IDELA Achievement by Settlement Type in Chongwe

Settlement-based disparities in Chongwe district were statistically significant, with planned areas exceeding unplanned areas by .148 - .273 points across domains (all $p < .001$). Literacy displayed the highest inequality ($M = .538$ vs $.265$, $p < .001$), followed by numeracy ($\Delta M = .221$). The overall developmental gap was substantial ($M = .628$ vs $.429$, $p < .001$), with unplanned settlements showing low variability in literacy ($SD = .198$), indicating uniformly poor outcomes.

Children's School Readiness in Lusaka and Chongwe Cities

Children are expected to be prepared for school enrolment prior to graduating from early childhood programmes. In this evaluation, school readiness was assessed through performance on the four primary developmental domains of the IDELA. A child was deemed school-ready if they scored above 70% on each subscale; those scoring below this threshold were classified as not ready. Furthermore, developmental progression was categorized as struggling ($< 40\%$), emerging (between 40% and 70%), or had attained mastery ($> 70\%$) for each domain. Results are presented by city in Figures 2 and 3.

School Readiness in Lusaka District

In Lusaka district, when we statistically evaluated the between school ready children and those who are not, results revealed significant disparities in developmental scores between children deemed school-ready and those not classified as such across all assessed domains. Children identified as school-ready demonstrated higher mean scores in several developmental areas: motor development ($M = .85$, $SD = .11$), socio-emotional skills ($M = .80$, $SD = .13$), emergent literacy ($M = .75$, $SD = .16$), and emergent numeracy ($M = .82$, $SD = .11$). Conversely, children categorized as not school-ready exhibited lower mean scores across these domains: motor development ($M = .47$, $SD = .29$), socio-emotional skills ($M = .48$, $SD = .24$), literacy ($M = .30$, $SD = .19$), and numeracy ($M = .44$, $SD = .21$). The differences in means between school-ready and not school-ready children were statistically significant ($p < .001$) across developmental domains.

Additionally, school readiness was evaluated to determine the percentage of children prepared for school based on their IDELA performance in Lusaka district. Figure 1 highlights significant disparities across the five developmental domains. Overall, only 36.45% of children are deemed school ready,

leaving 63.55% inadequately prepared to enter grade school. Among the domains assessed, motor skills development had the highest readiness rate at 51.32%, while emergent literacy showed the lowest at 25.42%. A moderate percentage of children met the minimum expectations in socio-emotional development, and emergent numeracy was similarly concerning, with only 39.81% considered school ready.

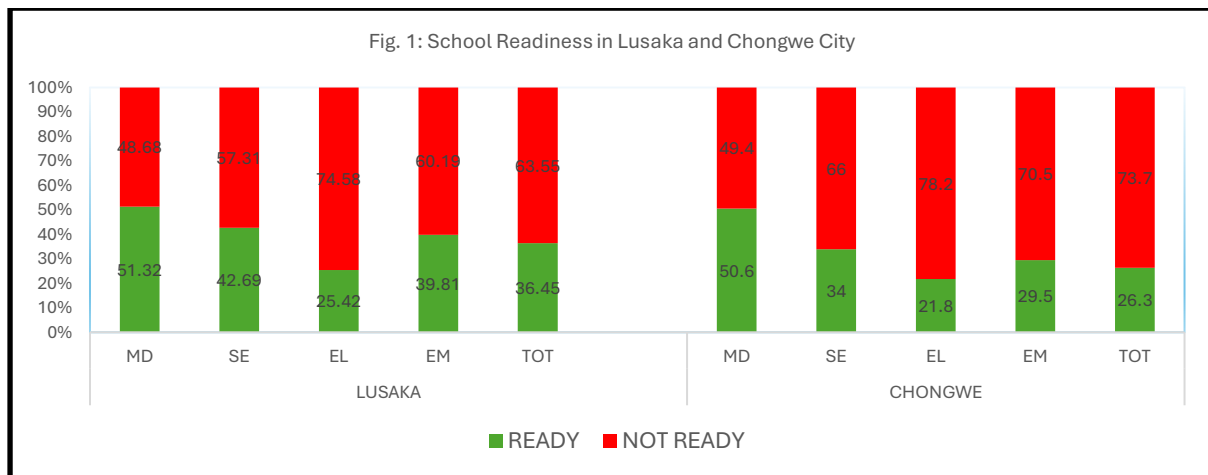


Figure 7: School Readiness in Lusaka and Chongwe City

Note: MD = motor development, SE = socio-emotional, EL = emergent literacy, EM = emergent numeracy, TOT = overall total

In summary, the majority of children in the Lusaka district are not prepared for school, indicating a pressing need for early intervention across all developmental domains. This situation raises serious concerns, particularly in emergent literacy and numeracy, which are critical for children's success in school. Given these findings, over 63% of children entering the school system are unprepared and require targeted interventions focused on emergent literacy and numeracy to fully benefit from the educational process. Without these foundational skills, they are likely to struggle.

Given the low levels of emergent literacy school readiness in Lusaka, we recommend the implementation of structured literacy programmes at the ECE level, emphasizing phonemic awareness and vocabulary acquisition. Additionally, to bolster emergent numeracy, the city should adopt movement-based numeracy programmes that leverage children's stronger motor skills for teaching counting, shapes, and patterns through physical activities and manipulatives.

School Readiness in Chongwe District

In Chongwe, significant mean differences were noted between school-ready and non-school-ready children across all developmental domains ($p < .001$). School-ready children demonstrated significantly higher mean scores in motor development ($M = .868$ vs $M = .544$), socio-emotional skills ($M = .789$ vs $.443$), emergent literacy ($M = .744$ vs $.277$), and emergent numeracy ($M = .799$ vs $.455$). These findings highlight critical developmental disparities in literacy skills, necessitating targeted early interventions.

In Chongwe district, 73.7% of children were unprepared for school (see Figure 1 above), with only 26% demonstrating essential skills across various developmental domains. The evaluation found that motor development was the strongest area, with over 50% of the children showing readiness. However, only 34% were considered school ready in socio-emotional development, revealing that 66% lacked the foundational skills needed for grade school. Figure 7 shows the distribution of children's school readiness in Chongwe district.

In comparison, academic foundations in emergent literacy and numeracy were notably insufficient. Readiness for emergent literacy was just 21.8%, while emergent numeracy readiness was slightly better

at 29.5%. The fact that over 78.2% and more than 70% of the sampled children may enter the school system inadequately prepared highlights the urgent need for policy review and intervention in Chongwe.

The city can organize village reading camps and mobile library services to improve the low literacy readiness rate, particularly in rural areas of Chongwe, to improve performance. Additionally, recruiting and training local volunteers to lead parent-child shared reading sessions at home is recommended, as many parents may lack the literacy skills to do this themselves.

School Readiness Progression Levels in Lusaka District

The mean score differences at progression stage (mastery, emerging, and struggling) were compared within the developmental domains. Lusaka exhibited statistically significant differences across progression stages in all domains ($p < .001$). Numeracy emerged as the strongest domain for mastery-level students ($p < .001$). Numeracy emerged as the most proficient domain for mastery-level students ($M = .82$, $SD = .11$), while literacy again showed the most pronounced disparity between mastery ($M = .75$) and struggling ($M = .16$) students.

In Lusaka, the percentage of school-ready children, categorized as struggling, emerging, or at mastery, is notably low across all domains (see Figure 2). Only 36.50% of participants achieved comprehensive mastery, while 35.00% were classified as emerging, and over 28% were struggling. In specific domains, children demonstrated the highest mastery in motor skills development at 51.30%, with approximately 26% struggling. In the socio-emotional domain, about 43% were striving. Significant deficits were observed in academic foundational skills, with nearly 46% struggling in emergent literacy and 29% in numeracy. Only 25.40% reached mastery in literacy, and 39.80% in emergent numeracy, indicating substantial challenges. Figure 8 shows a pictorial distribution of children's school readiness in terms of their level of mastery, emerging and struggling levels.

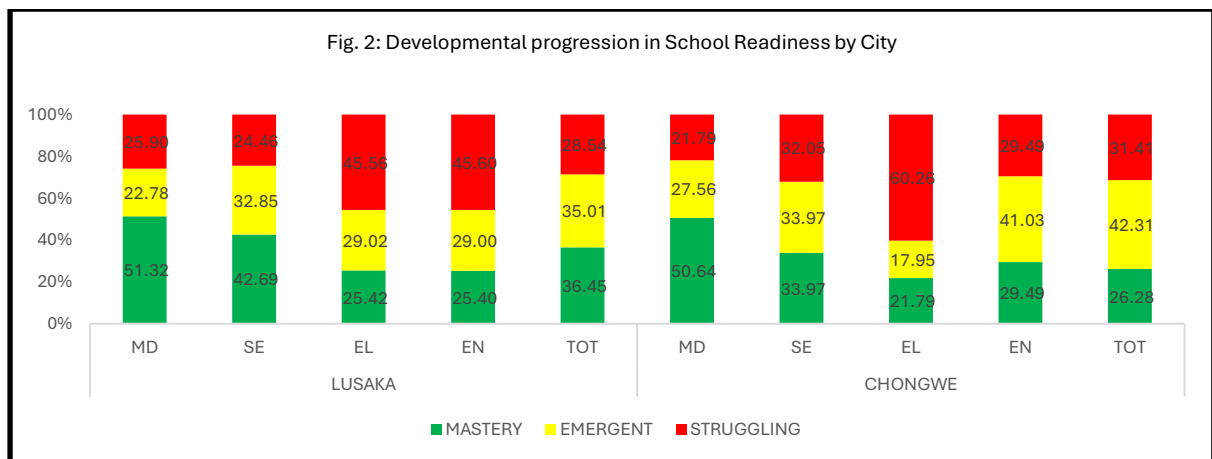


Figure 8: School Readiness in Lusaka and Chongwe City

Note: MD = motor development, SE = socio-emotional, EL = emergent literacy, EM = emergent numeracy, TOT = overall total

School Readiness Progression in Chongwe District

Students classified in the Mastery category exhibited significantly superior performance across all developmental domains when compared to their Emerging and Struggling counterparts, and the mean differences across these categories are statistically significant ($p < .001$). The motor domain demonstrated the highest overall mean score ($p < .001$), with a mean of $M = .63$ ($SD = .27$), while the literacy domain exhibited the lowest mean score ($M = .40$, $SD = .27$). Literacy showed the most significant performance gap between Mastery students ($M = .74$) and Struggling students ($M = .15$). In

the socio-emotional domain, Mastery students ($M = .79$) significantly outperformed struggling students ($M = .24$) by a factor exceeding three.

In Chongwe district, just over 26% of children demonstrated overall mastery across the IDELA developmental domains. While a significant portion, approximately one-third (31.4%), of the sample struggled, an alarming 42.31% remained in the emerging stage of development. Specifically, motor development was the best mastered, with 5.60% achieving proficiency, and only 21.8% struggling. This was followed by socio-emotional development, where 34.97% mastered the skills and 32% faced challenges. Figure 8 above shows proportions of children at mastery, emerging, and struggling stage across domains.

Similar to Lusaka district, emergent literacy and numeracy posed significant difficulties for children, resulting in notable achievement gaps. Only 21.79% of children mastered basic literacy skills, while 29.49% achieved proficiency in numeracy. Unfortunately, the number of children at the emerging stage in emergent literacy was also lamentably low. This lag in critical developmental milestones should concern stakeholders in the district, as it indicates that the majority of children require substantial support to get back on track.

Both Lusaka and Chongwe districts exhibit significant similarities. In both areas, children demonstrated higher mastery levels in motor development compared to other domains, with emergent literacy posing the greatest challenge. Overall, children in Lusaka showed superior mastery across all domains; over 60% of children in Chongwe experienced difficulties with emergent literacy, compared to 45.60% in Lusaka. These disparities highlight notable inter-district variations likely influenced by differences in socioeconomic status (SES), suggesting the necessity for district-specific interventions. While emergent literacy and, to a lesser extent, emergent numeracy require urgent attention in both districts, Chongwe necessitates more comprehensive and targeted interventions. Motor skills were the most mastered at 5.60%, with Lusaka slightly outperforming Chongwe. In summary, only 26.28% achieved high mastery levels, while 31.41% were struggling across domains, with Chongwe exhibiting the highest proportion of children at the emerging level at 42.31%, indicating a significant need for targeted support.

Predictors of School Readiness

Multivariate regression analyses were performed to identify the factors that predict school readiness in each city. The outcome variable for school readiness was children's IDELA achievement, while the predictor variables included household vulnerability and settlement type (planned vs. unplanned). Sibling relationship, HOME inventory, caregiver education level, and caregiver age were included as covariates. The results are presented in Table 18.

Predictors of School Readiness in Lusaka District

Household (HH) vulnerability and settlement type were significant predictors of school readiness in Lusaka district. Increased HH vulnerability had a negative impact on school readiness ($\beta = -.185, p < .01, 95\% CI [-.014, -.002]$). While children from planned settlements were associated with better school readiness outcomes ($\beta = -.167, p < .05, CI [-.145, -.017]$) than their counterparts residing in unplanned residences. The two covariates [sibling relationships ($\beta = .212, p < .001$) and HOME Inventory scores ($\beta = .215, p < .001$)] also had significant and positive influence on school readiness. Older caregivers also demonstrated a statistically significant, albeit modest, influence on children's readiness for formal schooling ($\beta = .146, p < .05$) in the Lusaka district.

Predictors of School Readiness in Chongwe District

In Chongwe district, school readiness was a function of settlement type ($\beta = -.518, p < .001, CI [-.318, -.133]$). Unlike Lusaka city, settlement type was three times more effective in Chongwe in preparing children school life. Contrastingly, the influence of HH vulnerability in predicting school readiness was insignificantly minimal and also not statistically significant ($p > .05$). Education level ($\beta = .267, p < .01$) and age of caregiver ($\beta = .336, p < .001$) had very high and positive influence on school readiness. On the

other hand, sibling relationships ($\beta = .063-.182, p < .05$) was a minimally effective covariate, while the home environment did not have significant contribution.

Table 18: Predictors of School Readiness in Lusaka and Chongwe Cities

	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
HH Vulnerability	.003	-.185**	-.014	-.002	.005	.086	-.006	.013
planned vs unplanned	.032	-.167*	-.145	-.017	.046	-.518***	-.318	-.133
Model 2								
HH Vulnerability	.003	-.108	-.010	.001	.004	.041	-.006	.010
planned vs unplanned	.032	-.148*	-.135	-.009	.045	-.428***	-.277	-.096
Sib Relationship	.001	.212***	.001	.003	.001	.182**	.000	.004
HOME Inventory	.002	.215***	.003	.013	.003	.063	-.004	.008
Caregiver educ level	.005	.106*	-.002	.017	.005	.267**	.004	.023
Age of caregiver	.002	.146*	.001	.007	.002	.336***	.004	.012

* = < .05 ** = < .01 *** = < .001

Thematic area 2: Household Caregiving Practices

This section will address seven research questions related parenting practices, knowledge, attitudes and beliefs. The questions to answered are as follows:

1. What is the status of 0-3-year-old children's parental and caregivers' knowledge, attitudes, behaviours, and practices that promote or hinder children's development and learning in Zambia?
2. To what extent is responsive parenting practiced at home?
3. What do Zambian households consider to be, nurturing care and early stimulation of children?
4. To what extent do caregivers practice responsive social protection and ensure child safeguarding in Zambia?
5. What proportion of children experience psychological aggression or physical punishment in Zambia?
6. How can nurturing care be promoted among parents and caregivers in Zambia?
7. How do socio-economic factors impact caregiving practices and child development outcomes?

Caregiver Knowledge on Child Development

Knowledge of child development was assessed using the Caregiver Knowledge of Child Development Questionnaire by Ertem, which has a maximum possible score of 40, with higher scores indicating greater knowledge. On average, caregivers in Lusaka scored 20.14, while those in Chongwe scored

20.01. These results suggest that caregivers in both areas possess only a basic understanding of child development. The table below sums up this information.

Table 19: Descriptive Statistics, Caregiving Practices

Statistic	All Cities	City		P
		Lusaka	Chongwe	
Mean	20.09	20.14	20.01	> 0.05
Sd	5.88	5.92	5.80	

Further analysis was conducted to examine caregiver knowledge of child development by settlement type. The findings indicate no significant difference in knowledge levels between caregivers living in planned and unplanned settlements across both cities. This suggests a critical need for parenting coaching programs that target all caregivers, regardless of settlement type or location. See the table below:

Table 20: Caregiving Practices by Settlement Type and Districts

	Lusaka City			Chongwe City					
	Planned	Unplanned	P	Planned	Unplanned	P	Rural	Urban	P
Mn	19.99	20.35	>0.05	20.22	19.76	>0.05	19.71	20.25	>0.05
SD	5.95	5.89		5.41	6.24		6.28	5.38	

This is higher compared to other studies that have used this tool in Zambia (Choma and Pemba) - M=16.71, SD=5.51).²⁵

The specific areas of child development knowledge assessed are presented in the table below. Overall, there were no major differences between the two cities, except in six areas. These include knowledge about when a child's brain begins to develop and learn, when children begin to make sounds in response to someone talking to them, when they start reaching for toys in front of them, when mothers should begin talking to their children, when they should start giving children clean and safe objects or toys to mouth, and when they should begin looking at pictures or picture books with their children. These findings have important implications for caregiver practices related to early stimulation and brain development.

Table 21: Knowledge of Child Development at Item Level by Districts

Knowledge domain	No.	%	Chongwe	Lusaka	Sig
When child's brain begins to develop and learn	752	49	45.3	51.2	.001
When children begin to see	225	14.7	14.7	14.7	

²⁵ Chipili, M. (2020). The use of participatory theatre in early childhood development interventions: a focus on mother groups in Choma and Pemba, Zambia (Doctoral dissertation, The University of Zambia).

Knowledge domain	No.	%	Chongwe	Lusaka	Sig
When children begin to follow a moving person or toy, with their eyes	564	30.3	31.7	29.5	
When children begin to make sounds in response to someone talking to them	183	12	13.2	11.3	
When children begin to smile socially, that is smile into the face of another person	359	23.4	26.7	21.7	.025
When children begin to say single meaningful words	796	52	52.7	51.6	
When children begin to play imaginary play	483	31.5	31.8	31.2	
When children begin to reach for a toy in front of them	792	51.7	57.3	48.7	.003
When children begin to pick up tiny things like ground nuts, with their fingertip	620	40.5	40.6	40.4	
When children begin to walk alone with good co-ordination	875	57.2	55.8	57.9	
When mothers should begin to talk to children	714	46.6	42.3	49	.034
When mothers should begin to show colourful objects to children to help them practise reaching	250	16.3	14.3	17.4	
When mothers should begin to teach children to count?	835	54.5	53.4	55.1	
When mothers should begin to teach children colours?	805	52.6	52.9	52.4	
When mothers should start to give children a spoon to eat or to let them eat by themselves	701`	45.8	45.5	46	
When mothers should begin to let children draw on the ground with a stick or draw on paper with a pencil?	845	55	54.7	55.4	
When mothers should begin to let children sit with support?	574	37.5	34.7	39	
When mothers should begin to give children clean and safe objects or toys which they can mouth?	539	35.2	32.3	36.8	.009
When mothers should begin to look at pictures or picture books with their children?	252	16.5	15.2	17.1	.037
When mothers should begin to give children clean and safe household items like spoons or cups to play with?	347	22.7	21.5	23.3	

The baseline study also examined the predictors of caregiver knowledge of child development in the two cities. In Lusaka, the primary predictors—settlement type and household vulnerability—were not initially significant. However, after introducing covariates into the model, living in an unplanned settlement (compared to a planned one) was associated with lower levels of caregiver knowledge. Among the four covariates included, only caregiver education emerged as a significant predictor, with

higher education linked to greater knowledge of child development. In contrast, in Chongwe, high household vulnerability remained a significant predictor of lower caregiver knowledge, even after adjusting for covariates. Additionally, both older caregiver age and higher education were significantly associated with better knowledge of child development. For more details, see the table below.

Table 22: Predictors of Caregiver Knowledge of Child Development by District

Predictors	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
Planned vs unplanned	.39	-.04	-1.28	.24	.53	.01	-1.02	1.05
HH vulnerability	.04	-.06 [†]	-.15	.01	.05	-.12**	-.25	-.04
Model 2								
Planned vs unplanned	.40	-.08*	-1.73	-1.16	2.58	-.25	-7.95	2.18
HH vulnerability	.04	-.03	-.12	.04	.05	-.10*	-.23	-.02
Age of caregiver	.19	.03	-.02	.06	.03	.09*	.00	.11
Caregiver education	.05	.14***	.11	.32	.07	.16**	.11	.38
Being female	.92	.03	-.91	2.69	.98	.03	-1.26	2.59
Urban vs rural living					2.57	.21	-2.63	7.49

Note: *<0.05; **<0.01; ***<0.001; [†] borderline

Early Parenting Attitudes and Beliefs

Parenting attitudes and beliefs were measured by the Early Parenting Attitudes and Beliefs Questionnaire by Hembacher & Frank (2020)²⁶. The EPAB questionnaire has a maximum possible score of 96, with higher scores indicating more positive parenting attitudes and beliefs. It comprises three subscales reflecting different dimensions of parenting. The overall average EPAB score was 72.16, suggesting that approximately one-third of parents demonstrated strong early parenting attitudes and beliefs. In Lusaka, caregivers living in planned settlements had more positive EPAB scores compared to those in unplanned settlements. In contrast, in Chongwe, caregivers from unplanned settlements had better EPAB scores than those from planned areas. See details below:

²⁶ The Early Parenting Attitudes Questionnaire: Measuring Intuitive Theories of Parenting and Child Development. Collabra: Psychology, 6(1): 16. DOI: <https://doi.org/10.1525/collabra.190>

Table 23: Early Parenting Attitudes and Beliefs by Settlement Type

City	Settlement	Affection & attachment	Early learning	Rules respect	and EPAB
All	All	23.14±4.31	23.47±4.33	25.56±3.96	72.16±10.24
Lusaka	Unplanned	22.92±3.63	23.05±4.02	25.79±3.82	71.76±9.22
	Planned	23.51±3.61	24.41±4.44	26.07±3.68	73.99±9.54
	<i>P</i>	<0.01	<0.001	>0.05	<0.001
Chongwe	Unplanned	23.15±3.02	22.98±4.17	25.28±3.64	71.41±8.70
	Planned	22.54±3.94	22.61±4.33	24.41±4.66	69.57±10.78
	<i>P</i>	<0.05	>0.05	<0.01	<0.01

An interesting observation was made in Chongwe, where caregivers in rural areas demonstrated more positive EPAB scores compared to those in urban areas. The details are summarized in the table below.

Table 24: Early Parenting Attitudes and Beliefs by Rural/Urban, Chongwe

Location Chongwe	Affection & attachment	Early learning	Rules respect	and EPAB
Rural	23.20±3.01	23.0±4.20	25.26±4.64	71.46±8.75
Urban	22.52±3.93	22.59±4.30	24.45±4.64	69.56±10.71
	<0.05	>0.05	<0.05	<0.05

In both cities, settlement type and household vulnerability were significant predictors of EPAB scores. In Lusaka, living in a planned settlement was associated with more positive EPAB scores compared to living in an unplanned settlement. Conversely, in Chongwe, caregivers in unplanned settlements had better EPAB scores than those in planned areas. Higher household vulnerability in both cities was linked to poorer EPAB scores. Among the five covariates included in the model, only older caregiver age was significantly associated with better EPAB in Lusaka, but not in Chongwe. The details are presented in the table below.

Table 25: Predictors Early Parenting Attitudes and Beliefs, Lusaka and Chongwe

Predictors	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
Planned vs unplanned	.60	.07*	.06	2.41	.84	-.22***	-5.95	-2.64

HH vulnerability	.06	-.25***	-.96	-.36	.09	-.38***	-.94	-.60
Model 2								
Planned vs unplanned	.62	.04	-.38	2.05	4.18	-.05	-9.24	7.19
HH vulnerability	.06	-.24***	-.59	-.35	.09	-.37***	-.92	-.58
CKCD	.05	.09**	-.24	-.04	.07	.06	-.04	.23
Age of caregiver	.03	.06*	.00	.12	.05	.01	-.08	.10
Caregiver education	.08	.06†	-.01	.32	.11	.04	-.13	.31
Being female	1.42	.11***	2.20	7.76	1.59	.05	-1.35	4.88
Urban vs rural living					4.17	-.18	-11.82	4.58

Note: *<0.05; **<0.01; ***<0.001; † borderline

Caregiving Practices

Caregiving practices were assessed using the HOME Inventory, which measures the quality and quantity of support, structure, and stimulation provided in the home environment. Higher scores (58) reflect better caregiving practices and more positive caregiver-child interactions. The HOME has six dimension that were used in this study.²⁷ The Acceptance dimension of the HOME Inventory has a maximum possible score of 6. However, in this baseline study, it recorded one of the lowest scores, with a mean of 1.69, indicating very poor caregiving practices in this area. This pattern was consistent across both cities and across settlement types within each city. The Organization dimension, which has a maximum score of 11, showed slightly above-average scores and did not vary significantly by city or settlement type.

For the Learning Materials dimension (maximum score of 9), households scored very low overall, and significant differences were observed between settlement types in both Lusaka and Chongwe. The Involvement dimension (maximum score of 5) had an average score of 2.32 across both cities. While no significant difference was found between planned and unplanned settlements in Lusaka, caregivers in planned settlements in Chongwe scored significantly higher. The Variety dimension, with a maximum score of 15, was one of the highest scoring areas, with an average of 10.81. In Lusaka, households in planned settlements scored significantly higher than those in unplanned areas, whereas no differences were found between settlement types in Chongwe. For the Responsiveness dimension (maximum score of 12), the average score was 8.43. There were no significant differences between settlement types in Lusaka, but in Chongwe, households in planned settlements scored significantly higher than those in unplanned settlements.

Overall, the average total HOME environment score was 32.59 out of 58. Differences emerged between settlement types in the two cities: in Lusaka, unplanned settlements scored higher, whereas in Chongwe, planned settlements had higher scores. See the table below for details.

²⁷ **Responsiveness** – degree to which a caregiver is attuned and responds to the physical and emotional needs of a child; **Acceptance** – covers discipline and rules practiced by a caregiver ; **organization** – degrees of organization in relation to child's routine, safety and caregiver structure; **involvement** – measure quality of caregiver-child interaction and stimulation; **variety** – captures diverse, frequency and enrichment of child's daily experiences; **learning** – degree to which the environment stimulates and promotes learning.

Table 26: HOME Environment Dimensions by Settlement Type, Lusaka and Chongwe

Dimension	All	Lusaka			Chongwe		
		Unplanned	Planned	P	Unplanned	Planned	P
Acceptance	1.69±.99	1.67±.99	1.67±.95	ns	1.66±1.03	1.75±1.75	ns
Organisation	6.78±1.65	6.77±1.67	6.65±1.67	ns	6.99±1.69	6.86±1.68	ns
Learning materials	2.57±2.63	2.08±2.27	2.77±2.65	***	1.84±2.26	3.51±3.00	***
Involvement	2.32±1.51	2.37±1.44	2.21±1.60	ns	2.16±1.36	2.59±1.51	***
Variety	10.81±1.60	10.55±1.46	10.92±1.66	***	10.91±1.62	10.87±1.58	ns
Responsiveness	8.43±2.73	8.41±2.67	8.45±2.70	ns	8.17±2.87	8.61±2.76	*
HOME score	32.59±6.36	32.59±6.36	31.85±5.69	*	31.74±6.15	34.18±6.52	***

Note: *<0.05; **<0.01; ***<0.001; † borderline; ns – non significant

In Chongwe, no significant differences were observed between rural and urban households in the Acceptance, Organization, and Responsiveness dimensions. However, significant differences emerged in the overall HOME score, as well as in the Involvement and Learning Materials dimensions, where urban households scored significantly higher than rural ones. Conversely, rural households scored higher than their urban counterparts in the Variety dimension. See details below.

Table 27: HOME Environment Dimensions by Rural/Urban, Chongwe

Dimension	All	Chongwe		
		Rural	Urban	P
Acceptance	1.69±.99	1.66±1.03	1.75±1.04	ns
Organisation	6.78±1.65	6.97±1.70	6.88±1.67	ns
Learning materials	2.57±2.63	1.97±2.45	3.53±2.99	***
Involvement	2.32±1.51	2.16±1.37	2.58±1.51	***
Variety	10.81±1.60	10.89±1.63	10.78±1.58	***
Responsiveness	8.43±2.73	8.20±2.88	8.58±2.75	ns
HOME score	32.59±6.36	31.67±6.17	34.20±6.48	***

Note: *<0.05; **<0.01; ***<0.001; † borderline; ns – non significant

Predictors of the HOME Environment

In Lusaka, lower household vulnerability was associated with a better HOME environment, while in Chongwe, living in a planned settlement predicted a better HOME environment. However, the predictive strength of these factors weakened after covariates were introduced into the models. In both cities, caregiver knowledge of child development was positively associated with the quality of the home environment. Additionally, higher caregiver education levels consistently predicted a better HOME environment. See details below:

Table 28: Regression on the Predictors of a Good Home Environment

Predictors	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
Planned vs unplanned	.41	.04	-.33	1.27	.58	.19***	1.25	3.53
HH vulnerability	.04	-.14***	-.26	-.10	.06	-.01	-.13	.10
Model 2								
Planned vs unplanned	.41	-.02	-.99	.62	2.81	-.13	-7.19	3.83
HH vulnerability	.04	-.08	-.19	-.02	.06	.04	-.07	.18
EPAB	.02	-.01	-.04	.04	.03	.06	-.02	.10
CKCD	.03	.15***	.10	.23	.05	.10*	.02	.21
Age of caregiver	.02	.03	-.02	.06	.03	.08	-.01	.11
Caregiver education	.05	.22***	.26	.50	.08	.17***	.14	.44
Urban vs rural living					2.80	.29	-1.78	9.23

Note: * <0.05 ; ** <0.01 ; *** <0.001 ; † borderline; EPAB -Early Parenting Attitudes and Beliefs; CKCD – Caregiver Knowledge on Child Development

Responsive Parenting

The baseline study examined the extent to which responsive parenting was practiced in the two cities. Overall, there were no significant differences between Lusaka and Chongwe in most indicators of responsive parenting. However, one notable exception was the indicator of talking to a child, where caregivers in Lusaka appeared to engage more frequently in verbal interaction. Despite this difference, the overall proportion of caregivers practicing this behavior remained well below desirable levels as is the case with most of the indicators in the table below. Only two indicators of responsive parentings showed better results – “Mother’s speech distinct, clear and audible to child” and “Mother’s voice conveys positive feeling.” For specific details see table below:

Table 29: Responsive Parenting Indicators by District, Lusaka and Chongwe

Responsiveness indicator	All	District		P
		Lusaka	Chongwe	
Mother spontaneously talks to child	1183(77)	761(76.6)	422(78.3)	>0.05
Mother verbally responds to child	1133(73.9)	739(74.3)	394(73.1)	>0.05
Mother tells child names of objects or visiting person	676(44.1)	421(42.4)	255(47.3)	>0.05
Mother's speech distinct, clear and audible to child	1432(93.4)	930(93.6)	502(93.6)	>0.05
Mother talks more about child	918(50)	613(61.7)	305(56.6)	<0.05
Mother permits child to play freely	1179(76.9)	765(77)	414(76.8)	>0.05
Mother spontaneously praises the child	972(63.3)	626(63)	346(64.2)	>0.05
Mother's voice conveys positive feeling	1388(90.5)	903(90.8)	485(90)	>0.05
Mother cares, strokes head or kisses	776(50.6)	512(51.5)	264(49)	>0.05
Mother positively responds to child praises by others	1090(71.1)	724(72.8)	366(67.8)	<0.05)
Mother makes effort to provide play objects	713(46.5)	445(44.8)	268(49.7)	>0.05

With regards to responsive practices in planned and unplanned settlement, the overall picture from these results suggests that caregivers across both Lusaka and Chongwe demonstrate generally positive levels of responsiveness, especially in basic emotional and verbal engagement with children. However, there are some variations by city and settlement type that reveal areas for improvement and targeted intervention. For details show the table below:

Table 30: Responsive Parenting Indicators by Settlement Type, Lusaka and Chongwe

Responsiveness indicator	Lusaka		Chongwe	
	Unplanned	Planned	Unplanned	Planned
Mother spontaneously talks to child	317(75.5)	447(77.3)	189(76.3)	233(79.8)
Mother verbally responds to child	307(73.1)	435(75.3)	179(72.5)	215(73.6)
Mother tells child names of objects or visiting person	176(41.9)	246(42.6)	104(42.1)	151(55.7)

Responsiveness indicator	Lusaka		Chongwe	
	Unplanned	Planned	Unplanned	Planned
Mother's speech distinct, clear and audible to child	398(94.8)	537(92.9)	322(93.9)	270(92.5)
Mother talks more about child	247(58.8)	371(64.2)	122(49.4)	180(61.6)
Mother permits child to play freely	331(78.8)	437(75.6)	195(78.9)	219(75)
Mother spontaneously praises the child	268(63.8)	360(62.3)	151(61.1)	195(66.6)
Mother's voice conveys positive feeling	383(91.2)	526(91)	223(90.3)	262(89.7)
Mother cares, strokes head or kisses	197(46.9)	317(54.8)	112(45.3)	152(52)
Mother positively responds to child praises by others	315(75)	411(71.1)	168(68)	198(67.8)
Mother makes effort to provide play objects	194(46.2)	252(43.6)	107(43.30)	161(55.1)

Early Stimulation

The study also explored perceptions of caregivers on early stimulation. Three themes emerged on early stimulation. These were limited access to structured early stimulation opportunities, reliance on informal and home-based learning, Cultural beliefs and low awareness limit stimulation, economic constraints affect stimulation activities and stakeholder response to improve stimulation.

Early stimulation for children under five remains inadequately addressed across Lusaka and Chongwe. Key informants highlighted a lack of structured play environments and insufficient integration of early stimulation in community and health settings. ECD spaces that promote early stimulation—such as playparks, community halls, or ECE classrooms—were either inadequate, underutilized, or poorly maintained.

“There are no proper play areas or learning centres for children under five.” – FGD, Lusaka Male

“We don't have play parks for children; the only one we had is no longer operational.” – In-depth interview, Chongwe WDC.

The key informants emphasized the need to repurpose existing infrastructure into child-friendly spaces and advocated for low-cost, community-driven play solutions using local materials.

Parents in focus group discussions expressed that children predominantly engage in informal play and media consumption due to unaffordable ECD centres and a lack of public play areas. For instance, one parent remarked, “There are no proper play areas or learning centres for children under five,” reflecting widespread gaps in early learning access. However, there is minimal intentionality in stimulating cognitive development, and these methods often provide structure and variety:

“Children play with balls, stones, and handmade toys.” – FGD, Lusaka High-Cost Female

In-depth interviews with community leaders further revealed that cultural beliefs and economic constraints reduce parental investment in early learning. Many caregivers do not perceive structured

early learning as essential, with one leader stating, “Some parents believe their children are still young and do not need to go to school.”

Despite these challenges, stakeholders converged on opportunities for action: embedding play spaces in under-five clinics, expanding parental awareness campaigns, and implementing community-based early learning initiatives. These approaches, if effectively coordinated and scaled, offer a pathway to strengthening early stimulation in the early years. Innovative low-cost models and multi-sectoral collaboration are seen as key opportunities to improve early stimulation. To this end, suggestions included integrating child-friendly spaces in health centres, repurposing community infrastructure, and involving traditional leaders and fathers in caregiving and stimulation activities.

“Using local materials like tree trunks, sand, and tires can create affordable and sustainable play areas.”
– Key informant

“Integrating child-friendly spaces in health centres could improve early stimulation.” – Key informant

“As a network, we support integrated ECD platforms that provide early stimulation, nutrition support, and caregiver education.” – Key informant

Response Social Protection and Safeguarding

Overall access is low, with only 13.3% of households receiving any social protection services. Chongwe shows stark contrasts. In rural Chongwe, 36% of households receive social protection, compared to just 7.2% in planned (urban) Chongwe ($p < 0.01$). Similarly, rural areas in Chongwe report significantly higher access (36%) than urban (7.7%) ($p < 0.001$).

Lusaka, by contrast, shows no significant differences between planned and unplanned settlements, both with low access (around 9–10%). Social protection services appear more accessible in rural Chongwe than in urban settings, possibly due to targeted rural safety net programs. Lusaka shows uniformly low coverage, suggesting urban gaps in social welfare outreach.

Across all settings, government is the primary support provider (over 90%), with minimal support from NGOs and community groups. No statistically significant differences were observed between city or settlement types. There is heavy reliance on government for social protection, with little contribution from community-based or civil society actors, suggesting a need to diversify service providers and strengthen grassroots support networks.

Chongwe planned settlements report significantly better availability of services (29.5%) than unplanned areas (16.6%) ($p < 0.001$). In the rural–urban breakdown, urban Chongwe again shows higher availability than rural ($p < 0.01$). While service provision seems more concentrated in urban Chongwe, actual access favors rural areas, pointing to possible awareness gaps or differences in eligibility/enrollment in urban settings.

Safety concerns are widespread, especially in unplanned and rural areas, highlighting the need for improved child protection and neighborhood safety infrastructure. In Lusaka, both planned and unplanned settlements show high concern (57–63%) ($p < 0.05$). In Chongwe, concern is even higher: 86.2% in unplanned and 76.4% in planned areas ($p < 0.01$).

Rural Chongwe households also report higher perceived risk ($p < 0.01$). Perceptions of environmental danger—particularly in unplanned settlements and urban areas—suggest that urban infrastructure and planning deficits contribute to a sense of insecurity. Physical environment (e.g., hazards) is more often perceived as unsafe in. Unplanned Lusaka (79.7%) vs. planned (62.6%) ($p < 0.001$). Urban Chongwe (71.4%) vs. rural (51.5%) ($p < 0.01$). Road safety concerns are significantly higher in Lusaka ($p < 0.01$) but not significantly different in Chongwe. Social environment risks (e.g., crime, violence) show no significant differences in any comparison. Urban populations—especially in Lusaka—appear more receptive to structured safety interventions, possibly due to higher exposure to road-related dangers.

There is broad consensus on the need for enhanced safety measures, especially in areas with higher perceived risks, reinforcing the urgency for policy and community-level action. These findings call for expanded outreach, better service delivery models tailored to urban and rural needs, and greater involvement of non-government actors in child protection. For details see the tables below

Table 31: Child Protection Services by Settlement Type, Lusaka and Chongwe

variables	All n(%)	Lusaka		P	Chongwe		P
		Unplanned	Planned		Unplanned	Planned	
HH receives social protection services	204(13.3)	38(9%)	58(10%)	>0.05	89(36%)	7(7.2%)	<0.01
Who supports you:							
Government	189(12.3)	34(90%)	53(91.4%)	>0.05	84(94.4%)	20(95.2%)	>0.05
Community groups	10(0.7)	2(5%)	5(8.6%)	>0.05	2(2.2%)	1(4.8%)	>0.05
NGO	5(0.3)	2(5%)			3(3.5%)		
Availability of social protection services (legal aid, SCT, helpline)	305(19.8)	73(17.4%)	108(18.7%)	>0.05	41(16.6%)	86(29.5%)	<0.001
HHs covered under insurance	523 (34.1)	261 (45.6)	87 (20.8)	<0.01	138 (47.3)	36 (14.6)	<0.01
Consider neighbourhood unsafe for children	493(32.2)	238(56.7%)	367(63.5%)	<0.05	213(86.2%)	223(76.4%)	<0.01
Finds neighbourhood roads unsafe	262(17.1)	86(47.3%)	139(65.9%)	<0.01	11(32.4%)	28(40.6%)	>0.05
Finds neighbourhood physical environment unsafe	342(22.3)	145(79.7%)	132(62.6)	<0.001	17(50%)	50(72.5%)	<0.05
Find neighbourhood social environment unsafe	340(22.2)	135(74.2%)	148(70.1%)	>0.05	17(50%)	42(60.9)	>0.05
Support road closure (certain roads and times)	815(53.2)	125(61.2%)	354(61.2)	>0.05	74(30%)	133(45.5%)	<0.001
Want to see more effort in create safe environment	1472(96)	403(96%)	566(97.8%)	>0.05)	241(97.6%)	268(91.8%)	<0.01

Table 32: Child Protection Services by Rural/Urban, Chongwe

Variables	All	Chongwe		P
		Rural	Urban	
HH receives social protection services	204(13.3%)	87(36%)	23(7.7%)	<0.001
Who supports you:				
Government	189(12.3%)	83(94.3%)	22(95.7%)	>0.05
Community groups	10(0.7%)	2(2.3%)	1(4.3%)	>0.05
NGO	5(0.3%)	3(3.4%)		
Availability of social protection services (legal aid, SCT, helpline)	305(19.8%)	41(16.9%)	86(29%)	<0.01
Consider neighbourhood unsafe for children	493(32.2%)	209(86.4%)	227(76.4%)	<0.01
Finds neighbourhood roads unsafe	262(17.1%)	10(30.4%)	29(41.4%)	>0.05
Finds neighbourhood physical environment unsafe	342(22.3%)	17(51.5%)	50(71.4%)	<0.01
Find neighbourhood social environment unsafe	340(22.2%)	17(51.5%)	42(60%)	>0.05
Support road closure (certain roads and times)	815(53.2%)	73(30%)	134(45%)	<0.001
Want to see more effort in create safe environment	1472(96%)	236(97.5%)	272(91.9%)	<0.01

Psychological Aggression or Physical Punishment

Table 33 shows that in Lusaka, more than half of the children (56.1%) did not have a birth certificate, 14% of the children were spoken to by their caregivers using abusive language (18% in unplanned settlement compared to 11% in planned settlement), about 8% of children in the unplanned settlement had experienced repeated physical abuse and had a meal withheld from them as a form of punishment compared to about 6% in a planned settlement.

In unplanned settlements, one in ten children were withdrawn, consistently sad, unhappy or depressed, and not able to participate in daily activities including playing with friends and family compared to 5% in planned settlements.

Table 33: Psychological and Physical Aggression Among Children, Lusaka

Type of Psychological Aggression	Type of Settlement		Overall
	Unplanned	Planned	
Repeated physical abuse	8.4%	5.4%	7.0%
Withheld a meal to punish	7.8%	5.8%	6.9%
Involved in Child Labor	3.1%	2.2%	2.9%
Family separation (ran away, chased)/Neglected	5.0%	3.8%	4.5%
Sexually abused, defiled, raped, forced	2.5%	2.2%	2.4%
Stigmatized/ discriminated due to illness, disability or otherwise	1.7%	4.5%	3.0%
Using abusive words	17.6%	10.9%	14.4%
Has no birth certificate	50.7%	62.3%	56.1%
In contact/conflict with the law	3.3%	2.9%	3.1%
Are there any children in this household who are withdrawn or consistently sad, unhappy or depressed, not able to participate in daily activities including playing with friends and family	11.4%	5.4%	7.9%

Table 34 presents the results of psychological and physical aggression among children in Chongwe. More than half of the children in Chongwe have no birth certificate and more than two-thirds of children in unplanned settlements and rural Chongwe have no birth certificates. More than one in ten children was spoken to in an abusive language, the proportions are higher among children in planned and urban Chongwe. Similarly, relatively more children in planned settlements and urban Chongwe experienced

repeated physical abuse or had a meal withheld as a form of punishment than in rural or unplanned settlements of Chongwe.

Both Lusaka and Chongwe reported that some children had experienced sexual abuse, defilement or rape, in Chongwe the proportions were slightly higher in planned settlements and urban areas compared with rural or unplanned settlements (Table 34).

Table 34 : Psychological and Physical Aggression Among Children, Chongwe

Type of Psychological Aggression	Type of Settlement		Residence		Overall
	Unplanned	Planned	Rural	Urban	
Repeated physical abuse	2.6%	7.8%	2.7%	7.5%	5.5%
Withheld a meal to punish	5.8%	8.8%	5.4%	9.0%	7.5%
Involved in Child Labor	2.6%	3.1%	2.7%	3.0%	2.9%
Family separation (ran away, chased)/Neglected	3.9%	7.3%	3.4%	7.5%	5.8%
Sexually abused, defiled, raped, forced	1.3%	3.1%	1.4%	3.0%	2.3%
Stigmatized/ discriminated due to illness, disability or otherwise	4.6%	2.6%	4.8%	2.5%	3.5%
Using abusive words	10.4%	15.5%	10.9%	15.0%	13.3%
Has no birth certificate	66.9%	47.2%	66.7%	48.0%	55.9%
In contact/conflict with the law	2.0%	4.7%	2.0%	4.5%	3.5%
Are there any children in this household who are withdrawn or consistently sad, unhappy or depressed, not able to participate in daily activities including playing with friends and family	8.5%	8.9%	8.7%	8.8%	8.7%

Positive Parenting and Discipline Practiced at Household Level

The results suggest that harsh and punitive discipline practices are still relatively common, though used by a minority of caregivers overall. However, there are notable differences by location (Lusaka vs. Chongwe), settlement type (planned vs. unplanned), and urban vs. rural settings, particularly in Chongwe. About 1 in 4 caregivers (25.7%) reported using physical punishment or food denial, which is the most frequently reported harsh discipline method. Other punitive practices such as shouting (13.1%), restricting activities (18.1%), and threatening or criticizing (9.5%) are also present, though to a lesser extent. Physical aggression (hitting, pushing, shaking) was reported by 5.1% of respondents — still concerning due to its potential harm to children.

In Lusaka, there are no significant differences in harsh discipline practices between planned and unplanned settlements — suggesting a fairly uniform use of such practices. In Chongwe, planned

settlements report significantly higher use of several negative discipline practices: Shouting: 16.8% in planned vs. 8.9% in unplanned ($p < 0.01$). Complaining/speaking badly about the child: 11.3% in planned vs. 6.1% in unplanned ($p < 0.05$). Hitting/pushing/shaking: 7.9% in planned vs. 3.6% in unplanned ($p < 0.05$). Physical punishment/food denial: 19.2% in planned vs. 26.3% in unplanned (interestingly, lower in planned settlements). Findings also show that Urban caregivers in Chongwe are more likely to use verbal and physical aggression, while rural caregivers are more likely to use traditional physical punishment methods such as food denial or hitting. For details review the tables below:

Table 35: Predictors of Positive Parenting and Discipline by Settlement Type, Lusaka and Chongwe

Indicators	All	Lusaka		P	Chongwe		P
		Unplanned	Planned		Unplanned	Planned	
Mother shouts at child	201(13.1%)	48(11.4%)	82(14.2%)	>0.05	22(8.9%)	49(16.8%)	<0.01
Mother complains about child and says bad things	114(7.4%)	27(6.4%)	39(6.7%)	>0.05	15(6.1%)	33(11.3%)	<0.05
Mother hits, pushes or shakes child	78(5.1%)	17(4%)	29(5%)	>0.05	9(3.6%)	23(7.9%)	<0.05
Gives physical punishment or denies food	394(25.7%)	125(29.8%)	150(26%)	>0.05	65(26.3%)	56(19.2%)	<0.05
Mother threatens punishment or criticize child	145(9.5%)	38(9%)	46(8%)	>0.05	28(11.3%)	33(11.3%)	>0.05
Mother restricts or interferes with child activities	277(18.1%)	68(16.2%)	110(19%)	>0.05	49(19.8%)	50(17.1%)	>0.05

Table 36: Predictors of Positive Parenting and Discipline by Settlement Type, Chongwe

Indicators	All	Chongwe		P
		Rural	Urban	
Mother shouts at child	201(13.1%)	21(8.7%)	50(16.8%)	<0.01
Mother complains about child and says bad things	114(7.4%)	14(5.8%)	34(11.4%)	<.05
Mother hits, pushes or shakes child	78(5.1%)	9(3.7%)	23(7.7%)	<.05
Gives physical punishment or denies food	394(25.7%)	61(25.2%)	60(20.2%)	<.05
Mother threatens punishment or criticize child	145(9.5%)	27(11.2%)	34(11.4%)	>0.05
Mother restricts or interferes with child activities	277(18.1%)	49(20.2%)	50(16.2%)	>0.05

Predictors of positive parenting and discipline

For Lusaka city, in model 1, neither settlement type (planned vs. unplanned) nor household vulnerability is a significant predictor of positive parenting practices. In model 2, Caregiver age is a significant positive predictor ($\beta = .08$, $p < .01$): older caregivers are more likely to use positive parenting practices. Caregiver education shows a borderline negative relationship ($\beta = -.07^{\dagger}$), which is unexpected and weak — possibly reflecting confounding variables or measurement issues. EPAB, CKCD, and HH vulnerability do not significantly predict positive parenting in Lusaka. For Chongwe city, in model 1, Household vulnerability is a significant negative predictor of positive parenting ($\beta = .14$, $p < .01$): more vulnerable households are less likely to use positive parenting practices. In model 2, Household vulnerability remains a significant negative predictor ($\beta = .10$, $p < .05$). EPAB (parenting attitudes) has a strong negative association ($\beta = -.15$, $p < .01$), indicating that poorer attitudes are associated with harsher parenting. Caregiver education is a positive predictor ($\beta = .12$, $p < .01$): higher education is associated with more positive parenting. Other predictors like caregiver age, CKCD, and urban vs. rural residence are not significant. Details are in the table below:

Table 37: Regression on predictors of positive parenting and discipline, Lusaka and Chongwe

Predictors	Lusaka city				Chongwe city			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Model 1								
Planned vs unplanned	.06	.01	-.10	.15	.09	.09	-.01	.36
HH vulnerability	.01	.02	-.01	.02	.01	.14**	.01	.05
Model 2								
Planned vs unplanned	.07	-.03	-.16	.10	.46	-.14	-1.19	.62
HH vulnerability	.01	.03	-.01	.02	.01	.10*	.00	.04
EPAB	.01	.01	-.01	.01	.01	-.15**	-.03	-.01
CKCD	.01	-.02	-.01	.01	.01	-.01	-.02	.01
Age of caregiver	.00	.08**	.00	.02	.01	.01	-.01	.01
Caregiver education	.01	-.07†	.00	.03	.01	.12**	.01	.06
Urban vs rural living					.46	.16	-.56	1.25

Note: * <0.05 ; ** <0.01 ; *** <0.001 ; † borderline; EPAB -Early Parenting Attitudes and Beliefs; CKCD – Caregiver Knowledge on Child Development

Barriers to Positive Parenting

Five themes emerged on barriers to positive parenting. These include gender norms and cultural beliefs. Lack of parenting knowledge, economic hardships, social issues affecting parenting capacities and inadequate support systems.

The findings reveal multiple intersecting barriers to positive parenting, as expressed by community members, key informants, and local leaders. Focus group participants consistently pointed to entrenched gender norms as a key constraint. Fathers were seen primarily as financial providers, while caregiving was culturally coded as “women’s work.” A Lusaka father noted, “*Caregiving is mostly left to the mothers,*” while a mother added, “*Men are discouraged from caregiving due to cultural beliefs.*”

Economic hardship emerged as a major barrier across all data sources. Parents in FGDs lamented that their primary focus was on providing food and shelter, with little time or energy left for play or emotional bonding. As one community leader put it, “*Most caregivers focus on food and shelter, ensuring the children are safe at home.*” Another remarked that “*Some parents leave children unattended while they go to work.*” – In-depth interview, WDC Chongwe

Key informants and in-depth interviews highlighted the lack of structured parental support and the widespread absence of parenting education. A ward committee member remarked, “*We do not have structured parental coaching programs, but we need them.*” This aligns with views from the IMCI coordinator, who emphasized that parental education must be expanded and embedded into existing health and community platforms. There is a limited understanding of child development among parents, particularly regarding the importance of responsive caregiving.

“We need someone to teach us how to look after children.” – FGD, Lusaka Female

These findings align with the quantitative data. Less than 30% of household heads were aware of parenting programs. Among those who were aware, only 17.6% reported that such programs were available in their communities, and just 13.5% were actively participating in them.

Despite this low awareness and participation, over 80% of respondents agreed that parenting programs are important, and more than 85% expressed willingness to participate if such programs were offered. Notably, interest was significantly higher among parents living in unplanned settlements compared to those in planned areas. For further details, refer to the table below:

Table 38: Barriers to Positive Parenting by Settlement Type, Lusaka and Chongwe

Variables	All n (%)	Lusaka		P	Chongwe		P
		Planned	Unplanned		Planned	Unplanned	
Caregivers aware of parenting program/coaching	400(26.1)	135(23.4)	108(25.7)	>0.05	71 (24.3)	86(34.8)	<0.01
Availability of parenting programs/coaching in community	270(17.6)	61(16.3)	83 (19.8)	>0.05	207(16)	62(25.6)	<0.01
Currently in parenting/coaching program	207(13.5)	68 (11.8)	61 (14.5)	>0.05	31(10.6)	47(19.0)	<0.05
Importance of parenting/coaching programs	1265(82.5)	470(81.3)	364 (86.7)	<0.05	217(74.3)	217(87.9)	<0.05
Want to participate in parenting programs	1,111(87.6)	396(84.3)	325(89.3)	<0.05	188(86.6)	202(93.1)	<0.01

Finally, social issues such as alcoholism and emotional stress, especially among male caregivers, were seen to diminish parenting capacity. As one father admitted, *“Alcoholism prevents some fathers from supporting their children.”*

Stakeholders across levels pointed out a lack of structured support systems like parental training, peer groups, or integration of parenting programs into ECD platforms. The following quotes collaborates this findings:

“There are currently no parental coaching programs, but there is a recognized need for them.” – In-depth interview, WDC Lusaka

“Parental education should be expanded, engaging religious and traditional leaders.” – IMCI Coordinator

Despite these challenges, stakeholders also expressed optimism about integrating parenting education into community-based ECD services, health clinics, and traditional leadership structures—offering a potential pathway for improving positive parenting practices.

Perception of Nurturing care

Nurturing care in this context encompasses responsive caregiving, early learning, health, nutrition, and safety/security. These themes emerged across all the respondents and their views are discussed below:

Responsive caregiving covered parental roles and gender roles. It also included aspect of parenting skills and coaching. Fathers primarily see themselves as financial providers, while caregiving is left to mothers, with cultural stigma deterring male involvement.

“Caregiving is mostly left to the mothers.” – FGD Male, Lusaka.

Parents in focus groups acknowledged caregiving was culturally assigned to mothers. Male caregiving was received with stigma which affected male involvement in child care activities.

“Men rarely help with caregiving... it’s considered a woman’s duty.” – FGD Female, Lusaka.

Respondents acknowledged that parenting programs that enhance parenting skills or coach parents on how to care for children are ensure to promote nurturing care. However, most communities lack formal parenting programs. *“We do not have structured parental coaching programs, but we need them.”* – Ward Development Committee, Chongwe.

Early Learning and Stimulation is another theme that emerged from the data. This theme covered learning opportunities, access to structured play and informal learning. All stakeholders agreed that early learning opportunities were limited. Children across all socioeconomic groups have limited access to play spaces or learning centers. *“There are no proper play areas or learning centers for children under five.”* – FGD Male, Lusaka.

Even where ECD services exist, they are overcrowded or lack trained teachers. *“One teacher handles up to 70 children.”* – ECE Specialist.

However, informal learning was identified as prevalent among children under the age of 5. Home-based learning through media and chores was commonly mentioned at the informal learning. *“Children learn through songs, stories, and cartoons like Cocomelon.”* – FGD Female, Lusaka.

Play parks were acknowledge as critical for child development and early learning. However, community leaders expressed concern over lack of play facilities:

“We don’t have play parks for children; the only one we had is no longer operational.” – Ward Development Committee, Chongwe.

Health and nutrition were another theme that emerged from the data on nurturing care. This theme covered aspects such as breastfeeding challenges, poor nutrition and food security. Economic pressures interrupt exclusive breastfeeding even in situation where parents are aware of the importance of exclusive breastfeeding. The following quote collaborates this finding - *“Exclusive breastfeeding is difficult because mothers return to work early.”* – FGD Male, Lusaka.

Inadequate nutrition is widespread. This is due to lack of nutritional skills and how to prepare nutritious meals but more importantly due to economic challenges as validated by this quote from parents *“Children mostly eat nshima... we can’t always provide balanced meals because of financial challenges.”* – FGD Male, Chongwe.

However, there was acknowledgement of the importance of feeding programs at community level. Community leaders observed that feeding programs improved child attendance in ECD centres:

“Every child at Village of Hope receives food daily. And it makes them come to school” – Ward Development Committee, Chongwe.

Safety and security of the children was another theme that emerged under perception of nurturing care in the two cities. This theme includes aspects such as safety of play environments and sanitation issues. On unsafe play environments, respondents mentioned that children frequently play near roads or unsafe areas risking their safety. *“Children often play near dangerous areas like roads and drains.”* – FGD Female, Lusaka. This is consistent with quantitative findings which showed 32% of neighborhood were unsafe and 17% of roads to be unsafe for children. This figure increased when respondents were from planned settlements in Lusaka (63.5%) and 40% in Chongwe planned settlement.

In terms of poor sanitation, inadequate waste disposal and open defecation are common were mentioned particularly in Chongwe. *“Garbage disposal is poor, with rubbish dumped in open spaces.”* – FGD Female, Chongwe.

How can Nurturing Care be Promoted among Parents and Caregivers?

We asked caregivers and key informants how nurturing care can be promoted among parents and caregivers. Responses were analyzed using the Bronfenbrenner's ecological framework.

Microsystem (Direct Interactions)

Nurturing care at the microsystem level is most directly influenced by parental training, male involvement, and responsive parenting. Male involvement remains limited due to entrenched social norms.

"If a father takes a child to the clinic or makes porridge, people laugh at him." – Parent

Parents expressed a desire for practical tools to support play-based learning at home. - "Even at home, parents need simple learning tools, not just toys, to support brain development." – Stakeholder

Both stakeholders and parents emphasized a lack of structured support for parents to engage meaningfully with their children. The following quotes corroborates these findings:

"We don't have parental coaching programs, but I would really benefit from learning how to care better for my child." – Parent

"We leverage existing programs to educate parents, but more targeted training is needed, especially on nutrition and stimulation." – Stakeholder

Mesosystem (Connections Between Settings)

Linking ECD services across sectors was seen as a promising strategy to improve nurturing care. Integration with health services and community networks was repeatedly emphasized.

"At the clinic, they teach us how to cook for children using local food. It really helps." – Parent

"Integrating ECD training into health education sessions would expand reach and improve consistency." – Stakeholder

Community-based support was also cited as critical, especially for vulnerable groups like single mothers - "Single mothers struggle the most, and there are no support groups here." – Parent

Exosystem (Indirect Influences on Caregivers)

Broader institutional systems affect caregiving indirectly. Stakeholders highlighted the need to expand affordable ECD services and improve access to school feeding and childcare programs. Below are example of quotes from stakeholders:

"There are no daycares nearby, and I can't leave my child alone when I go to the market." – Parent

"We need workplace policies that support parents, including market-based childcare options." – Stakeholder

Macrosystem (Societal and Policy-Level Factors)

At the societal level, cultural norms and policy enforcement play a pivotal role in shaping nurturing care environments. Stakeholders stressed the importance of changing harmful beliefs and enforcing supportive ECD policies. The following quotes substantiates this finding:

"If community leaders talked more about fathers helping at home, it would change how people think." – Parent

"Changing cultural narratives requires engagement with traditional and religious leaders." – Stakeholder

Chronosystem (Changes Over Time)

Finally, caregivers acknowledged that societal change is slow but necessary. Adapting to new realities—such as changing family roles, urbanization, and new knowledge about child development—was viewed as essential. The quotes below validate this finding:

"These days, we are learning more about children. In the past, we just followed what our elders did." – Parent

"Adapting to change means continuous training, media campaigns, and long-term investment in parenting programs." – Stakeholder

Promoting nurturing care in ECD requires interventions across all ecological levels—strengthening individual parenting practices, connecting caregivers to services, reforming systems, and shifting societal norms. Grounded in lived experiences, these findings offer a roadmap for multisectoral collaboration that centers both caregiver realities and systemic enablers. The table below summarizes these findings.

Ecological System	Factors Promoting Nurturing Care in ECD
Microsystem (Direct Interactions)	• Parental Training & Coaching • Male Involvement in Childcare • Responsive Parenting & Play-Based Learning
Mesosystem (Connections Between Settings)	• Integrating Nurturing Care into Health Services • Strengthening Community Childcare Support Networks
Exosystem (Indirect Influences on Caregivers)	• Expanding Access to Affordable ECD Services Enhancing School Feeding & Nutrition Programs • Encouraging Workplace & Market-Based Childcare
Macrosystem (Societal & Policy-Level Factors)	• Challenging Harmful Cultural Norms • Developing & Enforcing ECD Policies • Strengthening Government & Donor Investment
Chronosystem (Changes Over Time)	• Adapting to Societal Changes

Thematic area 3: Early Childhood Development Services

This section will address seven research questions related ECD services and who the primary ECD services providers in are. The questions to answered are as follows:

1. What is the status of ECD service provision in Zambia?
2. Who are the primary ECD service providers in Zambia and who comprises providers' ECD workforce?

ECE Centre Quality

ECE centre quality was measured in 84 centres across the two cities using the MELE. Of the 84 centres, 25 were selected from Chongwe. In Chongwe, 4 were from unplanned settlements while 21 from planned settlements. In Lusaka, 20 ECE centres were from unplanned settlements and 39 from planned settlement

Quality was measured on different aspects including learning, classroom interactions and approaches and the physical environment that include facilities and safety. The score of 4 on each aspect reflect

excellent quality. A cutoff point of 2.5 is used to determine ECE centres meeting quality. As can be seen from the overall scores, majority of centre me quality standards across all the domains. On average ECE centres score 2.17 in unplanned settlements compared to 2.5 in planned settlements of Lusaka. Interestingly, in Chongwe ECE centres in unplanned settlements score higher 2.5 compared to 2.35 in planned settlements. However, this difference was not significant.

In terms of classroom interactions, on average all centres in both unplanned and planned settlements in the two were above the cutoff point of 2.5 indicating quality. This was also observed with regards to physical environment and the overall score. The timing of the observation could have influenced some of these results an issue that will be argued further in the discussion section of this report. In Lusaka ECE centres in planned settlement were significantly observed to have quality compared to those in unplanned settlement. No significant difference was observed in Chongwe. For details review the Table below:

Table 39: ECE Centre Quality by Settlement Type, Lusaka and Chongwe

Domain	All	Lusaka		P	Chongwe		P
		Unplanned	Planned		Unplanned	Planned	
Learning	2.93±0.58 (1.38-3.74)	2.17±.43	2.51±.65	<0.01	2.50±.22	2.35±.55	>0.05
Classroom interactions and approaches	3.11±0.61 (1.56-4.00)	2.89±.67	3.24±.55	<0.001	2.93±.41	3.10±.67	>0.05
Physical environment	3.35±0.63 (1.50-4.00)	3.02±.59	3.61±.42	<0.05	3.17±.76	3.23±.77	>0.05
Overall ECE quality	2.95±0.5 (1.59-3.29)	2.69±.46	3.14±.43	<0.01	2.78±.28	2.90±.59	>0.05

When compared between rural and urban based ECE centres in Chongwe, high quality was significantly observed in the centred based in urban areas across all domains than those in rural areas The details are show in the table below:

Table 40: ECE Centre Quality by Settlement Type, Chongwe

Domain	All	Chongwe		P
		Urban	Rural	
Learning	2.93±0.58(1.38-3.74)	2.88±.48	2.04±.55	<0.05
Classroom interactions and approaches	3.11±0.61(1.56-4.00)	3.18±.63	2.79±.59	>0.05
Physical environment	3.35±0.63(1.50-4.00)	3.32±.76	2.93±.71	>0.05
Overall ECD quality	2.95±0.51(1.59-3.29)	3.00±.54	2.98±.49	<0.05

Below the findings show the proportion of ECE centres that were found to be of good and poor quality across the two cities. Consistent with mean scores, the proportion of ECE centres that were classified good were significantly more than those classified poor quality. Review the table below:

Table 41: ECE Quality Domain by Settlement Type, Lusaka and Chongwe

Domain	All	Chongwe				Lusaka	
		Unplanned	Planned	Urban	Rural	Unplanned	Planned
Learning	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)
Good	33(39.8)	2(66.7)	10(47.6)	10(55.6)	2(33.3)	3(15)	18(46.2)
Poor	50(60.2)	1(33.3)	11(52.4)	8(44.4)	4(66.7)	17(85)	(53.8)
Physical environment							
Good	70(84.3)	3(75)	16(76.2)	15(83.3)	4(57.1)	14(70)	37(97.4)
Poor	13(15.7)	1(25)	5(23.8)	3(16.7)	3(42.9)	6(30)	1(2.6)
Classroom interactions							
Good	68(81)	3(75)	16(76.2)	15(83.3)	4(57.1)	15(75)	34(87.2)
Poor	16(19)	1(25)	5(23.8)	3(16.7)	3(42.9)	5(25)	5(12.8)
Overall ECE Quality							
Good	69(84.1)	3(100)	15(71.4)	15(83.3)	3(50)	15(75)	36(94.7)
Poor	13(15.9)		6(28.6)	3(16.7)	3(50)	5(25)	2(5.3)

Of the 84 ECE centres that were observed, 10 were community owned, 32 government, 1 NGO and 41 were private owned. For the purpose of analysis, NGO and private owned was combined. Based on this categorization, Kruskal Wallis test was conduct to determine the mean-rank differences across the three types of ownership. Across all the three domains, there was a significant difference in quality in the three types of ECE ownership with private owned ECE centres scoring high on quality followed by government and last community owned. For details review the table below:

Table 42: Quality by Ownership of ECE Center, Lusaka and Chongwe

Domain	All	Ownership			P
		Community	Government	Private	
Learning	2.93±0.58 (1.38-3.74)	1.88±0.37 (1.38-2.75)	2.43±0.46 (1.50-3.38)	2.49±0.64 (1.38-3.75)	<0.001

Domain	All	Ownership			P
		Community	Government	Private	
Classroom interactions and approaches	3.11±0.61 (1.56-4.00)	2.55±0.72 (1.56-3.44)	3.17±0.59(2.00-4.00)	3.21±0.53 (1.89-4.00)	<0.01
Physical environment	3.35±0.63 (1.50-4.00)	2.55±0.66 (1.50-3.67)	3.28±0.56 (1.83-4.00)	3.63±0.43 (2.33-4.00)	<0.001
Overall ECD quality	2.95±0.51 (1.59-3.29)	2.32±0.49 (1.59-2.94)	2.96±0.42 (1.93-3.55)	3.13±0.45 (1.98-3.92)	<0.001

Teacher Characteristics Predicting ECE Centre Quality

Having tertiary than secondary education was a better predictor of ECE quality and long duration working as an ECE teachers was also a predictor of better ECE centre quality. Other predictors were not significantly associated with ECE centre quality. For details see the table below:

Table 43: Predictors of Teacher Characteristics Predicting ECE Centre Quality

Predictors	SE	Standardised Coefficient Beta	95% CI		P
			LL	UL	
Trained as ECE teacher	4.62	-.04	-10.81	7.59	>0.05
Being remunerated	4.51	.15	-2.25	15.71	>0.05
Having tertiary education	5.89	.36	8.78	32.24	<0.001
Being a female teacher	6.76	.07	-17.95	8.97	>0.05
Duration of being ECE teacher	0.18	.37	.27	1.00	<0.001

Children Characteristics Associated with ECE Centre Quality

The baseline also explored characteristics related to children that were associated with ECE centre quality. Number of children enrolled in an ECE centre were not associated with ECE centre quality. But there was a positive association between number of children with special needs and ECE centre quality. Details are shown below:

Table 44: Predictors of Children Characteristics Associated with ECE Centre Quality

Predictors	SE	Standardised Coefficient Beta	95% CI		P
			LL	UL	

No. of enrolled children	.40	-.20	-.14	.05	>0.05
No. of children with special needs	2.14	.43	.15	9.06	<0.05

Play Parks and Recreation

The survey mapped play parks and recreational facilities open to the public on a fee or free in the two cities. In Lusaka, 31 private facilities were identified. Most of these were located in malls. In Lusaka, 33 public play parks were also identified. Of the 33 public, 2 were functional and 9 were under lease to private operator. In Chongwe, only three privates were identified. One was a theme park, the largest in Chongwe and Lusaka province while two others were located in malls. The Figure below shows the summary:

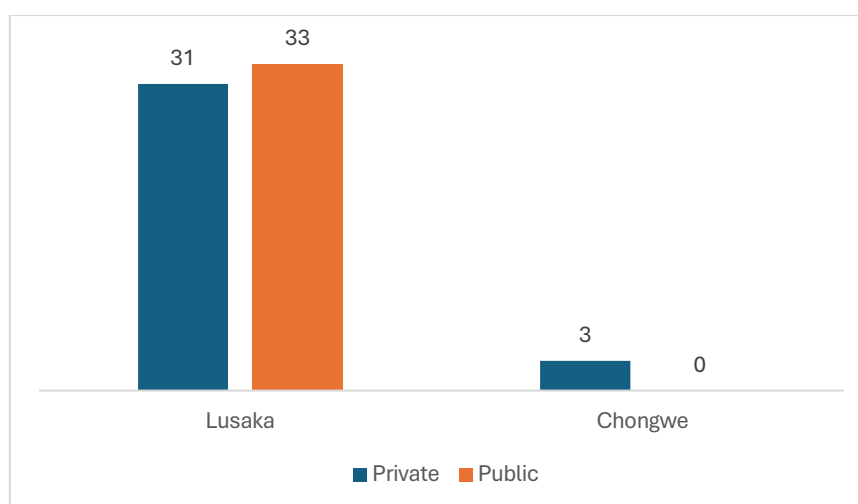


Figure 9: Ownership of Play Parks, Lusaka and Chongwe

The survey asked the household caregivers if safe age-appropriate play facilities were available in their communities. Only 33.6% of the sample mentioned having such facilities in their communities. In both Lusaka and Chongwe, those living in planned settlements significantly mentioned having such facilities. Findings also showed that some of these facilities were accessed at a fee. In Lusaka about 35% in unplanned and 36 % in planned settlement mentioned play facilities were accessed at a fee. In Chongwe significantly more HHs reported paying to access play facilities compared to unplanned settlement.

Over 80% of respondents in the two cities mentioned that they were aware of the importance of play for children. However, when asked why play was important, evidence suggest on average most of the caregivers are not aware of the importance of play. Regardless of their knowledge levels, over 90% of caregivers reported they needed more of play and recreational facilities for children in their communities. Of these 93% that mentioned they need more recreational facilities, about 66% were willing to pay to utilise these play or recreational facilities.

Availability of green space was reported to be significantly lower in both cities. However planned settlements were respondents who significantly reported availability of green spaces. Usage of the green space among those who reported it availability was also low. For details review the table below:

Table 45: Availability of Green Spaces for Play, Lusaka and Chongwe

Variables	All n(%)	Lusaka		P	Chongwe		P
		Unplanned	Planned		Unplanned	Planned	
Safe age-appropriate play facilities	515(33.6)	120(28.6)	213(36.9)	<0.01	54(21.9)	133(45.5)	<0.001
Pay play facilities	529(34.5)	147(35)	209(36.2)	>0.05	45(18.2)	132(45.2)	<0.001
Aware of importance of play	1333(86.7)	360(85.7)	512(88.6)	>0.05	221(89.5)	240(82.2)	<0.05
Knowledge of importance of play:							
i. Fosters play	589(44.2)	167(46.4)	227(44.3)	>0.05	95(43)	100(41.7)	>0.05
ii. Fosters collaboration	633(47.5)	159(44.2)	259(50.6)	>0.05	91(41.2)	124(51.7)	<0.05
iii. Learning & exploration							
iv. Exercising	960(72)	247(68.6)	365(71.3)	>0.05	159(71.9)	189(78.8)	>0.05
v. Build socioemot.skills	802(60.2)	210(58.3)	304(58.3)	>0.05	132(59.7)	157(65.4)	>0.05
vi. Builds motor skills							
	723(54.2)	172(47.8)	203(60.4)	<0.05	104(47.1)	122(50.8)	>0.05
	536(40.2)	136(37.8)	212(41.4)	>0.05	80(36.2)	108(45)	>0.05
Need for more play parks/recreational	1421(92.7)	389(92.6)	552(95.5)	>0.05	227(91.9)	257(88)	>0.05
Willing to pay recreation facilities	1011(65.9)	292(69.5)	368(63.7)	>0.05	133(53.8)	224(76.7)	<0.001
Available public green space	246(16)	42(10)	107(17.6)	<0.001	33(13.4)	70(24)	<0.01
Usage of play and green spaces:							
Regularly	96(6.4)	17(4)	46(8)	<0.01	5(2)	28(9.6)	<0.001
Occasionally	224(14.6)	52(12.4)	94(16.2)		17(6.9)	63(21.6)	
Rarely	273(17.8)	76(18)	115(19.9)		30(12.2)	54(18.5)	
Never	938(61.2)	275(65.6)	323(55.9)		195(78.9)	147(50.3)	

When compared between rural and urban areas in Chongwe city, significant differences were observed across most of the areas of investigation with more responses observed among urban respondents than rural respondents. For details see the table below:

Table 46: Availability of Green Spaces for Play, Chongwe

Variables	Chongwe		P
	Rural n(%)	Urban n(%)	
Safe age-appropriate play facilities	53(21.9)	134(45.1)	<0.001
Pay play facilities	45(18.6)	136(44.4)	<0.001
Aware of importance of play	217(89.7)	227(82.2)	<0.05
Knowledge of importance of play:			
vii. Fosters play	95(43.8)	100(41)	>0.05
viii. Fosters collaboration	89(41)	126(51.6)	<0.05
ix. Learning & exploration	155(71.4)	193(79.1)	>0.05
x. Exercising	131(60.4)	158(64.8)	>0.05
xi. Build socioemotional skills	102(47)	124(50.8)	>0.05
xii. Builds motor skills	78(35.9)	110(45.1)	<0.05
Need for more play parks/recreational	222(91.7)	262(76.4)	>0.05
Willing to pay recreation facilities	130(53.2)	227(76.4)	<0.001
Available public green space	32(13.2)	71(23.9)	<0.01
Usage of play and green spaces:			
Regularly	5(2.1)	28(9.4)	<0.001
Occasionally	17(7)	63(21.2)	
Rarely	30(12.4)	54(18.2)	
Never	190(78.5)	152(51.2)	

Geo-mapping of ECD-related Facilities

The survey conducted a mapping of ECD related facilities including ECE facilities such as daycare centre and preschools, health facilities, police stations, play and recreational facilities. The ECD related facilities that were mapped in the two cities are not in any way a comprehensive picture of the two cities. Here we only mapped facilities that were located in the settlements we had sampled for the baseline study. Click [map](#) to see the full map. Therefore, this mapping is just an indicative picture of the distribution of facilities. To access the maps, click on map. For details see the table below.

Table 47: Mapping of ECE Centers, Lusaka and Chongwe

Facility type	Lusaka		Chongwe	
	No.	Link to map	No.	Link to map
Daycare and preschool	69	Click Map	16	Click Map
Daycare only	2	Click Map	3	Click Map
Preschool only	80	Click Map	28	Click Map
Health facilities	31	Click Map	10	Click Map
Indoor recreational facilities	2	Click Map		
Outdoor	2	Click Map	1	Click Map
Police stations	21	Click Map	6	Click Map
Public Green space	4	Click Map	3	Click Map

Willingness to Pay for ECD Services

To be effective, ECD should take a holistic approach to child development. This means holistic ECD services must be accessed and utilised to support child development. Most of the ECD services come at a cost. This baseline examined caregivers' willingness to pay for ECD services and factors that are associated with caregiver willingness to pay for ECD services.

The table shows the percentage of households that are willing to pay for each of ECD services. The data were disaggregated for Lusaka and Chongwe and further desegregated into planned and unplanned settlements. In both Lusaka and Chongwe, pre-school for children aged 3 to 5 years was the ECD services for which respondents expressed the highest willingness to pay (63.3% of all household). This is consistent in both districts and also for the different settlement types (planned and unplanned settlements). Respondents indicated their least willingness to pay for coaching on responsive parenting (28.3% overall). Comparing between planned and unplanned settlements, we found statistical differences between the willingness to pay for specific ECD services. In Lusaka, respondents from unplanned settlements indicated higher willingness to pay for physical growth monitoring, hygiene practices and nutritious food. The differences were statistically significant. For planned settlements, willingness to pay was much higher for indoor recreation centre and outdoor recreation centres compared to unplanned settlements in Lusaka. In Chongwe, respondents from planned settlements expressed higher willingness to pay for physical growth monitoring, social, psychological and emotional development, diagnosis and treatment, day care services, children's toys, indoor and outdoor recreation, and these differences are statistically different.

Table 48: Willingness to Pay for ECE Services, Lusaka and Chongwe

Variables	All	Lusaka		p	Chongwe		p
		Planned	Unplanned		Planned	Unplanned	
ECD Services	n(%)	n(%)	n(%)		n(%)	n(%)	
Physical Growth monitoring	690(44.8)	241(41.7)	207 (49.3)	0.017**	144(49.3)	98 (39.6)	0.03**
Social, psychological and emotional develop	646(42.03)	244(42.2)	178 (42.4)	0.958	133(45.5)	91 (36.8)	0.04**
Diagnosis and treatment	799 (52.0)	295(51.0)	229 (54.5)	0.277	163 (55.8)	112 (45.3)	0.02**
Hygiene products	659 (42.9)	233(40.3)	200 (47.6)	0.022**	123(42.1)	103 (41.7)	0.921
Bottled milk (0-6 months)	406 (31.8)	134(28.9)	125 (34.9)	0.068*	87 (34.7)	60 (29.3)	0.221
Nutrition food for children (6-59 months)	703 (45.7)	369(42.4)	334 (50.1)	0.003***	140(48.0)	109 (44.1)	0.377
Day care for children (0-3 yrs)	692 (45.0)	389(44.7)	303 (45.4)	0.780	143(49.0)	99 (40.1)	0.039
Children's books	845 (55.0)	464(53.3)	381 (57.1)	0.139	170(58.2)	142 (57.5)	0.864
Children's toys	706 (45.9)	706(45.9)	414 (47.6)	0.138	155(53.1)	97 (39.2)	0.001***
Pre-school (3-5 yrs)	984 (64.0)	551(63.3)	433 (64.9)	0.521	192(65.8)	154 (62.3)	0.412
Indoor recreation centre	625 (40.7)	388(44.6)	237 (35.5)	0.000***	134(45.9)	73 (29.6)	0.000***
Outdoor recreation	835 (54.3)	501(57.6)	334 (50.1)	0.000***	170(58.2)	105 (42.5)	0.000***
Coaching on responsive parenting	435 (28.3)	247(28.4)	188 (28.2)	0.930	87 (29.8)	67 (27.1)	0.495

*** = p < 0.001, ** p < 0.05, * p < 0.1

Table below shows regression results separately for Lusaka and Chongwe City Councils. The dependent variable is the number of ECD services that a household is willing to pay for. The main predictors are planned settlement and overall vulnerability score. The covariates include overall developmental score from the CREDI analysis, age of the caregiver, the gender of the caregiver being female and the education of the caregiver. From both the Lusaka and Chongwe regressions, none of the predictors are statistically significant. However, from Lusaka only the education of the caregiver is statistically significant and indicates that education of caregivers is negatively correlated with their willingness to pay for ECD services. Thus, more experienced caregivers may attach more value to ECD services and thus more willing to pay for the services. Similar to Lusaka, we find a negative correlation between education of the caregiver and willingness to pay for ECD services in Chongwe.

Table 49: Predictors of Willingness to Pay for ECE Services, Lusaka and Chongwe

Predictors	Lusaka				Chongwe			
	SE	Std β	CI-L	CI-U	SE	Std β	CI-L	CI-U
Predictors								
Planned settlement	0.017	0.025	-0.002	0.005	0.208	-0.249	-0.658	0.160
Vulnerability score	0.002	0.001	-0.002	0.005	0.021	0.024	-0.016	0.065
Covariates								
Age of caregiver (yrs)´	0.001	-0.000	-0.002	0.001	0.011	0.021*	0.000	0.042
Female caregiver	0.04	0.005	-0.073	0.08	0.073	0.378	0.668	0.815
Education of caregiver	0.002	-0.004*	-0.008	0.000	0.026	-0.046*	-0.100	0.004

*** = $p < 0.001$, ** $p < 0.05$, * $p < 0.1$

Primary ECD Service Providers in Zambia

The study findings indicate that the implementation of ECD services follows both vertical and horizontal integration. Policy development, job aids, and guiding frameworks are largely driven by actors at the vertical level, involving policy-makers and decision-makers who shape the overall direction and support systems for ECD service delivery. In contrast, primary service providers are predominantly engaged at the horizontal level, delivering services within communities.

Vertical Service Providers

These are secondary and have direct bearing on the horizontal service providers who are the primary ECD providers in Zambia. In Zambia, the presidents is a champion for foundational learning at the African continental level and this appointment has renewed his support for ECE in Zambia. At national level, there are also steering committees that include ministers of respective ministries, permanent secretary, directors and members of the private sectors. Example of these include one on nutrition, enhancement of Early learning (ZEEL). Technical working groups that directly or indirectly affect ECD services in Zambia. Technical working groups such as knowledge translation on health, ECD projects under the ministry of health and education are example of such structures.

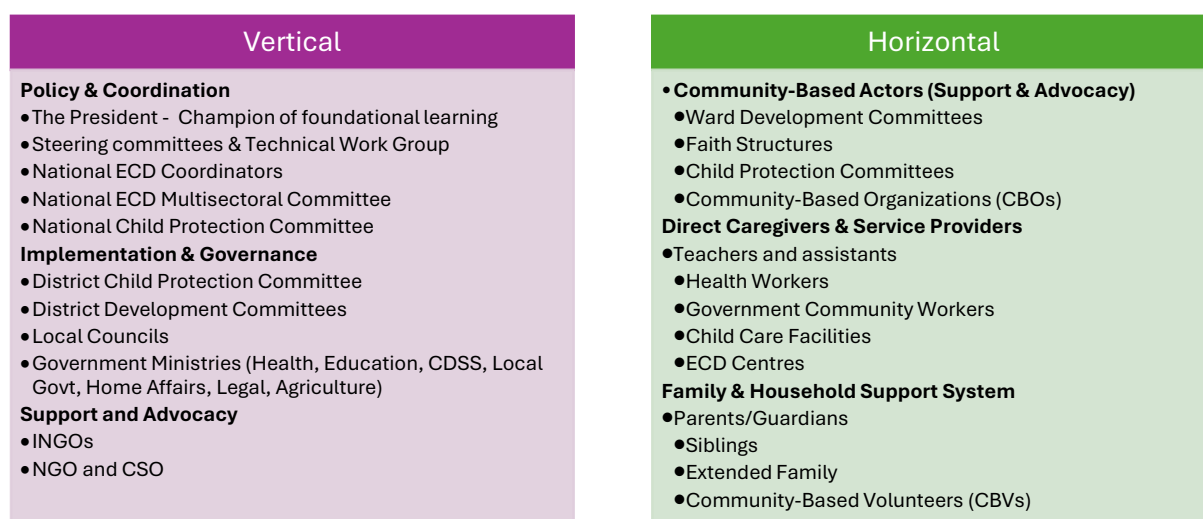
A national ECD coordination office at the ministry of health is also in place to coordinate and support national ECD multisectoral programs. A similar committee in child protection coordinated by the Ministry of Community Development and Social Services is also in place. To ensure coordinated implementation of ECD services, these committees are also at district level. However, in Chongwe, the study found that there was no district ECD multisectoral committee. District Development Committees (DDCs) are also instrumental in planning for developmental projects and services that affect ECD services. At the centre of these DDCs are the city councils who coordinate district development committees through their mayors and district commissioners.

ECD services are directly or indirectly implemented by the various government ministries. The principal ministries include Local Government and Rural Development, Health, Education, and Community Development and Social Services. Others are agriculture, home affairs and internal securities which is crucial for child protection enforcement and issuance of birth certificates.

Across all vertical levels of service provision, international nongovernmental organization, national NGOs and CSO are complementing government efforts through ECD support programs and advocating for quality service provision to the end users.

Horizontal Service Provision

Primary service providers are categorized at three levels. These are community-based actors, direct service providers and family and household support systems. Community based actors support or advocate for ECD services. These include ward development committees, faith structures, child protection committees and the role of community-based organisations who work with families to improve child outcomes. Direct caregivers or service providers include those mainly employed by the state or other actors such as NGO to provide direct services to families and children. These include teachers, health workers, community-based workers such as child protection officer and welfare officers, child care facilities and ECD centres such as day care service providers. Children are also affected by the support of their parents, siblings, extended families and community-based volunteers who provide different services from play and recreation, nutrition, health support, protection and safeguarding and providing a stimulating early learning environment. The figure below summarises the services providers and who they are at different levels.



Thematic area 4: Structural and Policy Landscape

This section will address seven research questions related to the policy and regulatory landscape of ECD, enablers and barriers of policy implementation and how ECD are coordinated in Zambia. The questions to answered are as follows:

1. What is the policy and regulatory landscape of ECD in Zambia? How does this interact with the wider country policy ecosystem (if applicable)?
2. How do these policies and regulations serve as either barriers or enablers of ECD programming?
3. How is coordination of ECD done at the multisectoral level?

Policy and Regulatory Landscape

An extended desk review showed that Zambia has various policies and frameworks that directly or indirectly affect child development. This ecosystem spans multiple sectors—child wellbeing, education, health, nutrition, play and recreation, and child protection—reflecting a multisectoral approach to ECD. Below the tables summarises these policies and frameworks.

Thematic area	Policy Name	Description
Overall child wellbeing	National Child Policy	This policy addresses a range of issues that affect children, including early childhood development, maternal and child health, and food and nutrition
Early Childhood Education	ECE Policy and Policy Implementation Plan (PIP)	This policy address early childhood stimulation and nutrition priorities.
	Curriculum and Implementation Guidelines	This document spells out the ECE curriculum and provides guidelines on how to implement it.
	ECE Standard Guidelines and Assessment	This is guideline of minimum acceptable ECE standards and how to conduct assessment
	Advocacy, Sensitization and Community Mobilization	This is strategy for advocating, sensitizing and mobilizing communities to participate in ECE activities.
	School Health and Nutrition policy, 2006	This policy address issues covering school health and hygiene and nutrition requirement
	Monitoring and Evaluation	This is a framework for monitoring and evaluating ECE and school activities
	The National WinS Strategy Zambia 2019–2030	Guides schools on how to provide health, well-being, education and dignity to all schoolchildren by providing safe water, sanitation and hygiene in schools paying special attention to the needs of girls and those in vulnerable situations such as ECE children
Health	National Health Strategic Plan (NHSP)	A five-year plan that guides the health sector from 2022–2026. The plan focuses on achieving national health goals, Universal Health Coverage (UHC)
	Zambia National Health Policy	A policy that recognizes the importance of food safety, environmental health, climate change, and port health services
	Essential Newborn Care Guidelines	These guidelines encompass evidence-based, newborn interventions for essential and immediate care, which are designed to address the most important causes of newborn illness and death in Zambia
	Zambia Consolidated Guidelines for Treatment and Prevention of HIV Infection	These are guidelines on how to treat and prevent HIV. It has a special section on prevention of Mother to Child Transmission (PMTCT), treatment and care
	Zambia Family Planning Guidelines and Protocols	Providers offer comprehensive, up-to-date information

Thematic area	Policy Name	Description
		and guidance to empower healthcare professionals in meeting the needs of clients. The guidelines are also important for providing information regarding contraception choice and supporting the use of contraceptives
	Health Sector Monitoring and Evaluation Framework NHSP 2022-2026	Framework for monitoring and evaluation of health indicators including ECD indicators (0-3 years)
	Community Based Volunteers' Integrated Care Handbook	This handbook has been designed to provide the necessary knowledge and skills required by a CBV to provide services in line with the Community Health Services Package (CHSP).
	Guidelines For Health Data: Access, Sharing and Use	This provides guidelines of accessing, sharing and using data on health, including child health and early childhood development indicators.
Nutrition	Food-based dietary guidelines	Provides guidelines aimed at promoting healthy dietary patterns to contribute towards addressing the high levels of undernutrition and the emerging problem of overweight and diet-related NCDs
	An Investment Framework for Nutrition in Zambia: Reducing Stunting and Other Forms of Child Malnutrition (2015)	This framework emphasizes the importance of investing in nutrition interventions to combat child malnutrition and stunting, outlining the necessary strategic investments to achieve these goals.
	Guidelines for the Implementation of the School Health and Nutrition Programme (2011)	Issued by the Ministry of General Education, these guidelines promote healthy eating habits and nutrition-sensitive school meals, integrating nutrition education into the school curriculum.
Play and recreation	The Local Government Act, 2019	Article 6 of the local government act state that a local authority shall, in relation to public amenities— (a) establish and maintain parks, zoos, gardens, pleasure grounds, camping grounds, caravan sites and open spaces
	Constitution of Zambia (Amendment) of 2016	Among other things, section C of the amended constitution states that local authority functions include developing, running and maintaining local play parks and recreation
	National Volunteer Policy	Provide a framework to guide persons who seek to engage in volunteer work as well as all

Thematic area	Policy Name	Description
Social and child protection		stakeholders that engage services of volunteers in order to build an effective and efficient volunteer sector
	The Minimum Standards for Child Care Facilities	Provides holistic guidelines on care and support to children in centres and institutions caring for children.
	Child Safeguarding Policy	Provides guidance on principles, values, standards, responsibilities and commitment to child safeguarding
	Children's Code Act of 2022	Aims to protect children and ensure their rights are upheld
	UN convention on the Rights of Children	Zambia is a signatory to the UN Convention on the Rights of Children. The UNCRC is a legally-binding international agreement setting out the civil, political, economic, social and cultural rights of every child
	National Civil Registration and Vital Statistics Strategy	Aims at promoting a holistic approach to civil registration and identity management from birth to death

How do these policies interact with the wider child development policy? The National Child Policy (NCP) serves as the overarching policy framework that guides all efforts aimed at ensuring the wellbeing, protection, development, and participation of children in Zambia. All sector-specific policies—whether in education, health, nutrition, protection, or recreation—are intended to align with and operationalize the vision and objectives of this central policy. Although Zambia has a rich policy ecosystem as it relates to ECD, their implementation is fragmented because at the time of the review there was no national multisectoral ECD policy in enforcement. National ECD coordination committee was also very strengthened to effectively spearhead the coordination of ECD programs because the committee was not an act of parliament. However, at the time of the review a draft national multisectoral ECD policy had been developed awaiting approval.

Barriers or Enablers of ECD Programming

Interviews from Key informants reviewed several enablers and barriers to ECD programming. The issues identified as barriers and enablers and their associated descriptions have been described in the table below:

Category	Issue/Factor	Description
Barriers to ECD programming	Urban-Rural ECD Access Gap	Urban centres have better access to ECD services, while rural areas face long travel distances.
	Shortage of Trained ECD Workforce	Few trained ECD professionals; government and university programs do not meet demand.

	Traditional Childcare Over Formal ECD	Many families rely on traditional caregiving, leaving young children in sibling care.
	Limited National ECD Coverage	ECD hubs have been piloted but lack nationwide scaling due to funding limitations.
	Parental Financial Constraints	Parents prioritize older children's education over ECD due to financial constraints.
	Cultural Myths on Child Nutrition	Cultural beliefs, such as withholding certain foods, negatively impact child nutrition.
	Limited ECD Infrastructure & Funding	ECD centres have limited infrastructure and funding, restricting service expansion.
	Low Parental Awareness of ECD	Many parents do not understand the long-term benefits of ECD, limiting engagement.
	Limited budget allocation	Limited budget is allocated to ECD related activities making it challenging program ECD interventions
	Overdependence on donor support	Overdependence on donor support make it hard to develop home grown sustainable and scalable programs after end of donor support
Enablers to ECD programming	High level political will	High level political well and engagement in ECD
	High Parental Interest in ECD	Once informed about ECD, parents appreciate its benefits and engage actively.
	ECD policy ecosystem	Interconnectedness and interdependent of different policies and framework create an entry point for multisectoral implementation of ECD services to achieve desired outcomes
	Community-Based ECD Model INSAKA	The INSAKA model provides integrated health, nutrition, and early learning support.
	Self-Sustaining ECD Centre Models	Models that integrate sustainability measures reduce reliance on donor funding.
	High Demand for ECD Services	The free education policy has significantly increased ECD enrolment rates.
	Constituency Development Fund (CDF) Support	Engagement with Parliament and local leaders has enabled CDF funding for ECD hubs.
	Public-Private Partnership Model	Partnerships with private entities help develop and maintain ECD infrastructure.
	Parental Willingness to Pay Fees	Some parents are willing to pay for structured ECD services if they perceive value.

	Government & NGO ECD Partnerships	Collaboration between NGOs and the government strengthens ECD implementation.
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ECD Multisectoral Coordination

At national, provincial and district level, ECD multisectoral coordination committees do exist. The main stakeholders in this committees are the Ministry of Health, Ministry of Education and Ministry of Community Development and Social Services. Others are the Ministry of Homes Affairs, Department of Birth Records, Victim Support Unit of the Police Ministry of Agriculture and the Local Government.

Challenges in Coordination

1. **Lack of Unified ECD Policy** – There is no central policy guiding ECD, leading to inefficiencies. Multisectoral ECD policy in draft form
2. **Weak Multisectoral Coordination** – Ministries and stakeholders work independently, limiting synergy.
3. **Fragmentation of ECD Implementation** – Sectors operate in silos instead of integrating efforts across health, education, and social services.
4. **Sectoral Silos in ECD Planning** – District Development Committees separate child-related programs rather than merging them into a unified ECD framework.
5. **Limited Local ECD Committees** – At the district level, there are no structured committees to coordinate ECD services effectively.
6. **Uncoordinated NGO Efforts** – NGOs provide support but are not well integrated with government programs.

Opportunities for Improved Coordination

1. **Government and NGO ECD Partnerships** – Engagement between the government, NGOs, and private actors can enhance program implementation.
2. **District Commissioner-Led Coordination** – Giving the District Commissioner oversight can improve accountability and efficiency.
3. **Inter-ministerial ECD Committee** – A national steering committee with representation from key ministries (Education, Health, Community Development) can streamline coordination.
4. **Strengthening NGO-Government Collaboration** – Networks like ECDAN Zambia facilitate collaboration for advocacy and program delivery.
5. **Local Authority-Led ECD Coordination** – Mandating local governments to oversee ECD services can improve integration.
6. **Community-Led ECD Monitoring & Evaluation** – Traditional leaders, community figures, and parents can be involved in ensuring quality service delivery

Discussion of Findings

This section of the report will discuss findings reported in the results section. The results will be discussed according to the study objective. These are around early childhood development outcomes, HH caregiving practices, early childhood development services and on policy and regulatory landscape.

Thematic 1: Early Childhood Development Outcomes

Overall Child Development

Overall child development in Lusaka and Chongwe follows a generally linear trajectory with age. However, the study findings reveal that in both cities, children's development scores are significantly higher than the norm reference for children aged 0–17 months, but drop notably after this age, falling below the reference scores. Several possible explanations may account for this trend. Firstly, the number of children delivered in health facilities by skilled healthcare providers has significantly improved—from approximately 70% in 2014 to over 93% in 2024.²⁸ This improvement in early healthcare access likely contributes to better developmental outcomes in the early months of life. During the same period, the proportion of children under five sleeping under a mosquito net increased from 68% in 2014 to 78% in 2024. Improved access to medical care and hygienic conditions during childbirth reduces the risk of complications and infections that can affect both the mother and baby. If not properly managed, these complications may lead to serious illness, ultimately impacting child development outcomes.²⁹ Secondly, between birth and approximately 24 months, biological mothers are often present to breastfeed and care for their young children, providing essential physical, socioemotional, and cognitive stimulation. However, after weaning—typically around 24 months—many mothers begin leaving their children in the care of siblings or other caregivers. This shift often results in reduced mother–child interaction and less consistent stimulation, which may negatively impact the child's developmental trajectory, hence the observed drop in our study.³⁰ Therefore, child care after this period begins to suffer due to the absence of a biological parent. UNICEF estimated 78% of all children under 5 years in Zambia, are at risk of poor developmental outcomes. The study findings show that, after 17 months of age, children in both Lusaka and Chongwe score significantly lower than the norm reference, highlighting the need for more concerted efforts to improve child development outcomes. This trend has important implications for the design of future Early Childhood Development (ECD) interventions. Specifically, interventions should not focus solely on biological parents. In the Zambian context, secondary caregivers often spend more time with children, as primary caregivers—particularly mothers—are heavily involved in income-generating activities.

While factors such as household vulnerability and settlement type do play a role in predicting child development, this study found that the home environment was a more significant predictor of child outcomes in both cities. A nurturing home environment supports cognitive, emotional, and social development. Research consistently shows that higher scores on the HOME Inventory are strongly associated with positive developmental outcomes in children.³¹ The HOME Inventory assesses various aspects of the home environment that support child development, which can be grouped into three main areas. First is the physical environment, which includes the availability of space and access to materials that encourage learning and play. Second is parental interaction, referring to the warmth, responsiveness, and engagement of caregivers—factors that are essential for a child's emotional and

²⁸ Zambia Demographic and Health Survey 2024

²⁹ Mselle, L. T., Moland, K. M., Mvungi, A., Evjen-Olsen, B., & Kohi, T. W. (2013). Why give birth in health facility? Users' and providers' accounts of poor quality of birth care in Tanzania. *BMC health services research*, 13, 1-12.

³⁰ Hapunda, G. (2022). Caregiving roles of female guardians, older siblings, and time spent on child activities. *Family Relations*, Doi: <https://doi.org/10.1111/fare.12762>

³¹ Lansford, J. E., Odgers, C. L., Bradley, R. H., Godwin, J., Copeland, W. E., Rothenberg, W. A., & Dodge, K. A. (2023). The HOME-21: A revised measure of the home environment for the 21st century tested in two independent samples. *Psychological assessment*, 35(1), 1.

social development. A nurturing home environment also includes educational opportunities, such as access to learning materials and activities that stimulate cognitive development—examples include books, educational toys, and interactive experiences. Analysis of the different aspects of the HOME environment revealed a positive correlation between these factors and child development outcomes. Therefore, interventions should focus on supporting households to enhance the home environment, particularly by promoting the use of locally available materials for play and by encouraging parents and other caregivers to actively engage in stimulating interactions with their children.

Nutritional Status

In the last two decades, Zambia has made notable strides in improving child nutrition indicators, particularly in reducing stunting rates from 53% in 2001 to 32% in 2024³². However, despite these significant strides, Chongwe continues to struggle with higher-than-average child nutrition indicators such as stunting rates, reported at 37.8%, compared to 27% in Lusaka and the national estimate of 32.1%^{ibid}. This suggests that while national efforts have yielded positive results, local disparities remain critical challenges and are a risk to optimal early childhood development and the general well-being of children.

The worrying trends in Chongwe and Lusaka can be partially attributed to the adverse drought of the 2023/2024 farming season³³, which brought about food scarcity and potential rationing in households. The significant rise in the cost of living over the past 6 years has undoubtedly strained households' budgets, making it increasingly difficult for many to afford nutritious food. The JCTR (Jesuit Centre for Theological Reflection) Basic Needs and Nutrition Basket (BNNB) for a family of five in Lusaka is currently at ZMW 11,546.87 as of March 2025³⁴. This reflects a persistent rise in the cost of living in Zambia. This economic pressure is compounded by other factors such as inadequate knowledge about proper food preparation and poor feeding practices, which may further exacerbate the nutritional status of children³⁵.

Further, the 2024 ZDHS shows that the proportion of women practising exclusive breastfeeding has decreased from 73% in 2013 to 64% in 2024. This decline may reflect several reasons regarding infant feeding practices. The reduction in exclusive breastfeeding practices can have long-term implications on child health, contributing to underweight and wasting, which also predisposes children to common child illnesses such as diarrhoea, ultimately affecting their physical, motor, and cognitive development³⁶.

Moreover, this survey indicates that slightly over half of the households in both Lusaka and Chongwe expressed a willingness to pay for nutritious food. This highlights a critical gap in both awareness and accessibility, suggesting that even when nutritious options are available, financial constraints deter families from prioritizing them.

School Readiness

The results from the IDELA's framework provided important insights regarding the status of early childhood development (ECD) and school readiness in Lusaka and Chongwe districts. This discussion section focuses on three key themes: (1) developmental trends for children aged 3.5 to 6 years, (3)

³² Zambia Statistics Agency, Ministry of Health (MoH) [Zambia], and ICF. 2024. Zambia Demographic and Health Survey 2024: Key Indicators Report. Lusaka, Zambia, and Rockville, Maryland, USA: Zambia Statistics Agency, MoH, and ICF.

³³ UNICEF (2024) Immediate action required to prevent malnutrition crisis among children in Zambia, <https://www.unicef.org/zambia/press-releases/immediate-action-required-prevent-malnutrition-crisis-among-children-zambia-unicef>

³⁴ JCTR March 2024 BNNB Bulletin- https://repository.jctr.org.zm/bitstream/handle/20.500.14274/1864/20250326-bnnb_march_bnnb_statement-V02.pdf?sequence=1&isAllowed=y

³⁵ Andrew Banda., Elizabeth T. Nyirenda., Chabila C. Mapoma, Bwalya B. Bwalya., Nkuye Moyo. (2022). Influence of Infant and Young Child Feeding Practices on Stunting in ged 6 – 23 months in Zambia. *Adv Nutr Food Sci*, 7(1), 117-127.

³⁶ Sichimba, F., Nakazwe, K. C., Halawi, L., & Khalaf, A. (2025). Exploring exclusive breastfeeding readiness: experiences of first-time mothers in Lusaka, Zambia. *International Journal of Qualitative Studies on Health and Well-Being*, 20(1). <https://doi.org/10.1080/17482631.2025.2463159>

important variations disparities in school readiness between districts, and (4) implications for policy and intervention.

Despite inherent SES variations among districts, results revealed significant age-related progression trends with critical policy implications. Overall, children exhibited poor performance across all domains, reflecting prevailing global trends at the ECD level.^{37,38,39} Emergent numeracy and literacy presented greater challenges than other developmental domains.⁴⁰ Children demonstrated rapid developmental growth between ages 4 and 5, with domain-specific analysis indicating the highest development rates in motor skills; Chongwe children outperformed those in Lusaka. Motor skills development peaked at age 6, consistent with global trends showing that physical development precedes cognitive milestones. The observed lag in emergent literacy, particularly in Chongwe at age 6, may result from systemic challenges in access to and quality of ECD services in Zambia, a pattern common across sub-Saharan Africa.⁴¹

Children demonstrated strong motor skills compared to other areas, yet their overall development levels fall short of expectations. Effective interventions in numeracy and literacy are necessary for children in Lusaka and Chongwe. However, the stagnation in literacy scores after age 5 in Chongwe indicates that current preschool programmes are not sufficiently reinforcing foundational skills. The significant literacy gaps—74.6% unprepared in Lusaka and 78.2% in Chongwe—reflect findings from Zambia’s National Assessment Survey⁴², highlighting a national crisis in early literacy.

Girls outperformed boys in most areas, although there are achievement disparities between the two cities. In Chongwe, girls scored higher in motor skills, emergent numeracy, and emergent literacy. The superior motor skills of Chongwe girls may be attributed to their engagement in physical activities, such as fetching water and outdoor chores, which are more common among rural children. This trend of girls outperforming boys contradicts global norms⁴³, but may be explained by cultural factors in rural areas, including early maturation and caregiver expectations. In contrast, gender gaps are minimal in Lusaka⁴⁴. In terms of school readiness, the Lusaka district outperformed Chongwe, with 36.5% of children ready for school compared to 26.3% in Chongwe. This difference may be due to Lusaka's advantages, including better resources, more early childhood education centers, and higher parental education levels.⁴⁵

The number of struggling children to achieve mastery of basic skills is staggering in both cities. The proportion of the struggling was much larger in Chongwe, which shows how disadvantaged rural-based children are overall. The main reason for this lag may be lack of resources and ECE opportunities.⁴⁶ Additionally, urban schools have access to trained and well-paid caregivers and teachers, while rural schools depend lowly-educated school leavers without appropriate pedagogical content knowledge. There are fewer children who graduate to mastery level across domain, but the number is much higher in emergent literacy.

³⁷ OECD. (2023). *Education at a Glance 2023: OECD Indicators*. OECD Publishing.

³⁸ UNESCO. (2022). *Global Education Monitoring Report: Non-state actors in education*. UNESCO Publishing.
<https://doi.org/1.54676/YDLT1694>

³⁹ World Health Organization. (2018). *Nurturing care for early childhood development: A framework for helping children survive and thrive to transform health and human potential*. WHO. <https://apps.who.int/iris/handle/10665/272603>

⁴⁰ Jukes, M. C. H., et al., (2018). "Respect is an investment": Community perceptions of social and emotional competencies in early childhood from Mtwara, Tanzania. *Global Education Review*, 5(2), 160-188.

⁴¹ Matafwali, B. (2020). Early literacy in Zambia: Challenges and opportunities. *Zambia Journal of Education*, 8(1), 12–30

⁴² MoGE. (2019). *National Assessment Survey of learning achievement at Grade 2: Baseline report*. Lusaka: Author.

⁴³ OECD. (2023). *Education at a Glance 2023: OECD Indicators*. OECD Publishing.

⁴⁴ Lancy, D. F. (2016). *Anthropological perspectives on children as helpers, workers, and artisans*. Palgrave Macmillan.

⁴⁵ Dembo, M. H. (1994). *Applying educational psychology* (5th Ed.). Longman.

⁴⁶ Dembo, M. H. (1994). *Applying educational psychology* (5th Ed.). Longman.

The findings indicate significant developmental disparities based on settlement type, aligning with previous research on the impact of neighbourhoods on child development⁴⁷. The notable literacy gap observed in Chongwe's unplanned settlements ($\Delta M = .273$) highlights that early language development is particularly susceptible to environmental disadvantages⁴⁸. The minimal disparities in motor skills observed between Lusaka ($\Delta M = .060$) and Chongwe ($\Delta M = .154$) lend support to neurodevelopmental models that propose physical milestones are less susceptible to environmental influences than cognitive skills.⁴⁹ These findings underscore the necessity for targeted early childhood interventions in unplanned settlements, with a particular emphasis on the development of foundational literacy and numeracy skills.⁵⁰

School readiness is influenced by a number of factors in the two cities. In Lusaka, household (HH) vulnerability has significant effects on school readiness; however, its effects diminish when other factors are accounted for. Research suggest that this reduction in influence may be due to the indirect effects of mediating factors such as the home environment and caregiver characteristics.⁵¹ These factors are more susceptible to the influence of economic instability, which is inherently an urban phenomenon. This could also explain why HH vulnerability has less influence in more rural Chongwe, where stronger community support and lower stability in socioeconomic status (SES) exist⁵².

The type of settlement affects school readiness in both cities. The lack of facilities such as schools and WASH facilities in unplanned settlements, compared to planned settlements, explains the disproportionate variations in children's school preparedness.⁵³ The influence of siblings has a positive effect because of the emotional and cognitive stimulation they provide to younger children. The strong effect of sibling relationships in urban areas may be attributed to closely knit interpersonal interactions facilitated by the prevalence of nuclear families.⁵⁴ In contrast, in Chongwe, the limited influence of siblings may reflect the diluting impact of extended family structures.⁵⁵

The significant influence of the home environment in Lusaka, compared to Chongwe, may suggest that the quality of the home learning environment is more critical in urban areas. Urban parents tend to recognize the benefits of a good education for their children and thus are more likely to provide the necessary support by equipping their homes with educationally stimulating resources to compete in resource-competitive settings.⁵⁶ Rural children often seek alternative pathways for stimulation through outdoor play and community interactions.⁵⁷

The significant effects of caregiver education in Chongwe, as opposed to Lusaka, may be due to the smaller number of educated individuals; thus, the influence of the few educated caregivers may explain their disproportionate impact.⁵⁸ In Lusaka, however, other factors such as preschool availability and

⁴⁷ Leventhal, T., & Brooks-Gunn, J. (2000). The neighbourhoods they live in: The effects of neighbourhood residence on child and adolescent outcomes. *Psychological Bulletin*, 126(2), 309-337. <https://doi.org/10.1037/0033-2909.126.2.309>

⁴⁸ Shonkoff, J. P., et al. (2012). The lifelong effects of early childhood adversity and toxic stress. *Paediatrics*, 129(1), e232-e246. <https://doi.org/10.1542/peds.2011-2663>

⁴⁹ Nelson, C. A., et al. (2007). Cognitive recovery in socially deprived young children: *The Bucharest Early Intervention Project*. *Science*, 318(5858), 1937-194. <https://doi.org/10.1126/science.1143921>

⁵⁰ Black, M. M., et al. (2017). Early childhood development coming of age: Science through the life course. *The Lancet*, 389(10064), 77-9. [https://doi.org/10.1016/S0140-6736\(16\)31389-7](https://doi.org/10.1016/S0140-6736(16)31389-7)

⁵¹ Bradley, R. H., & Corwyn, R. F. (2002). Socioeconomic status and child development. *Annual Review of Psychology*, 53, 371-399.

⁵² Grantham-McGregor, S., et al. (2007). Developmental potential in the first 5 years for children in developing countries. *The Lancet*, 369(9555), 60-7.

⁵³ Ejdemyr, S., Kramon, E., & Robinson, A. L. (2018). Segregation, polarization, and ethnic conflict in urban Africa. *American Political Science Review*, 112(3), 551-568.

⁵⁴ Downey, D. B. (2001). Number of siblings and intellectual development: The resource dilution explanation. *American Psychologist*, 56(6-7), 497-504. <https://doi.org/10.1037/0003-066X.56.6-7.497>

⁵⁵ Weisner, T. S. (2017). *The eco-cultural project of human development: Why ethnography and its findings matter*. University of Chicago Press.

⁵⁶ Bradley, R. H., & Corwyn, R. F. (2002). Socioeconomic status and child development. *Annual Review of Psychology*, 53, 371-399.

⁵⁷ Tudge, J. R. H., et al., (2015). Still misused after all these years? A reevaluation of the uses of Bronfenbrenner's bioecological theory of human development. *Journal of Family Theory & Review*, 8(4), 427-445. <https://doi.org/10.1111/jftr.12165>

⁵⁸ Serpell, R. (2017). Early literacy in Zambia. *International Journal of Educational Development*, 55, 117-13.

age-appropriate social media may be diminishing the influence of education⁵⁹. Additionally, it is not surprising that older caregivers exert positive effects on school readiness, primarily due to their parenting experience and stability.⁶⁰

Overall, these results illustrate context-specific drivers shaped by urban-rural disparities.⁶¹ It is clear that child development is influenced by layered environmental factors. Thus, policy interventions should specifically address the factors impacting school readiness. For instance, in Lusaka, intervention programmes should focus on improving home environments and leveraging sibling dynamics. In contrast, in Chongwe, intervention programmes will only be effective if they aim at enhancing caregiver education and settlement planning.⁶²

Thematic 2: Household Caregiving Practices

Caregiver knowledge of child development is essential for supporting healthy growth and well-being. By understanding typical developmental stages—such as physical, cognitive, and emotional milestones—caregivers can offer appropriate support and stimulation for each age. This awareness also helps them recognize signs of developmental delays or concerns early, allowing for timely intervention.⁶³ However, this study found that caregiver knowledge of child development was very low. On average, parents knew only about half of what is expected regarding key developmental milestones. For example, less than 20% of caregivers knew when children typically begin to smile socially—such as smiling in response to another person’s face—or when mothers should start looking at picture books with their children. These findings on low caregiver knowledge are consistent with results from similar studies conducted in Zambia’s Southern Province.⁶⁴ The low levels of caregiver knowledge on child development can largely be attributed to the lack of targeted programs at both community and household levels. In Chongwe, lower levels of household vulnerability were associated with higher caregiver knowledge, suggesting that when families are less vulnerable, caregivers are better able to focus on understanding optimal child development. Importantly, maternal education emerged as a strong predictor of caregiver knowledge. Educated mothers were more likely to have a better understanding of child development—an observation consistent with findings from studies among Turkish mothers. These findings have important implications for policy and intervention: there is a clear need to create more opportunities for educating mothers.

Parents and caregivers play a critical role in forging children’s early environment. Therefore, there attitudes and beliefs towards child development and children remain crucial.⁶⁵ However, in the two cities, positive parenting attitudes and beliefs were relatively low. Attitudes and beliefs around attachment, early learning and rules were explored and were found to be low. These have implication

⁵⁹ Heckman, J. J. (2006). Skill formation and the economics of investing in disadvantaged children. *Science*, 312(5782), 1900–1902.

⁶⁰ Bornstein, M. H., et al. (2006). Mother-child emotional availability in ecological perspective: Three countries, two regions, two genders. *Developmental Psychology*, 42(5), 831–84.

⁶¹ Bronfenbrenner, U., & Morris, P. A. (2006). *The bio-ecological model of human development*. *Handbook of Child Psychology*.

⁶² Engle, P. L., et al., (2011). Strategies for reducing inequalities and improving developmental outcomes for young children in low-income and middle-income countries. *The Lancet*, 378(9799), 1339–1353

⁶³ Ertem, I. O., Atay, G., Dogan, D. G., Bayhan, A., Bingoler, B. E., Gok, C. G., ... & Isikli, S. E. D. A. T. (2007). Mothers' knowledge of young child development in a developing country. *Child: care, health and development*, 33(6), 728–737.

⁶⁴ Chipili, M. (2020). The use of participatory theatre in early childhood development interventions: a focus on mother groups in Choma and Pemba, Zambia (Doctoral dissertation, The University of Zambia).

⁶⁵ Hembacher, E., & Frank, M. C. (2020). The early parenting attitudes questionnaire: Measuring intuitive theories of parenting and child development. *Collabra: Psychology*, 6(1), 16.

on child development. For instance, early relationships are crucial in shaping an individual's capacity for future connections, with implications for social adjustments.⁶⁶ The study found that lower household vulnerability plays a key role in strengthening early parenting attitudes and beliefs (EPAB). Therefore, interventions that reduce household vulnerability are essential for improving EPAB. Additionally, enhancing caregiver knowledge of child development is equally important in promoting positive parenting attitudes and beliefs. Due to low caregiver knowledge and prevalent negative attitudes toward early parenting, the home environments of most children in the two cities are not conducive to supporting positive child development. Overall, this study highlights that caregiving practices in the two cities are shaped by several key factors: gender norms and cultural beliefs, limited parenting knowledge, economic hardship, social challenges that affect parenting capacity, and the lack of adequate support systems.

The prevalence of physical and psychological aggression or abuse predisposes children to lower childhood productivity and inability to flourish, affecting their motor, cognitive and physical growth⁶⁷. Consistent with this report, the Violence Against Children Survey (VACS) 2014 showed that violence against children in Zambia is rife and persistently high⁶. Currently, there has been an increase in cases of child abuse such as incest, defilement, and child molestation, in some cases perpetrated by parents and close family members⁶⁸. Interventions should be targeted to enhance child nutrition and overall well-being. Because children who are malnourished are less likely to flourish and grow up into productive adults⁶⁹⁷⁰. Further while recognising that Zambia has made commendable progress in child nutrition over the past two decades, significant challenges remain, particularly in specific regions like Chongwe. Addressing these disparities requires targeted interventions that consider the local context, economic factors, and community-driven educational initiatives on nutrition with a multi-component approach to ensure that all children have the opportunity to flourish and enhance childhood

Thematic area 3: Early Childhood Development Services

Overall, the quality of Early Childhood Education (ECE) centers in the two cities was generally found to be good. However, the learning dimension fell below standard in Lusaka's unplanned settlements and in rural areas of Chongwe.

While the overall results suggest good quality, it is important to interpret these findings within the specific context and circumstances of the assessment period. At the time of the baseline, schools had just reopened, and a new curriculum had been introduced. Most teachers had recently undergone training or orientation on the new curriculum in the weeks leading up to the assessment—this recency and heightened preparedness may have positively influenced the findings. Additionally, the reopening of schools coincided with a cholera outbreak, prompting authorities to enforce strict mitigation measures. As a result, many schools had visibly improved their physical facilities and environments, which likely contributed to the high ratings observed during the assessment.

We also observed that the quality of ECE centers varied by ownership. Community-owned centers generally had lower quality compared to government and private centers, with private centers demonstrating the highest quality overall. Community-owned centers are often constrained by limited

⁶⁶ Jethava, V., Kadish, J., Kakonge, L., & Wiseman-Hakes, C. (2022). Early attachment and the development of social communication: A neuropsychological approach. *Frontiers in Psychiatry*, 13, 838950.

⁶⁷ Lee N, Massetti GM, Perry EW, Self-Brown S. Adverse Childhood Experiences and Associated Mental Distress and Suicide Risk: Results From the Zambia Violence Against Children Survey. *J Interpers Violence*. 2022 Nov;37(21-22):NP21244-NP21265. doi: 10.1177/08862605211056726

⁶⁸ World Vision Zambia (2025) Joining Forces Alliance Zambia Condemns Rising Cases of Rape and Defilement in Zambia – Calls for Urgent Action <https://www.wvi.org/stories/zambia/joining-forces-alliance-zambia-condemns-rising-cases-rape-and-defilement-zambia>

⁶⁹ Cavagnari, B. M., Guerrero-Vaca, D. J., Carpio-Arias, T. V., Duran-Aguero, S., Vinuesa-Veloz, A. F., Robalino-Valdivieso, M. P., Morejón-Terán, Y. A., & Vinuesa-Veloz, M. F. (2023). The double burden of malnutrition and gross motor development in infants: A cross-sectional study. *Clinical Nutrition*, 42(7), 1181–1188. <https://doi.org/10.1016/j.clnu.2023.05.001>

⁷⁰ De, P., & Chattopadhyay, N. (2019). Effects of malnutrition on child development: Evidence from a backward district of India. *Clinical Epidemiology and Global Health*, 7(3), 439–445. <https://doi.org/10.1016/j.cegh.2019.01.014>

resources and typically rely on less qualified staff. In contrast, private centers tend to maintain higher standards due to their ability to invest in quality teaching and learning materials, as well as attract highly qualified and experienced ECD teachers. These findings are therefore expected and align with existing narratives on ECE quality by ownership type.⁷¹ Teacher characteristics that appear to predict ECE quality include educational qualifications and length of experience. Both higher levels of education and longer durations of teaching experience were positively associated with better quality. This finding aligns with meta-analytic evidence showing a strong link between teacher qualifications and the quality of early childhood education.⁷² In addition, we found that high number of children with disabilities in a centre was associated with better quality. This was somewhat surprising. A possible explanation for the finding that a higher number of children with disabilities in an ECE class is associated with better early childhood center quality is that higher-quality centers are more likely to adopt inclusive education practices and have the capacity to support diverse learners. These centers often attract children with disabilities because they offer better infrastructure, trained teachers, and individualized support, making them more appealing to parents seeking appropriate learning environments for their children. Additionally, inclusive centers may receive additional resources, monitoring, or training from government or partner organizations aimed at supporting children with special needs, which in turn raises the overall quality of the center. Therefore, the presence of more children with disabilities may be an indicator of a center's inclusiveness, professionalism, and responsiveness, rather than a direct cause of higher quality.

The differences in respondents' willingness to pay for ECD services is reflection of their demand for these services. In Lusaka, residents unplanned settlements exhibited higher demand for services to meet basic necessities such as hygiene, nutrition and growth monitoring, while those from planned settlements demonstrate their aspirations towards recreational facilities. This is possibly because they are already meeting their basic needs. The demand patterns also correspond with public expenditures which have put more emphasis on health and nutrition⁷³. Generally, findings from Chongwe show that households from planned residences are more willing to pay for more ECD services than those from unplanned residences. The differences are largely an indication of low purchasing power among poor household in unplanned settlements in Chongwe compared to those coming from planned settlements. It is worth noting that some of the households in Chongwe are from rural areas, where poverty tends to be higher than their urban counterparts. Furthermore, ECD facilities are disproportionately more distributed in urban areas⁷⁴ making it difficult for residents in unplanned and especially rural areas not to appreciate the importance of ECD.

The availability of play and recreation facilities for children falls below desirable standards. In Lusaka—despite having one of the highest numbers of public and planned play parks—most were found to be in a dilapidated and nonfunctional state. Of the 33 identified parks, only 2 were operational. Although 9 parks were under private lease agreements, most private partners had not developed or improved the facilities.

Play facilities are predominantly in private hands and often located in affluent areas. Access usually comes at a fee, which disadvantages children from low-income households. While demand for play facilities is high, willingness to pay remains low, making it difficult to sustainably develop new play spaces. As a result, low-cost play parks built using locally available materials should be prioritized over expensive, high-end alternatives. Furthermore, awareness of the developmental importance of play is relatively low among caregivers, which reduces efforts to create or support play opportunities at home

⁷¹ Rougier, E. B. (2022). Does Private Ownership in the Kindergarten Sector Benefit the Children? Empirical Evidence from Norway Does the kindergarten's ownership form impact the quality? (Master's thesis).

⁷² Manning, M., Wong, G. T., Fleming, C. M., & Garvis, S. (2019). Is teacher qualification associated with the quality of the early childhood education and care environment? A meta-analytic review. *Review of educational research*, 89(3), 370-415.

⁷³ UNICEF.2023. Investing in Early Childhood to advance Human Capital Development: Early Childhood Development Budget Brief.

⁷⁴ Namponje, L.2017. The Status of Early Childhood Care, Development and Care in Zambia. Policy Monitoring Research Center.

and in communities. The lack of accessible play areas is further compounded by concerns about unsafe physical environments and traffic hazards, which discourage unstructured outdoor play, especially for children in high-density areas.

A mapping of ECD-related facilities revealed that Lusaka has a greater number of services—such as parks, health centers, police stations, and green spaces—compared to Chongwe. However, these services are unevenly distributed and densely concentrated in specific areas. This unequal access may contribute to lower levels of school readiness, particularly among children in underserved parts of Lusaka.

Overall, we find that caregivers are unwilling to pay for ECD services and products. Our finding of a negative correlation between caregivers' education and their willingness to pay for ECD services is surprising. Studies indicate that caregivers play a major role in children's ECD and thus their education is expected to increase their demand for ECD services⁷⁵. While this is a surprising result, it is plausible considering that more educated caregivers have less time to spend with children. This may be the reason for their low willingness to pay for ECD services. This may cause some of the caregivers not to value ECD services. Findings from Chongwe show a positive correlation between the age of the caregiver and willingness to pay. This is plausible because the age of the caregiver may denote experience. Thus, older caregivers may be more willing to pay for ECD services compared to their younger counterparts.

Thematic area 4: Structural and policy landscape

The evidence of the structural and policy landscape shows that Zambia has a fairly extensive policy and regulatory ecosystem that touches on various aspects of child development. This policy ecosystem spans multiple sectors, including child wellbeing, education, health, nutrition, play and recreation, and child protection, which indicates a commendable multisectoral approach to ECD. The National Child Policy (NCP) serves as the overarching framework to guide all efforts aimed at ensuring the wellbeing, protection, development, and participation of children in Zambia. Sector-specific policies, such as those in education, health, nutrition, protection, and recreation, are designed to align with and operationalize the NCP's vision and objectives.

However, despite the existence of this rich policy environment, the review indicates that implementation is fragmented. A key challenge is the absence of a national multisectoral ECD policy in force at the time of the review. This gap potentially leads to inefficiencies in the coordination and delivery of ECD programs and services. Furthermore, the effectiveness of the National ECD coordination committee has been hampered by its lack of grounding in an Act of Parliament, which weakens its capacity to spearhead the coordination of ECD programs.

The review also highlights several barriers and enablers to ECD programming in Zambia. Barriers include, urban-rural disparities in access to ECD services, shortages of trained ECD professionals, reliance on sometimes ineffective traditional childcare practices, limited national ECD coverage and infrastructure due to funding constraints, and parental financial constraints and cultural myths that affect child nutrition. These barriers pose significant challenges to the effective implementation and scaling up of ECD programs across the country.

Conversely, the review identifies several enablers that can be leveraged to improve implementation of ED services. These are itemized below:

- High-level political will and engagement in ECD.
- High parental interest in ECD, once parents are informed of its benefits.
- The interconnectedness of the existing policy framework.

⁷⁵ Ayob, Z., Christopher, C. and Naidoo, D., 2021. Caregivers' perception of their role in early childhood development and stimulation programmes in the early childhood development phase within a sub-Saharan African context: An integrative review. *South African Journal of Occupational Therapy*, 51(3), pp.84-92.

- Community-based ECD models (like the INSAKA model), a model that is self-sustaining ECD center models.
- Increased ECD enrollment rates due to free education policies.
- Support from the Constituency Development Fund (CDF).
- Public-private partnerships.
- Parental willingness to pay fees for ECD services.
- Government and NGO partnerships.
- These enablers provide a foundation for strengthening ECD programming in Zambia.

Coordination challenges are also evident at the national, provincial, and district levels. These challenges include the lack of a unified ECD policy, weak multisectoral coordination, fragmentation of ECD implementation, sectoral silos in ECD planning, limited local ECD committees, and uncoordinated NGO efforts. These coordination issues hinder the effective delivery of integrated ECD services. The evidence on the policy landscape shows that while Zambia has made strides in developing a comprehensive policy framework for ECD, challenges remain in implementation and coordination. Addressing these challenges, particularly the need for a unified multisectoral ECD policy and strengthened coordination mechanisms, is crucial for ensuring that all children in Zambia have access to quality ECD services.

Recommendations

Based on the study findings and literature reviewed, several recommendations have been proposed. While there are minor differences between the two cities, the recommendation applies to all the two cities. The recommendations are organized around four thematic areas:

Early Childhood Development Outcomes

- **Prioritize developmental support for children especially after 17 months:** Tailor interventions to focus on the critical drop in development scores observed after 17 months of age by intensifying stimulation and responsive caregiving during this window while maintaining momentum for the 1000 critical days.
- **Integrate developmental screening into routine child health visits:** Ensure regular developmental assessments in child delivery services to identify at-risk children early and guide targeted support.
- **Expand direct school readiness programs:** Increase structured ECD programs for children aged 3–6 to promote early cognitive, motor, and socio-emotional skills, particularly in underserved areas.
- **Invest in interventions targeting stunting and undernutrition:** Design nutrition-sensitive ECD programs—such as feeding schemes, parenting education on infant diets, and micronutrient supplementation—especially for children under five, where stunting affects up to 38% in Chongwe.
- **Strengthen household-level awareness of child milestones:** Develop community-based caregiver education campaigns to increase recognition of developmental milestones and the importance of early intervention.

Household Caregiving Practices

- **Promote responsive caregiving:** Train and support parents and secondary caregivers (siblings, grandmothers, neighbors) on positive parenting and stimulation using home-based learning materials.
- **Caregiver education on child development:** Provide simplified parenting materials and community support programs for caregivers with limited knowledge on child development.

- **Promote positive discipline methods:** Scale up parenting programs that teach alternatives to physical and psychological punishment, especially in unplanned and rural settlements where such practices are more prevalent.
- **Leverage fathers'provider role for caregiving awareness:** Engage fathers in ECD messaging to expand their role beyond economic provision to include stimulation and interaction with children.
- **Support informal caregivers:** Recognize the role of siblings and extended family in childcare by creating community-based learning hubs or playgroups that engage these caregivers.
- **Promote exclusive breastfeeding and nutrition practices:** Increase household-level nutrition education and breastfeeding promotion using local health and community workers, especially where early introduction to solids is common.

Early Childhood Development Services

- **Increase access in low-income and rural areas:** Construct low-cost, community-run ECD centers with support from local government and NGOs, prioritizing underserved areas.
- **Improve quality of ECD centers:** Sustain providing training for ECD teachers and caregivers, ensuring centers meet minimum standards for learning materials, safety, and stimulation.
- **Play and recreational facilities:** Construct play and recreational facilities in communities using locally available and sustainable materials. Designate ECD week in which various activities that support ECD will be championed and implemented using a community driven approach.
- **Establish workplace-linked daycare programs:** Incentivize employers to offer or co-fund ECD centers as part of workplace benefits, particularly in urban settings.

Structural and Policy Landscape

- **Strengthen multisectoral coordination:** Institutionalize inter-ministerial ECD coordination at national and city levels, integrating health, education, nutrition, child protection, and local governance.
- **Increase public funding for ECD:** Advocate for increased ECD-specific budget allocations, particularly for children under 3 years, whose funding is currently disproportionately low.
- **Decentralize ECD policy implementation:** Empower local governments with funds and technical support to operationalize ECD policies tailored to local needs and vulnerabilities.
- **Monitor policy enforcement:** Strengthen accountability and enforcement of child protection laws (e.g., Children's Code Act 2022) and ECE policies across all settlements.
- **Support policy integration and data systems:** Develop an integrated ECD information system that tracks service coverage, workforce training, and child outcomes from 0- 6 years to inform planning and evaluation.
- **Promote private and community partnerships:** Encourage partnerships with faith-based, private, and community organizations to scale ECD services while ensuring adherence to quality standards.

Conclusion

The baseline study highlights critical developmental deficits among children in Lusaka and Chongwe, particularly after the age of 17 months. These are compounded by weak service delivery platforms, limited caregiver capacity, and systemic policy gaps. While political commitment exists, there is an urgent need to expand and integrate ECD services equitably across settlement types, with a focus on

caregiver education, infrastructure, and policy enforcement. The findings lay a strong foundation for targeted interventions and monitoring frameworks to enhance developmental outcomes for Zambia's youngest citizens

Appendix

Inception report	
Code book	
Date set	